

SURVEILLANCE OF ANTIMICROBIAL RESISTANT BACTERIA IN BELGIAN HOSPITALS

National report
Data up to and including 2024

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EXECUTIVE SUMMARY



Introduction

Antibiotics have revolutionized modern medicine, but their overuse and misuse have reduced their effectiveness. Some bacteria have adapted to survive or neutralize certain antibiotics, a phenomenon known as antimicrobial resistance (AMR). This makes infections harder to treat, leading to increased illness, mortality, longer hospital stays, and higher healthcare costs. The World Health Organization identifies AMR as one of the top ten global public health threats.

In order to follow up the national evolution of the resistance proportion and incidence of multidrug resistant organisms (MDRO) in Belgian hospitals, Sciensano collects and analyses AMR surveillance data. By Royal Decree, all Belgian acute care hospitals mandatorily have to participate in the surveillance of methicillin-resistant *Staphylococcus aureus* (MRSA) and multi-resistant Gram-negative bacilli (MRGN). The surveillance of vancomycin-resistant enterococci (VRE) is one of four additional programs from which hospitals must choose one for participation.

This report presents the 2024 findings from the MRSA, MRGN, and VRE surveillance programs, highlighting trends in AMR across Belgian acute and chronic care hospital sites.

Methods

Surveillance data for 2024 were retrospectively collected in 2025 by the microbiology laboratories and/or infection prevention and control teams of the participating hospitals. The data, aggregated at the hospital level, included either annual or semester-specific figures, except for the VRE surveillance, which required annual data exclusively.

Data from acute and chronic care hospital sites were analysed and presented separately. Acute care hospital sites with an average length of stay of ≥ 16 days were reclassified as chronic care hospital sites.

Following microorganisms and resistances were explored:

- o *Staphylococcus aureus* (*S. aureus*) resistant to methicillin or oxacillin
- o *Escherichia coli* (*E. coli*) and *Klebsiella pneumoniae* (*K. pneumoniae*) resistant to 3rd generation cephalosporins (3GC-R) and/or resistant to meropenem (mero-R)
- o Carbapenemase producing (CPE+) *E. coli*, *K. pneumoniae* and other Enterobacterales
- o Mero-R in *Acinetobacter baumannii* (*A. baumannii*)
- o Multidrug-resistant (MDR) *Pseudomonas aeruginosa* (*P. aeruginosa*), i.e. resistant to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin or levofloxacin), aminoglycosides (gentamicin, tobramycin or amikacin), carbapenems (meropenem or imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime or cefepime)
- o *Enterococcus faecalis* (*E. faecalis*) and *Enterococcus faecium* (*E. faecium*) resistant to vancomycin (vanco-R), resistant to linezolid (linezolid-R) and resistant to both vancomycin and linezolid (vanco-R and linezolid-R)

Only hospitals that provided de-duplicated data, ensuring each patient was counted only once per hospitalization period (type D data), were included in the analyses. Unless otherwise specified (see MRSA and CPE details below), this report is based exclusively on data from clinical samples. Both invasive (e.g. blood) and non-invasive sample types (e.g. urine) were included, yet faecal samples were considered as screening specimen and therefore excluded from the category of clinical samples.

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The potential healthcare-associated character was assessed for MRSA only. Healthcare-associated (HA-)MRSA was defined as either colonization or infection with MRSA considered to be acquired in the hospital (first positive sample for MRSA collected more than 48 hours after admission), not present on admission and not known from the patient's history (in the past 12 months).

For each species, the resistance proportion (%), incidence (number of cases per 1 000 hospitalisations) and/or incidence density (cases per 1 000 patient-days) were calculated.

Historical data were used to present the evolution of resistance proportions and incidence (densities). We fitted a negative binomial regression model with hospital as cluster and year as fixed effect to explore and assess statistically significant ($p < 0.05$) changes in the incidence. The result was expressed as incidence rate ratio (IRR) and its 95% confidence interval (CI). To assess whether trends observed in resistance proportions were statistically significant ($p < 0.05$), we used linear regression with hospital as cluster.

Data were analysed in STATA 17 (StataCorp LP, College Station, Texas, USA) and presented by region and hospital level of specialty care.

Results

In 2024, participation in the national AMR surveillance systems remained very high across Belgian acute care hospital administrative groups. Coverage reached 95.1% for the mandatory MRSA surveillance and 98.0% for the MRGN surveillance, while the optional VRE surveillance also achieved strong uptake, with participation from 94.1% of hospital groups.

The 2024 surveillance of antimicrobial resistance (AMR) in Belgian acute care hospitals presents both encouraging trends and areas of concern. Table 1 highlights resistance proportions and incidences per 1 000 hospitalisations for key bacteria under observation.

Table 1. Resistance proportion and incidence per 1 000 hospitalisations of the bacteria included in the surveillance of antimicrobial resistance (clinical samples only), Belgian acute care hospital sites, 2024

		2024			
		Resistance proportion (%)		Incidence per 1 000 hospitalisations	
		Crude	Median	Crude	Median
<i>Staphylococcus aureus</i>	Methicillin R	9.0	8.2	1.34	1.16
Healthcare-associated <i>Staphylococcus aureus</i>	Methicillin R	26.3*	20.9*	0.35	0.23
<i>Enterococcus faecium</i>	Vancomycin R	3.08	0.00	0.139	0.000
<i>Enterococcus faecalis</i>	Vancomycin R	0.05	0.00	0.006	0.000
<i>Escherichia coli</i>	3GC-R	9.2	9.0	4.20	4.12
	Meropenem R	0.06	0.00	0.026	0.000
<i>Klebsiella pneumoniae</i>	3GC-R	18.2	15.7	1.96	1.76
	Meropenem R	1.21	0.57	0.130	0.068
<i>Acinetobacter baumannii</i>	Meropenem R	6.21	0.00	0.027	0.000
<i>Pseudomonas aeruginosa</i>	MDR	3.1	1.8	0.29	0.16

*Proportion healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) on total number of MRSA; R = resistant, 3GC = 3rd generation cephalosporins, MDR = resistance to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin or levofloxacin), aminoglycosides (tobramycin or amikacin), carbapenems (meropenem or imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime or ceftipime)

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In 2024, 9.0% of all *S. aureus* clinical isolates were MRSA, with a crude incidence of 1.34 cases per 1 000 hospitalisations. While MRSA resistance and incidence have declined significantly since 2004, the downward trend in resistance proportion has plateaued since 2022. Among patients testing positive for MRSA on admission, 32% had a prior history of MRSA, and nearly half of those without prior history had recent healthcare exposure. Healthcare-associated MRSA (HA-MRSA), defined as cases detected more than 48 hours after admission in patients without known prior MRSA, represented 26.3% of all MRSA-positive clinical samples, with an incidence of 0.35 per 1 000 hospitalisations. Although long-term surveillance shows a substantial reduction in HA-MRSA, the decline has not been statistically significant in recent years, with fluctuations in the proportion of HA-MRSA among all MRSA cases.

Among 8 729 *E. faecium* isolates from clinical samples, 3.08% were vanco-R, and the incidence was 0.139 per 1 000 hospitalisations. After years of decline, both resistance proportion and incidence have been increasing since 2021. Most vanco-R isolates carried *vanA* genes. Linezolid resistance remained rare at 0.25% of *E. faecium* isolates.

In *E. faecalis*, resistance to linezolid (0.22%) was comparable to that observed in *E. faecium* (0.25%), whereas vanco-R was much lower (0.05% vs. 3.08%). Combined resistance to both drugs was very uncommon.

Despite this more restrictive definition, a higher proportion of hospitals reported a VRE outbreak in 2024 (15.6%) compared with 2023 (13.8%).

Gram-negative bacteria showed mixed trends in resistance. For *E. coli*, 9.2% of isolates were 3GC-R, with an incidence of 4.2 per 1 000 hospitalisations. After a decrease from 2019 to 2022, the resistance proportion has increased slightly in recent years, though incidence has remained stable. Mero-R *E. coli* remained rare, with only 51 cases reported across 28 hospitals.

In *K. pneumoniae*, 18.2% of isolates were 3GC-R, showing a long-term decline in both resistance and incidence since 2018. Mero-R *K. pneumoniae* was more widespread, detected in over 60% of hospitals, though resistance levels have stabilized after prior fluctuations.

These trends in both *E. coli* and *K. pneumoniae* should be interpreted with caution, as the exclusion of the EUCAST “I” category in the resistance definition since 2021 may have influenced comparisons over time.

Notably, 2024 marked the reintroduction of carbapenemase-producing Enterobacterales (CPE) surveillance within the MRGN program. Sixty hospitals (53% of Belgian acute care hospitals) voluntarily contributed data, providing a national overview of CPE detection in both clinical and screening samples. Clinical isolates included 268 CPE-positive *K. pneumoniae*, 125 CPE-positive *E. coli*, and 156 CPE-positive other Enterobacterales, exceeding the number captured by meropenem-resistance alone, likely due to low-level resistance conferred by OXA-48-like enzymes. Screening detected even higher numbers, highlighting the importance of systematic CPE surveillance. OXA-48 was the predominant carbapenemase overall, particularly in *E. coli* (86%) and other Enterobacterales (80%), while *K. pneumoniae* showed more diverse carbapenemases, including KPC and NDM.

Other critical pathogens included *A. baumannii* and *P. aeruginosa*. Mero-R *A. baumannii* was detected in a minority of hospitals, with a crude resistance of 6.2% and a decreasing incidence trend. For MDR *P. aeruginosa*, resistance was reported at 3.1% of isolates, with stable incidence. Changes in resistance definitions over recent years make interpreting long-term trends for *P. aeruginosa* challenging, though overall resistance levels remain moderate.

The current data collection also explored MDRO screening practices in greater depth than in previous years, when only MRSA screening was assessed. Across the four organisms, screening strategies differed substantially, with more systematic and risk-based approaches for CPE, VRE, and MRSA than for ESBL. Universal admission screening on all wards was uncommon overall, reported by only 1.1% of hospitals for ESBL, CPE, and VRE, and by 6.6% for MRSA.

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Screening of high-risk wards upon admission was more frequent for MRSA (82.4% of hospitals), followed by CPE (68.2%), VRE (56.0%), and ESBL (50.0%). Similarly, screening of known carriers was nearly universal for MRSA (98.9%), CPE (96.6%), and VRE (93.4%), but less common for ESBL (43.2%). Screening of transferred patients showed the same pattern: MRSA (98.9%), CPE (94.3%), and VRE (79.1%) were screened far more frequently than ESBL (40.9%).

During hospitalisation, outbreak-driven screening was reported by the vast majority of hospitals for MRSA (96.7%), CPE (98.9%), and VRE (96.7%), compared with 70.8% for ESBL. Routine screening on high-risk wards was also more common for MRSA (85.7%) and CPE (76.4%) than for VRE (58.7%) or ESBL (49.4%). Similarly, screening of room contacts was widely implemented for MRSA (93.4%), CPE (98.9%), and VRE (91.3%), but less frequently for ESBL (42.7%).

Overall, screening practices were most comprehensive for MRSA, followed by CPE and VRE, while ESBL screening was more limited and selectively applied.

Table 2 presents the resistance proportion and incidence density per 1 000 patient-days for bacteria under surveillance in Belgian chronic care hospital sites in 2024. However, with only 11 hospitals participating, these findings should be interpreted with caution.

Table 2. Resistance proportion and incidence density per 1 000 patient-days of the bacteria included in the surveillance of antimicrobial resistance (clinical samples only), Belgian chronic care hospital sites, 2024

		2024			
		Resistance proportion (%)		Incidence per 1 000 patient-days	
		Crude	Median	Crude	Median
<i>Staphylococcus aureus</i>	Methicillin R	12.3	14.3	0.07	0.06
Healthcare-associated <i>Staphylococcus aureus</i>	Methicillin R	60.0*	100.0*	0.04	0.04
<i>Enterococcus faecium</i>	Vancomycin R	8.7	0.00	0.562	0.000
<i>Enterococcus faecalis</i>	Vancomycin R	0.0	0.00	0.000	0.000
<i>Escherichia coli</i>	3GC-R	9.0	8.5	0.22	0.20
	Meropenem R	0.08	0.00	0.002	0.000
<i>Klebsiella pneumoniae</i>	3GC-R	32.3	31.3	0.24	0.17
	Meropenem R	1.27	0.00	0.009	0.000
<i>Acinetobacter baumannii</i>	Meropenem R	0.00	0.00	0.000	0.000
<i>Pseudomonas aeruginosa</i>	MDR	2.8	0.0	0.02	0.00

*Proportion healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) on total number of MRSA; R = resistant, 3GC = 3rd generation cephalosporins, MDR = resistance to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin or levofloxacin), aminoglycosides (tobramycin or amikacin), carbapenems (meropenem or imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime or cefepime)

Conclusions

In 2024, Belgium's national AMR surveillance demonstrated high participation across hospitals, ensuring robust and representative data.

MRSA incidence continues to decline compared with historical levels, though HA-MRSA remains present, highlighting ongoing infection control challenges.

Resistance patterns among enterococci and Gram-negative bacteria show mixed trends. Vancomycin-resistant *E. faecium* is increasing, while *E. faecalis* remains largely susceptible. 3GC-R *E. coli* shows a slight upward trend, and carbapenem resistance remains rare. *K. pneumoniae* maintains high cephalosporin resistance, with a modest

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rise in carbapenem resistance. *A. baumannii* and multidrug-resistant *P. aeruginosa* display moderate resistance levels, with largely stable trends.

The reintroduction of CPE surveillance highlights the importance of monitoring emerging resistance, as many CPE isolates, particularly OXA-48 producers, may appear susceptible to meropenem during routine testing. Screening practices were most systematic for MRSA, followed by CPE and VRE, ensuring early detection in high-risk wards.

Belgium's AMR trends are aligned with EARS-BE data, confirming stable MRSA and MDR *P. aeruginosa* rates, and moderate but rising resistance in *K. pneumoniae*. The findings support continued infection prevention and control, antimicrobial stewardship, and adaptive surveillance strategies.

The upcoming One Health National Action Plan 2026–2030 aims to strengthen AMR monitoring by integrating the AMR and EARS-BE surveillance systems into a single platform (NSIH-AMREXT), and to expand coverage to nursing homes and primary care. This integrated approach will enhance data quality, representativeness, and the capacity to respond to emerging threats, while addressing important knowledge gaps on AMR outside the hospital setting.

While progress has been achieved, continued vigilance, adaptive strategies, and a comprehensive One Health approach remain essential to mitigate AMR and protect public health across Belgium.

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ABBREVIATIONS



3GC	Third generation cephalosporins
3GC-R	Resistance to third generation cephalosporines
4GC	Fourth generation cephalosporins
<i>A. baumannii</i>	<i>Acinetobacter baumannii</i>
AMR	Antimicrobial resistance
AST	Antimicrobial susceptibility testing
CLSI	Clinical and Laboratory Standard Institute, USA
CPE	Carbapenemase-producing <i>Enterobacterales</i>
CRE	Carbapenem-resistant <i>Enterobacterales</i>
<i>E. coli</i>	<i>Escherichia coli</i>
<i>E. faecalis</i>	<i>Enterococcus faecalis</i>
<i>E. faecium</i>	<i>Enterococcus faecium</i>
EARS-BE	European Antimicrobial Resistance Surveillance in Belgium
EARS-Net	European Antimicrobial Resistance Surveillance Network
ESBL	Extended spectrum beta-lactamase
EUCAST	European Committee on Antimicrobial Susceptibility Testing
I	Susceptible, increased exposure (the former intermediate susceptibility category)
IPC	Infection prevention and control
IQR	Interquartile Range
IRR	Incidence Rate Ratio
I/R	Non-susceptible (intermediate susceptible or resistant)
<i>K. pneumoniae</i>	<i>Klebsiella pneumoniae</i>
MDR	Multidrug-resistant
Mero-R	Resistance to meropenem
MIC	Minimal inhibitory concentration
MRGN	Multi-resistant Gram-negative bacteria
MRSA	Methicillin-resistant <i>Staphylococcus aureus</i>
NAP	National action plan
NGS	Next-generation sequencing
<i>P. aeruginosa</i>	<i>Pseudomonas aeruginosa</i>
R	Resistant or non-susceptible
<i>S. aureus</i>	<i>Staphylococcus aureus</i>
Type D	Data collection method with de-duplication of data: per period of hospitalisation and bacteria each patient is counted only once
Vanco-R	Resistance to vancomycin
VRE	Vancomycin-resistant enterococci
WGS	Whole-genome sequencing

INTRODUCTION



Antibiotics have revolutionized modern medicine, saving countless lives and making many medical procedures possible. However, their overuse and misuse have progressively undermined their effectiveness in treating infections. Over time, some bacteria have adapted to survive exposure to certain antibiotics or have developed mechanisms to inactivate them, a phenomenon known as acquired antimicrobial resistance (AMR). As a result, infections become more difficult to treat, leading to prolonged illness, disability, and increased mortality. In addition to these serious health consequences, AMR also imposes a substantial economic burden, including longer hospital stays and the need for more complex and expensive treatments. The World Health Organization therefore identifies AMR as one of the top ten global public health threats facing humanity.¹

The service “Healthcare-associated infections and antimicrobial resistance” of Sciensano organizes, collects and analyzes AMR surveillance data originating from Belgian hospitals. The Royal Decree of [25 April 2002](#) on the establishment of and budget for hospitals, Art 56, Par 2, amended on [8 January 2015](#) and [10 September 2020](#), stipulates that all Belgian general hospitals - with the exception of Sp hospitals for palliative care – mandatorily have to participate in the surveillance of methicillin-resistant *Staphylococcus aureus* (MRSA) and multiresistant Gram-negative bacteria (MRGN). In addition, these hospitals have to participate in at least one of four optional surveillance programs: vancomycin-resistant enterococci (VRE), *Clostridioides difficile* infections, surgical site infections, or healthcare-associated infections in intensive care units.

Note: Due to the COVID-19 pandemic, hospitals were exempt from mandatory MRSA and MRGN surveillance in 2020 and 2021, which impacted participation rates and potentially affected the data for 2019 and 2020.

The first national voluntary surveillance program for MRSA was initiated in 1994. This resistant Gram-positive bacterium causes difficult to treat infections, such as skin and soft tissue infections, surgical site infections, catheter-related infections, bloodstream infections and pneumonia. Participation in this surveillance became mandatory in 2006.

The second MRGN surveillance was set up in the late 1990s following the emergence of antimicrobial resistance in a wide range of *Enterobacteriales* as well as in nonfermenting Gram-negative bacteria (e.g. *Pseudomonas aeruginosa*, *Acinetobacter baumannii*). Multi-resistant *Enterobacter aerogenes* (later reclassified as *Klebsiella aerogenes*) was the first within the family of *Enterobacteriaceae* to be monitored (started in 2000, stopped in 2011) due to major healthcare-associated outbreaks with a subsequent endemic character in many Belgian hospitals. Because of the increased prevalence and incidence of extended-spectrum beta-lactamases (ESLBs) reported locally by several Belgian hospitals, this surveillance program was subsequently extended to several other *Enterobacteriaceae* species, including *Escherichia coli* (2005), *Klebsiella pneumoniae* (2005) and *Enterobacter cloacae* (2009, stopped in 2017), as well as to nonfermenting Gram-negative bacteria (*Pseudomonas aeruginosa* and *Acinetobacter baumannii*). In 2015, with the rise of carbapenem-resistant and carbapenemase-producing *Enterobacteriaceae* (CRE and CPE, respectively), participation in the MRGN surveillance became mandatory. The third program, VRE, was initiated in 2014 after multiple Belgian hospitals reported VRE outbreaks.

The primary objective of these three epidemiological AMR surveillance programs – MRSA, MRGN and VRE - is to monitor trends in resistance proportions and the incidence of (multi)drug-resistant bacteria in Belgian hospitals, thereby establishing comprehensive national data on these microorganisms. A secondary objective is to encourage participating hospitals to track and analyse their own results over time, including benchmarking their performance. However, data collection is not an end in itself. The ultimate goal is to guide and support targeted interventions to prevent and control antimicrobial resistance based on these insights.

The aim of this report is to present the 2024 results of the three AMR surveillance programs and to describe trends in AMR in Belgian acute and chronic care hospital sites.

METHODOLOGY



The surveillance results were collected and reported by the microbiology laboratories and/or the infection prevention and control teams of the participating hospitals to the service “Healthcare-associated infections and antimicrobial resistance” of Sciensano. The detailed modalities of the data collection for the AMR surveillance can be found in the study protocol.²

The data (year 2024) were collected retrospectively in the following year and were aggregated at hospital level. Hospitals could either provide annual figures or data for one semester, except for the VRE surveillance for which only annual data were allowed.

Following microorganisms and resistances were explored in the three surveillance programs:

- o **MRSA** ***Staphylococcus aureus*** (*S. aureus*) resistant to methicillin or oxacillin (determined as resistant to ceftazidime according to EUCAST*)

- o **MRGN** 1) Enterobacterales:
 Escherichia coli (*E. coli*) and ***Klebsiella pneumoniae*** (*K. pneumoniae*)
 - a) Resistance (R) to 3rd generation cephalosporines (cefotaxime, ceftriaxone, or ceftazidime) (3GC-R)
 - b) Resistance to meropenem (mero-R)
Carbapenemase producing (CPE+) *E. coli*, *K. pneumoniae* and other Enterobacterales
 2) Mero-R ***Acinetobacter baumannii*** (*A. baumannii*):

 3) Multidrug-resistant (MDR) ***Pseudomonas aeruginosa*** (*P. aeruginosa*):
Resistance (R) to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin or levofloxacin), aminoglycosides (tobramycin or amikacin), carbapenems (meropenem or imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, ceftazidime, ceftazidime)

o **VRE** ***Enterococcus faecalis*** (*E. faecalis*) and ***Enterococcus faecium*** (*E. faecium*)
 - a) Resistance to vancomycin (vanco-R)
 - b) Resistance to linezolid (linezolid-R)
 - c) Vanco-R and linezolid-R

*Note: In 2019, the European Committee on Antimicrobial Susceptibility Testing (EUCAST) revised the definition of the intermediate (I) category in antimicrobial susceptibility testing. Previously, "I" stood for "intermediate," implying uncertain clinical efficacy. However, the updated definition reinterprets "I" as “Susceptible, Increased exposure”, indicating that the infection can still be treated effectively if the antimicrobial agent is used at a higher dose, increased frequency, or through optimized delivery to achieve sufficient exposure at the infection site. This change aims to improve clinical decision-making by encouraging the use of optimized dosing regimens rather than defaulting to broader-spectrum or last-resort antibiotics. It can however also have implications for surveillance, as shifts in susceptibility reporting may affect trend analyses and interpretation of resistance data over time.

All sample types (e.g. blood, urine) had to be included. For MRSA and VRE, a distinction had to be made between clinical samples (i.e. all samples taken for diagnostic purposes) and screening samples (i.e. samples taken - in the absence of clinical signs/symptoms - to detect colonization with resistant bacteria). **Faecal samples could**

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not be considered as clinical samples in the MRGN and VRE surveillance programs, but had to be considered as screening samples.

There were five possibilities for data collection for each surveillance program:

- Type A: every positive sample was counted (screening samples and duplicates included)
- Type B: every positive clinical sample was counted (duplicates included)
- Type C: each sample originating for a different infection site was counted only once
- Type D: each patient was counted only once per period of hospitalisation (de-duplication)
- Type E: other

Duplicates were defined as isolates of the same species from the same patient with indistinguishable antibiograms or with the same resistance mechanism, regardless of the purpose for which the sample was taken.

Only hospitals providing Type D data (with de-duplication) were included in the analyses reported here.

The healthcare-associated character was explored for MRSA only. Healthcare-associated (or nosocomial) MRSA was defined as colonization or infection with MRSA, considered to be acquired in the hospital and not present on admission (first positive sample collected more than 48h after admission) or known in the patient's history (past 12 months).

Results are presented by hospital type (acute or chronic care hospital sites), by region (Flanders, Wallonia or Brussels), and by hospital level of specialty care site (not of the merger). The latter is defined as follows:

Level of specialty care	Definition ECDC ³	Definition FPS ⁴
Primary	- Often referred to as 'district hospital' or 'first-level' referral - Few specialties (mainly internal medicine, obstetrics-gynaecology, paediatrics, general surgery or only general practice) - Limited laboratory services for general, but not specialised, pathological analysis - Often corresponds to general hospital without teaching function	<i>Algemeen ziekenhuis</i> <i>Hôpital général</i> <i>Allgemein Krankenhaus</i>
Secondary	- Often referred to as 'provincial hospital' or 'second-level referral' - The hospital is highly differentiated by function with five to ten clinical specialties, such as haematology, oncology, nephrology, ICU - Takes some referrals from other (primary) hospitals - Often corresponds to general hospital with teaching function/mission	<i>Algemeen ziekenhuis met universitair karakter</i> <i>Hôpital général à caractère universitaire</i>
Tertiary	- Often referred to as 'central', 'regional' or 'tertiary-level' hospital - Highly specialised staff and technical equipment (ICU, haematology, transplantation, cardio-thoracic surgery, and neurosurgery) - Clinical services are highly differentiated by function - Specialised imaging units - Provides regional services and regularly takes referrals from other (primary and secondary) hospitals - Often a university hospital or associated to a university	<i>Universitair ziekenhuis</i> <i>Hôpital universitaire</i>
Specialised	- Single clinical specialty, possibly with sub-specialties - Highly specialised staff and technical equipment	<i>Gespecialiseerd ziekenhuis</i> <i>Geriatrisch- & Specialised</i> <i>Hôpital spécialisé</i> <i>Psychiatrisch ziekenhuis</i> <i>Hôpital psychiatrique</i>

ECDC = European Centre for Disease Prevention and Control; FPS = Federal Public Service Health, Food Chain Safety and Environment, ICU: intensive care unit

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For each bacterium, the resistance proportion was calculated by dividing the total number of resistant isolates by the total number of isolates reported by the hospital during the surveillance period. In addition, the incidence (number of cases per 1 000 hospitalisations) and incidence density (cases per 1 000 patient-days) were calculated for each resistant bacteria under surveillance and this by dividing the total number of resistant isolates by the total number of hospitalisations or patient-days reported by the hospital during the surveillance period.

Denominator data (hospitalisations and patient-days) are the same for all surveillances coordinated by the service “Healthcare-associated infections and antimicrobial resistance” of Sciensano. These data are collected via a separate module on the Healthdata platform (NSIH-DENO). For this purpose, the official definitions from the Belgian hospital administrations nomenclature (*résumé hospitalier minimal/minimale ziekenhuisgegevens - RHM/MZG*) are used. Nonetheless, in this system, hospitalisations are recorded based on discharge rather than admission, which impacts the registration of denominator data previously collected differently. Additionally, RHM/MZG operates on a biannual registration basis, while denominator data for surveillance are recorded monthly. Therefore, the RHM/MZG figures must be broken down by month to which they relate. More detailed information is available elsewhere.⁵

Following summary statistics were used in this report:

- Crude:
 - Crude resistance proportion: total number of bacterium X with resistance Y divided by the total number of bacterium X multiplied by 100
 - Crude incidence: total number of bacterium X with resistance Y divided by the total number of hospitalisations multiplied by 1 000
 - Crude incidence density: total number of bacterium X with resistance Y divided by the total number of patient-days multiplied by 1 000
- Mean: The sum of all scores (i.e. crude resistance proportions or crude incidences) divided by the number of scores
- Median (or P50): the middle score (i.e. crude resistance proportion or crude incidence) when scores are ranked in ascending/descending order
- P25 (or Q1, first quartile): the 25th percentile is the score (i.e. crude resistance proportion or crude incidence) below which 25% of the cases fall
- P75 (or Q3, third quartile): the 75th percentile is the score (i.e. crude resistance proportion or crude incidence) below which 75% of the cases fall
- Between P25 and P75 lies half of all scores (= interquartile range (IQR) = P75 - P25)

Because the median is less affected by outliers (e.g. hospitals experiencing an outbreak) and skewed data (e.g. many hospitals reporting zero resistance cases) than the mean, we recommended hospitals to use the median as the preferred measure of central tendency for benchmarking.

Historical data were used to present the evolution of resistance proportions and incidence (densities). To assess whether trends observed in resistance proportions were statistically significant ($p < 0.05$), we used linear regression with hospital as cluster. We fitted a negative binomial regression model with hospital as cluster and year as fixed effect to explore and assess statistically significant ($p < 0.05$) changes in the incidence (density). The result was expressed as incidence rate ratio (IRR) and its 95% confidence interval (CI). An IRR of 1.20 means an 20% increase in the incidence, while a IRR of 0.80 points to a 20% decrease.

METHODOLOGY

Data were analysed in STATA 17 (StataCorp LP, College Station, Texas, USA).

Hospitals that were part of an administrative hospital group could choose to participate as one hospital or to collect data by hospital site. Results were presented separately for acute care and chronic care hospital sites. **In this report, acute care hospital sites with an average length of stay of more than 16 days were considered as chronic care hospital sites.**

The present and historical results reported here may differ slightly from those published in previous reports. Some hospitals may revise or correct their data after publication of a report.

Individual hospital benchmark reports (2024 data) were distributed to participating hospitals in July 2025. Hospitals were given the opportunity to review their data and request corrections or clarifications until September 2025. This feedback process served as an internal validation step for the dataset used in the present analyses. The results presented here are based on the dataset after incorporation of any revisions received during this validation period.

PART 1: METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

In 2024, 95.1% (n=97/102) of all acute care hospital administrative groups (mergers) participated in the MRSA surveillance with at least one hospital site. Two hospital administrative groups were considered as chronic care hospitals (length of stay > 16 days).

Table 3 summarizes the 2024 participation in MRSA surveillance, categorized by hospital care type, region, and level of specialty care. One hospital was excluded from the analysis due to the absence of Type D data.

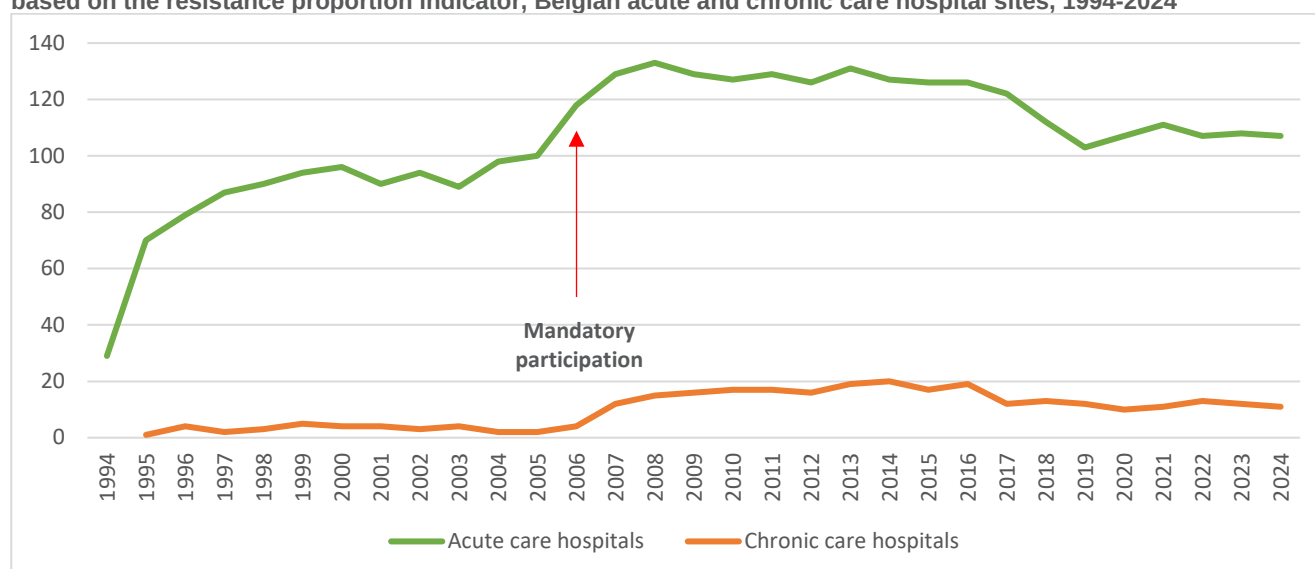
Table 3. Participation in the surveillance of methicillin-resistant *Staphylococcus aureus* D data by hospital care type, region and hospital level of specialty care (for acute care hospitals only), Belgian acute and chronic care hospital sites, 2024 (Type D data only)

	Flanders	Wallonia	Brussels	Belgium
N of acute care hospital sites	57	38	12	107
Primary hospital sites	46	27	7	80
Secondary hospital sites	8	10	2	20
Tertiary hospital sites	3	1	3	7
N of chronic care hospital sites	5	5	1	11

N = number

Figure 1 illustrates the number of participating hospital sites since the beginning of the surveillance (1994). Over the years, the number of participating hospital sites has decreased, primarily due to mergers resulting in site closures or hospitals opting to participate in surveillance as a merged entity. Due to the COVID-19 pandemic, hospitals were exempt from mandatory MRSA and MRGN surveillance in 2020 and 2021, which impacted participation rates for 2019 and 2020.

Figure 1. Evolution of the participation in the surveillance of methicillin-resistant *Staphylococcus aureus* (MRSA) based on the resistance proportion indicator, Belgian acute and chronic care hospital sites, 1994-2024



1. MRSA in acute care hospitals

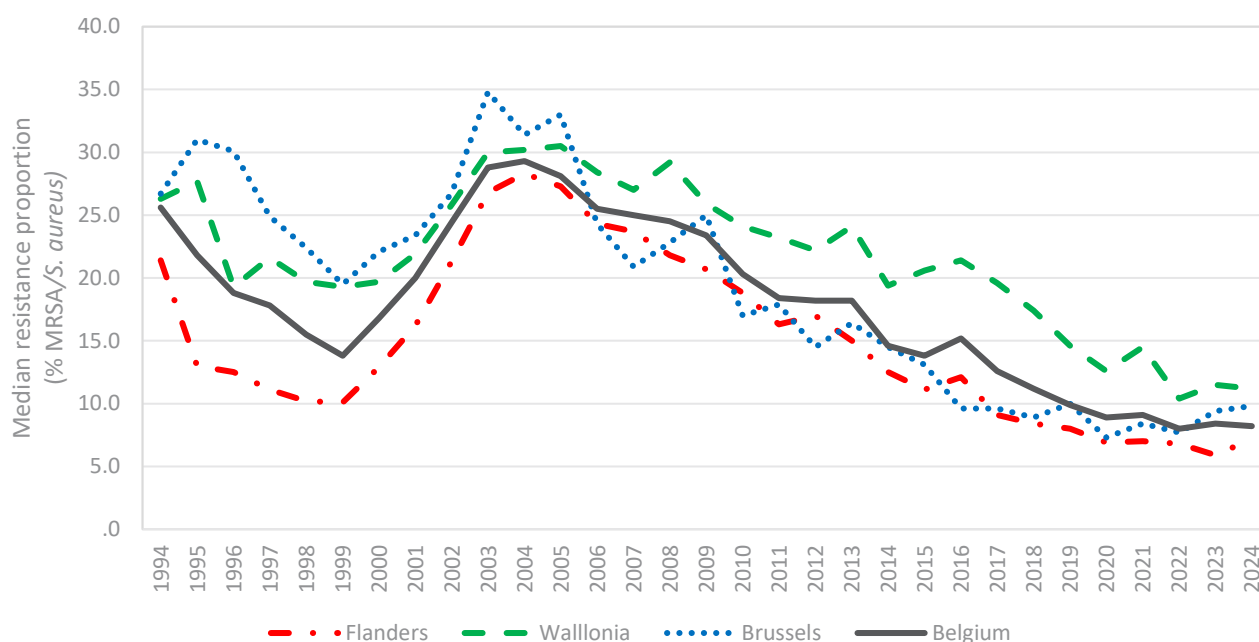
1.1 RESISTANCE IN *STAPHYLOCOCCUS AUREUS*

In 2024, 9.0% of all *S. aureus* isolates (from clinical samples only) were MRSA. The crude incidence of MRSA was 1.34 cases per 1 000 hospitalisations or 0.23 cases per 1 000 patient-days (Table 4).

Figures 2, 3, and 4 show the evolution of the median resistance proportion, incidence, and incidence density, respectively, both overall and by region. Since 2004, a significant downward trend has been observed in both the resistance proportion (-1.12% per year; $p < 0.001$) and the incidence of MRSA (IRR=0.916, 95% CI: 0.914–0.919, $p < 0.001$). However, from 2022 onward, the decline in the resistance proportion is no longer significant (-0.00% per year; $p = 0.991$).

The evolution of the median resistance proportion and incidence (per 1 000 hospitalisations) of MRSA by level of specialty care is available in the Annex (Figures A1–A2).

Figure 2. Evolution of the median proportion of methicillin-resistant *Staphylococcus aureus* (MRSA) on the total number of reported *S. aureus* by region (clinical samples only), Belgian acute care hospital sites, 1994-2024



PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

Table 4. Resistance proportion, incidence and incidence density of methicillin-resistant *Staphylococcus aureus* (MRSA) (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	MRSA (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	2 627	29 074	9.0	9.0	8.2	5.7-11.7
Flanders	1 166	15 545	7.5	7.3	7.2	5.1-8.8
Wallonia	979	8 978	10.9	10.8	11.2	8.5-13.9
Brussels	482	4 551	10.6	11.2	9.8	8.1-12.3
Primary hospitals	1 471	17 145	8.6	8.5	7.7	5.6-11.6
Secondary hospitals	659	6 566	10.0	10.6	9.9	7.5-12.2
Tertiary hospitals	497	5 363	9.3	9.4	8.4	8.0-10.3
Incidence per 1 000 hospitalisations						
Belgium	2 627	1 957 798	1.34	1.45	1.16	0.78-1.95
Flanders	1 166	1 179 548	0.99	0.99	0.89	0.63-1.23
Wallonia	979	563 465	1.74	1.88	1.58	1.11-2.52
Brussels	482	214 785	2.24	2.24	1.97	1.28-2.13
Primary hospitals	1 471	1 121 126	1.31	1.38	1.13	0.75-1.62
Secondary hospitals	659	466 880	1.41	1.65	1.49	0.90-2.25
Tertiary hospitals	497	369 792	1.34	1.59	1.63	1.05-2.08
Incidence density per 1 000 patient-days						
Belgium	2 627	11 263 066	0.23	0.23	0.20	0.13-0.30
Flanders	1 166	6 355 891	0.18	0.18	0.16	0.11-0.22
Wallonia	979	3 309 614	0.30	0.29	0.26	0.18-0.40
Brussels	482	1 597 561	0.30	0.30	0.23	0.15-0.33
Primary hospitals	1 471	6 611 361	0.22	0.23	0.19	0.12-0.28
Secondary hospitals	659	2 855 357	0.23	0.25	0.22	0.14-0.33
Tertiary hospitals	497	1 796 348	0.28	0.29	0.24	0.23-0.37

n = total number of methicillin-resistant *Staphylococcus aureus* (MRSA) isolates, N = total number of *Staphylococcus aureus* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

Figure 3. Evolution of the median incidence of methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 hospitalisations by region (clinical samples only), Belgian acute care hospital sites, 1994-2024

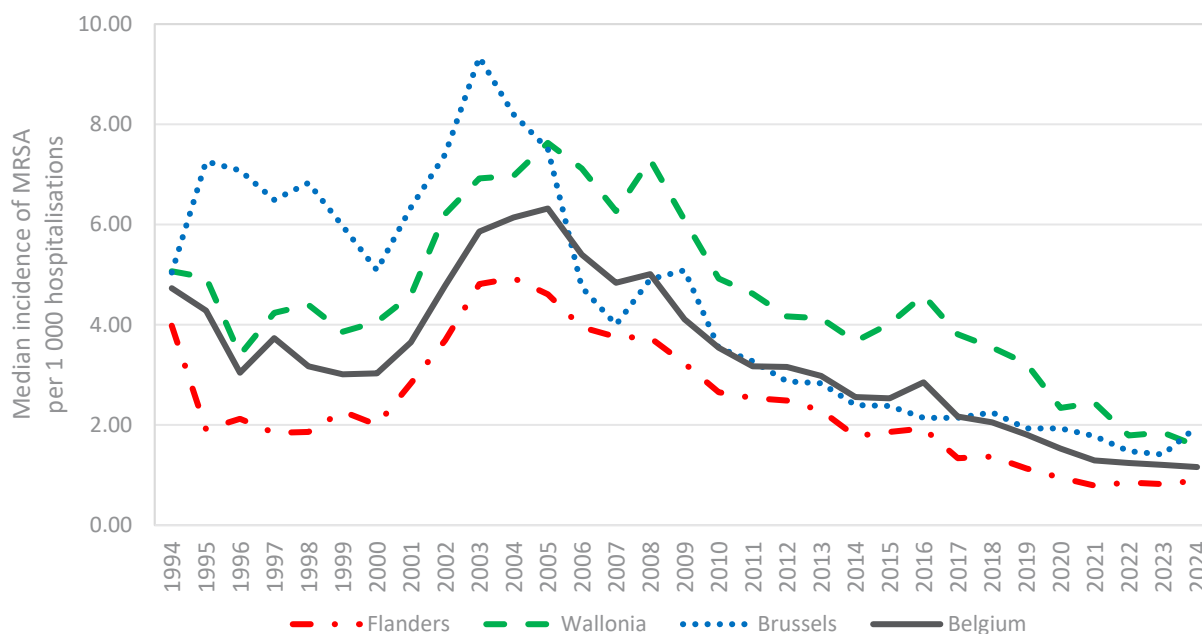
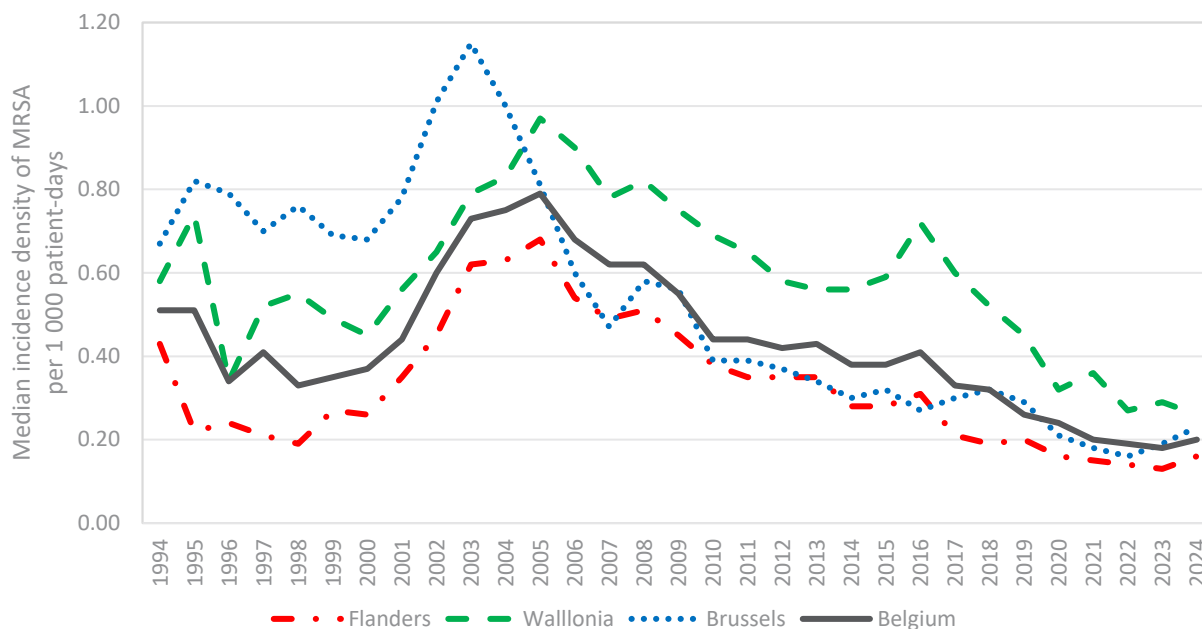


Figure 4. Evolution of the median incidence density of methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 patient-days by region (clinical samples only), Belgian acute care hospital sites, 1994-2024

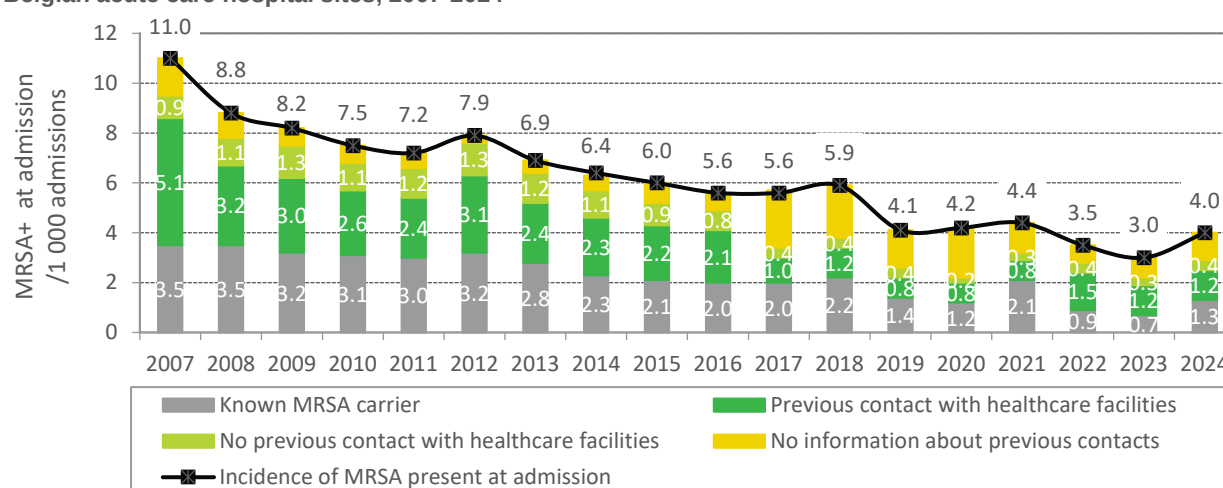


1.2 MRSA PRESENT AT ADMISSION

The incidence of patients who were MRSA-positive on admission (optional data) could only be calculated for 12 acute care hospital sites in 2024. Both clinical and screening samples that tested positive for MRSA within 48 hours after admission were considered.

The crude incidence of MRSA-positive patients on admission was 4.0 cases per 1 000 hospitalisations (n=805 /200 606 hospitalisations) in 2024 (Figure 5).

Figure 5. Evolution of the crude incidence of methicillin-resistant *Staphylococcus aureus* (MRSA) present at admission according to history of colonisation and previous contact (past 12 months) with healthcare facilities, Belgian acute care hospital sites, 2007-2024



In 2024, 32.0% of patients who tested MRSA-positive upon admission had a known history of MRSA colonization or infection in the previous 12 months (n=258/805). Of the patients without such a history (n=547), 45.2% (n=247) had recently been transferred to or had a recent stay in a healthcare facility (e.g., acute care hospital, day care hospital, nursing home). For 16.1% of patients (n=88), no contact with healthcare facilities in the previous 12 months was reported, while for 38.8% (n=212), information on prior healthcare facility contact was unknown.

1.3 HEALTHCARE-ASSOCIATED MRSA

1.3.1 HEALTHCARE-ASSOCIATED MRSA IN CLINICAL SAMPLES

In 2024, 26.3% of all MRSA-positive clinical samples were collected more than 48 hours after admission from patients with no known MRSA carriage in the past 12 months, representing cases of healthcare-associated (HA) MRSA. The crude incidence of HA-MRSA was 0.35 cases per 1 000 hospitalisations or 0.06 cases per 1 000 patient-days (Table 5).

Figures 6 and 7 show the evolution of these indicators (incidence and incidence density), respectively. Since 2004, the incidence of HA-MRSA has shown a consistent decreasing trend (IRR=0.881, 95% CI: 0.878-0.884; p<0.001). However, this decline is no longer statistically significant in the period from 2021 to 2024 (IRR=0.985, 95% CI: 0.938-1.035; p=0.555).

The evolution of the median incidence (per 1 000 hospitalisations) of HA-MRSA by level of specialty care is shown in the Annex (Figure A3).

PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

Table 5. Proportion, incidence and incidence density of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	Healthcare-associated MRSA (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Proportion healthcare-associated MRSA/MRSA (%)						
Belgium	690	2 627	26.3	24.5	20.9	10.0-35.2
Flanders	243	1 166	20.8	22.9	17.5	9.1-33.8
Wallonia	273	979	27.9	26.4	25.2	14.9-35.2
Brussels	174	482	36.1	26.0	23.4	16.3-32.1
Primary hospitals	384	1 471	26.1	24.1	20.0	10.0-31.7
Secondary hospitals	204	659	31.0	27.3	31.5	8.6-40.3
Tertiary hospitals	102	497	20.5	20.5	17.6	12.7-28.9
Incidence per 1 000 hospitalisations						
Belgium	690	1 957 798	0.35	0.37	0.23	0.10-0.47
Flanders	243	1 179 548	0.21	0.20	0.18	0.07-0.26
Wallonia	273	563 465	0.48	0.48	0.41	0.19-0.72
Brussels	174	214 785	0.81	0.84	0.44	0.25-0.52
Primary hospitals	384	1 121 126	0.34	0.36	0.21	0.08-0.45
Secondary hospitals	204	466 880	0.44	0.44	0.35	0.18-0.66
Tertiary hospitals	102	369 792	0.28	0.28	0.26	0.22-0.34
Incidence density per 1 000 patient-days						
Belgium	690	11 263 066	0.06	0.06	0.04	0.02-0.08
Flanders	243	6 355 891	0.04	0.04	0.03	0.01-0.05
Wallonia	273	3 309 614	0.08	0.08	0.08	0.03-0.11
Brussels	174	1 597 561	0.11	0.11	0.05	0.03-0.07
Primary hospitals	384	6 611 361	0.06	0.06	0.04	0.01-0.08
Secondary hospitals	204	2 855 357	0.07	0.07	0.05	0.03-0.09
Tertiary hospitals	102	1 796 348	0.06	0.06	0.05	0.04-0.08

n = total number of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) isolates, N = total number of MRSA isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

Figure 6. Evolution of the median incidence of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 hospitalisations by region (clinical samples only), Belgian acute care hospital sites, 1994-2024

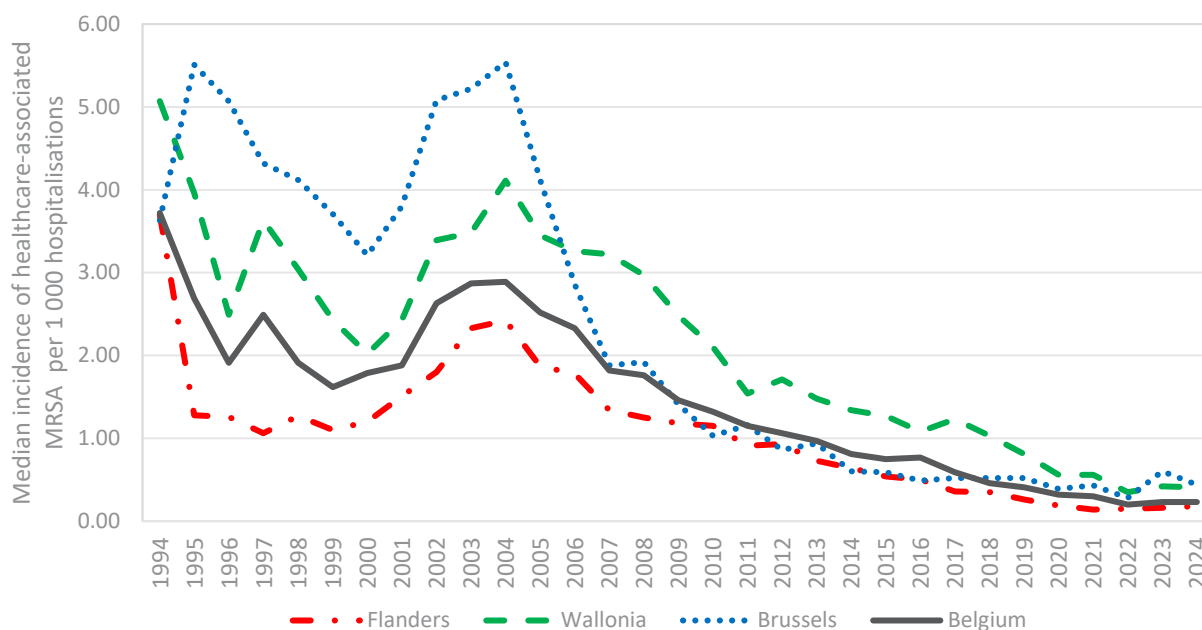


Figure 7. Evolution of the median incidence density of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 patient-days by region (clinical samples only), Belgian acute care hospital sites, 1994-2024

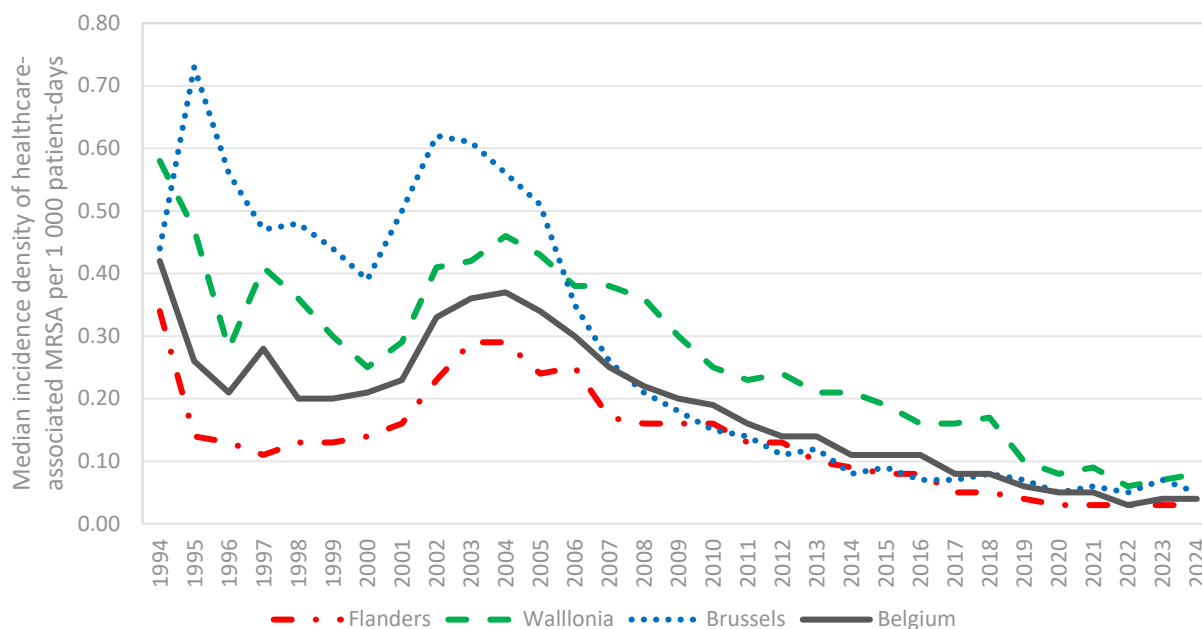
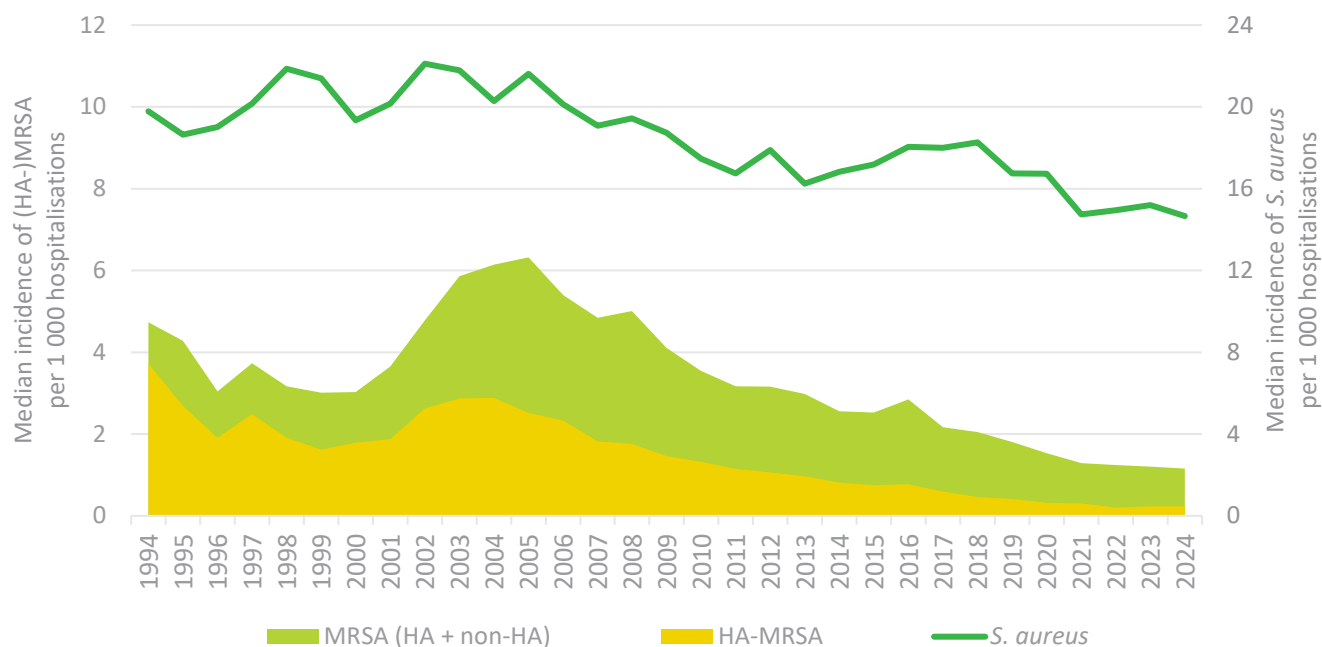


Figure 8 illustrates the overall evolution of the median incidence of *S. aureus*, MRSA, and HA-MRSA. Thanks to the combined efforts of infection prevention and control teams, as well as a range of actions - including recommendations for preventing MRSA transmission, more targeted screening policies, decolonisation and consecutive hand hygiene campaigns - the proportion of HA-MRSA among all MRSA cases decreased from 78.6% in 1994 (when surveillance began) to 39.9% in 2005 (the peak year for MRSA incidence) and 43.1% in 2006 (the year the surveillance became mandatory). In recent years, this percentage has fluctuated, decreasing from 22.7%

PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

in 2019 to 20.9% in 2020, followed by an increase in 2021 (23.3%) and a decrease in 2022 (16.1%). In 2023 and 2024, it rose again (19.2% and 19.8%, respectively).

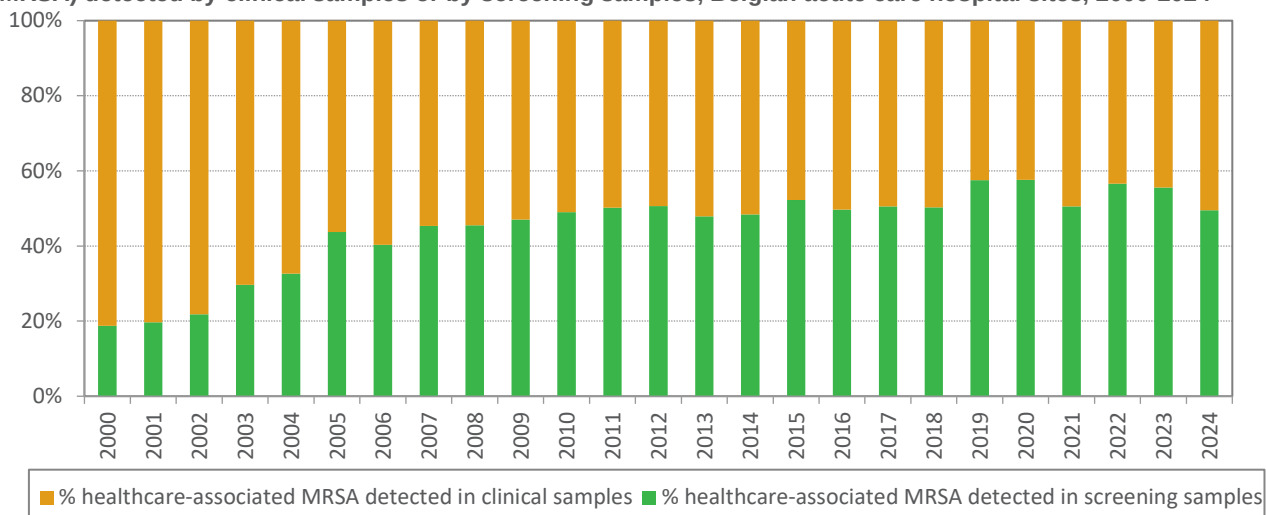
Figure 8. Evolution of the median incidence of *Staphylococcus (S.) aureus*, methicillin-resistant *S. aureus* (MRSA) and healthcare-associated (HA-)MRSA per 1 000 hospitalisations (clinical samples only), Belgian acute care hospital sites, 1994-2024



1.3.2 HEALTHCARE-ASSOCIATED MRSA IN SCREENING SAMPLES

In 2024, 677 screening samples were reported as MRSA-positive more than 48 hours after admission. The proportion of HA-MRSA cases detected through screening in 2024 (49.5%) was lower than in previous years (56.6% in 2022 and 55.6% in 2023). (Figure 9).

Figure 9. Evolution of the crude proportion of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) detected by clinical samples or by screening samples, Belgian acute care hospital sites, 2000-2024



1.4 MRSA SCREENING

Ninety-nine hospitals provided complete data on MRSA screening. In 2024, 263 575 patients of the 1 579 828 admitted patients (16.7%) were screened for MRSA carriage within 48 hours of admission (median: 1 942.5 screenings per hospital; IQR: 710.5-3448.5). On average, 20.2% of admitted patients were screened per hospital (median: 14.7%; IQR: 5.2-27.2%). This percentage varied by region: 18.9% in Flanders (median: 9.6%; IQR: 4.5-27.5%), 21.8% in Wallonia (median: 16.5%; IQR: 8.7-26.1%), and 21.6% in Brussels (median: 20.9%; IQR: 10.0-33.1%).

Ninety-one hospital sites provided information on their MRSA screening practices. Six hospital sites (6.6%) systematically screened all patients on all wards upon admission. Among the remaining hospitals, 75 (82.4%) screened all patients admitted to high-risk wards upon admission, according to locally defined criteria (e.g. intensive care units, onco-haematological wards). In addition, 90 hospitals (98.9%) screened all known MRSA carriers and all patients transferred from another Belgian hospital or from a foreign hospital with a known high prevalence upon admission.

During hospitalisation, 88 hospitals (96.7%) performed screening in the event of an outbreak on a specific ward, 78 (85.7%) conducted routine screening on high-risk wards, and 85 (93.4%) screened patients who had shared a room for more than 12 hours with a patient in whom MRSA infection or carriage had been identified.

2. MRSA in chronic care hospitals

2.1 RESISTANCE IN *STAPHYLOCOCCUS AUREUS*

The crude proportion of MRSA on the total number of reported *S. aureus* was 12.3% in 2024 (Table 6).

Table 6. Resistance proportion, incidence and incidence density of methicillin-resistant *Staphylococcus aureus* (MRSA) (clinical samples only) by region, Belgian chronic care hospital sites, 2024

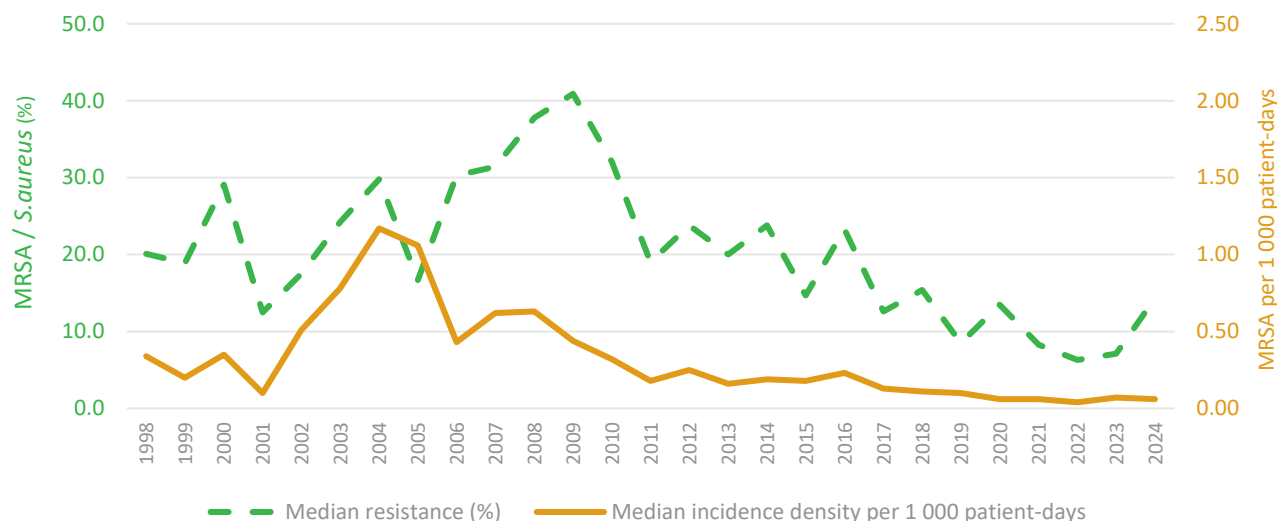
MRSA (clinical samples only) by region, Belgian chronic care hospital sites, 2024						
	n	N	MRSA (clinical samples only)			
			Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	39	318	12.3	27.5	14.3	7.2-20.0
Flanders	10	135	7.4	7.9	7.7	6.0-11.8
Wallonia	26	180	14.4	32.6	19.1	16.7-20.0
Brussels	3	3	100.0	-	-	-
Incidence per 1 000 hospitalisations						
Belgium	39	19 583	1.99	2.53	2.85	1.19-3.81
Flanders	10	5 521	1.81	2.64	2.97	0.71-4.56
Wallonia	26	13 275	1.96	2.19	2.85	1.19-2.85
Brussels	3	787	3.81	-	-	-
Incidence density per 1 000 patient-days						
Belgium	39	532 218	0.07	0.08	0.06	0.04-0.09
Flanders	10	207 837	0.05	0.04	0.04	0.04-0.06
Wallonia	26	286 538	0.09	0.12	0.09	0.04-0.20
Brussels	3	37 843	0.08	-	-	-

n = total number of methicillin-resistant *Staphylococcus aureus* (MRSA) isolates, N = total number of *Staphylococcus aureus* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

The overall evolution of the median MRSA resistance proportion and incidence density is shown in Figure 10.

Figure 10. Evolution of the median resistance proportion and incidence density per 1 000 patient-days of methicillin-resistant *Staphylococcus aureus* (MRSA; clinical samples only), Belgian chronic care hospital sites, 1998-2024



2.2 HEALTHCARE-ASSOCIATED MRSA

In 2024, 60.0% of all MRSA isolated from clinical samples were classified as healthcare-associated (Table 7). The crude incidence density of HA-MRSA was 0.04 cases per 1 000 patient-days.

Table 7. Resistance proportion, incidence and incidence density of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) (clinical samples only) by region, Belgian chronic care hospital sites, 2024

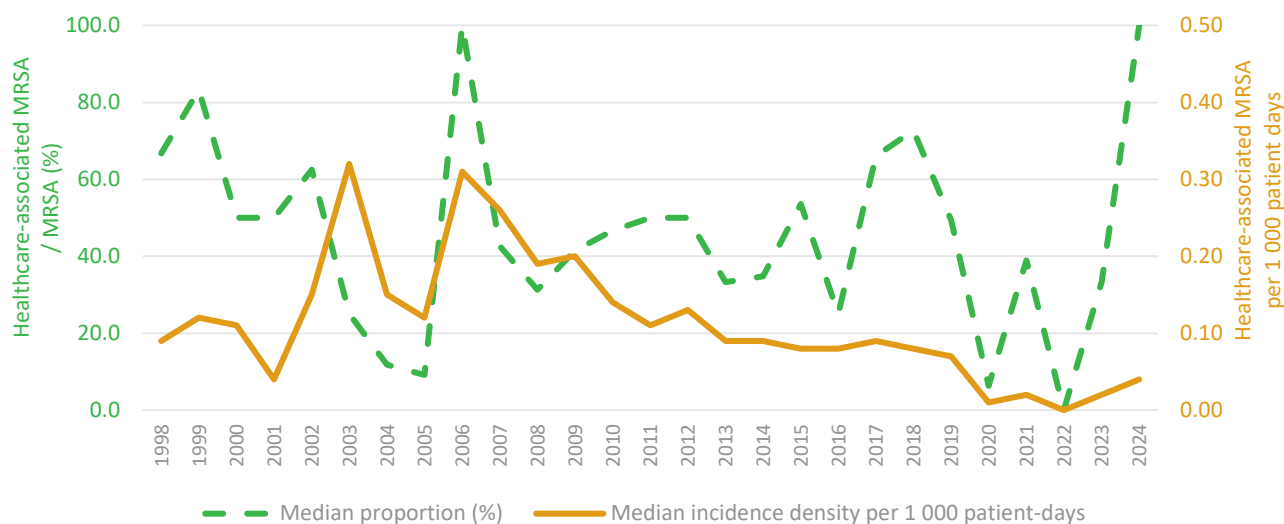
Staphylococcus aureus (MRSA) (clinical samples only) by region, Belgian chronic care hospital sites, 2024						
	Healthcare-associated MRSA (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Proportion healthcare-associated MRSA/MRSA (%)						
Belgium	21	35	60.0	66.7	100.0	33.3-100.0
Flanders	3	6	50.0	50.0	50.0	0.0-100.0
Wallonia	15	26	57.7	70.0	100.0	33.3-100.0
Brussels	3	3	100.0	-	-	-
Incidence per 1 000 hospitalisations						
Belgium	21	18 101	1.16	1.77	0.98	0.57-3.06
Flanders	3	4 039	0.74	1.45	0.64	0.28-2.61
Wallonia	15	13 275	1.13	1.61	1.01	0.95-2.85
Brussels	3	787	3.81	-	-	-
Incidence density per 1 000 patient-days						
Belgium	21	466 889	0.04	0.05	0.04	0.03-0.08
Flanders	3	142 508	0.02	0.03	0.03	0.01-0.04
Wallonia	15	286 538	0.05	0.05	0.04	0.03-0.08
Brussels	3	37 843	0.08	-	-	-

n = total number of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) isolates, N = total number of MRSA isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

PART 1. METHICILLIN-RESISTANT *STAPHYLOCOCCUS AUREUS* (MRSA)

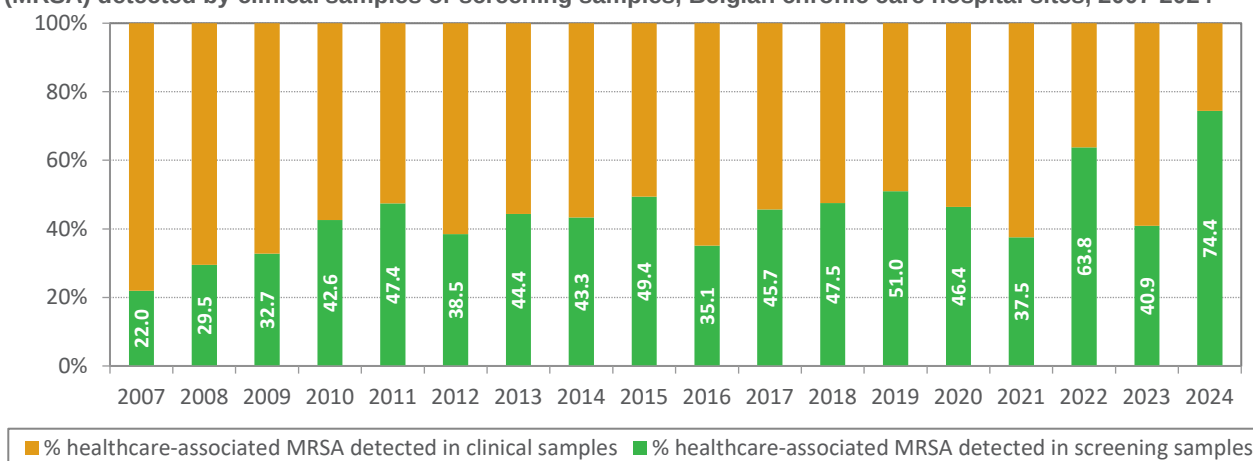
The overall evolution of these indicators (median proportion and incidence density) is presented in Figure 11.

Figure 11. Evolution of the median proportion and incidence density per 1 000 patient-days of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA; clinical samples only), Belgian chronic care hospital sites, 1998-2024



In addition, 61 cases (74.4%) of HA-MRSA were detected through screening in 2024 (Figure 12).

Figure 12. Evolution of the crude proportion of healthcare-associated methicillin-resistant *Staphylococcus aureus* (MRSA) detected by clinical samples or screening samples, Belgian chronic care hospital sites, 2007-2024



PART 2. VANCOMYCIN-RESISTANT ENTEROCOCCI (VRE)

Although participation in the VRE surveillance was optional, 94.1% (n=96/102, i.e. one less than the mandatory MRSA surveillance) of all Belgian acute care hospital administrative groups (mergers) participated with at least one hospital site in 2024.

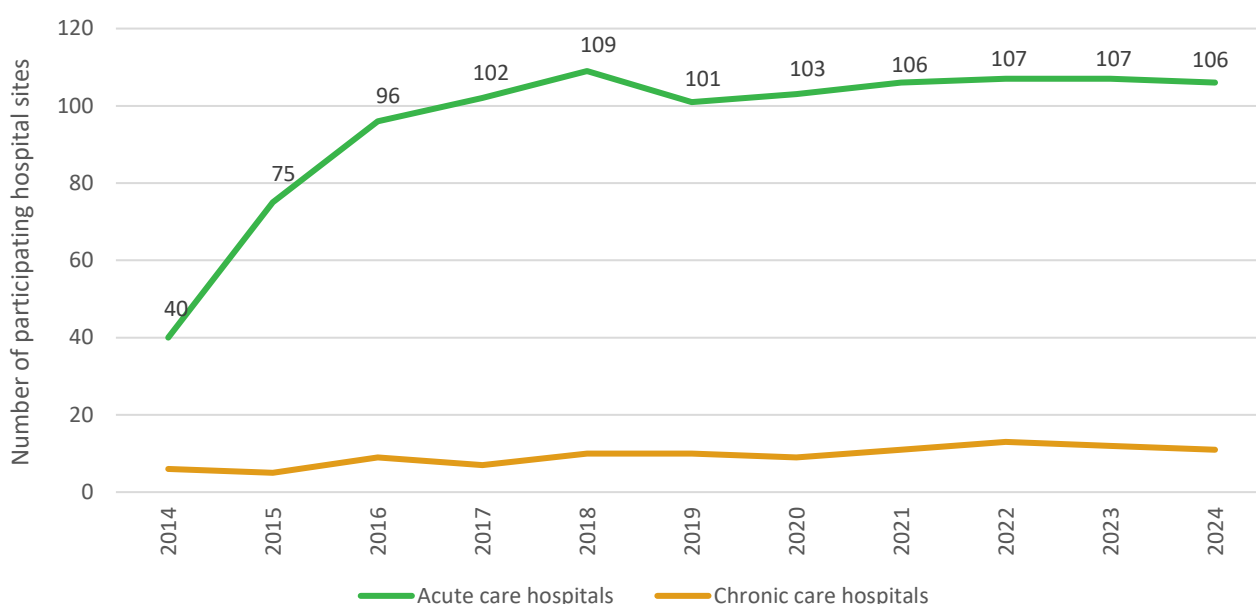
Table 8 presents the 2024 participation in the VRE surveillance by hospital care type, region and hospital level of specialty care. Figure 13 shows the number of participating hospital sites since the surveillance began in 2014.

Table 8. Participation in the surveillance of vancomycin-resistant enterococci by hospital care type, region and hospital level of specialty cares (for acute care hospitals only), Belgian acute and chronic care hospital sites, 2024 (Type D data only)

	2024			
	Flanders	Wallonia	Brussels	Belgium
N of acute care hospital sites	55	39	12	106
Primary hospital sites	44	28	7	79
Secondary hospital sites	8	10	2	20
Tertiary hospital sites	3	1	3	7
N of chronic care hospital sites	5	5	1	11

N = number

Figure 13. Evolution of the participation in the surveillance of vancomycin-resistant enterococci based on the resistance proportion indicator, Belgian acute and chronic care hospital sites, 1994-2024



1. VRE in acute care hospitals

1.1 ENTEROCOCCUS FAECIUM

There were 8 729 *E. faecium* (median: 50.5 isolates per hospital; IQR: 27-119) isolated from clinical samples (excluding faeces samples) reported in 2024. The proportion of hospitals reporting at least one vanco-R *E. faecium* case increased markedly, from 9.3% in 2023 to 49.1% in 2024. A total of 269 vanco-R *E. faecium* were reported by 52 acute care hospital sites, with a range of 1 to 50 isolates per hospital site. The crude resistance proportion and incidence of vanco-R *E. faecium* were 3.08% and 0.139 cases per 1 000 hospitalisations, respectively (Table 9).

Table 9. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) vancomycin-resistant *Enterococcus faecium* (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

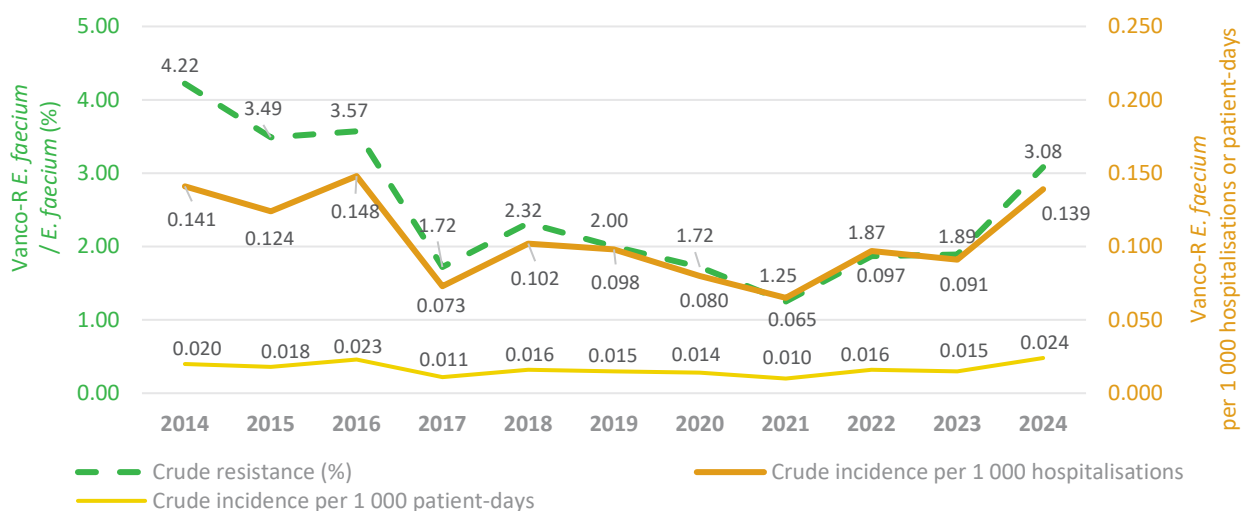
	Vancomycin-resistant <i>Enterococcus faecium</i> (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	269	8 729	3.08	3.55	0.00	0.00-2.94
Flanders	53	5 354	0.99	0.99	0.00	0.00-0.96
Wallonia	165	2 470	6.68	5.82	1.37	0.00-6.90
Brussels	51	905	5.64	8.06	3.97	1.09-8.69
Primary hospitals	140	4 539	3.08	3.24	0.00	0.00-2.78
Secondary hospitals	71	2 314	3.07	4.63	1.20	0.00-3.59
Tertiary hospitals	58	1 876	3.09	3.88	2.70	1.54-8.41
Incidence per 1 000 hospitalisations						
Belgium	269	1 937 141	0.139	0.177	0.000	0.000-0.125
Flanders	53	1 151 556	0.046	0.048	0.000	0.000-0.045
Wallonia	165	570 800	0.289	0.352	0.066	0.000-0.204
Brussels	51	214 785	0.237	0.196	0.111	0.019-0.351
Primary hospitals	140	1 100 469	0.127	0.143	0.000	0.000-0.101
Secondary hospitals	71	466 880	0.152	0.287	0.059	0.000-0.144
Tertiary hospitals	58	369 792	0.157	0.245	0.184	0.036-0.510
Incidence density per 1 000 patient-days						
Belgium	269	11 159 287	0.024	0.027	0.000	0.000-0.023
Flanders	53	6 216 136	0.009	0.009	0.000	0.000-0.007
Wallonia	165	3 345 590	0.049	0.053	0.010	0.000-0.032
Brussels	51	1 597 561	0.032	0.025	0.015	0.004-0.036
Primary hospitals	140	6 507 582	0.022	0.022	0.000	0.000-0.016
Secondary hospitals	71	2 855 357	0.025	0.041	0.008	0.000-0.024
Tertiary hospitals	58	1 796 348	0.032	0.038	0.027	0.010-0.056

n = total number of vancomycin-resistant *Enterococcus faecium* isolates, N = total number of *Enterococcus faecium* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

Whereas a significantly decreasing trend in resistance proportion was observed across the surveillance period up to 2023 (-0.13% per year; p=0.048), this trend was no longer evident in 2024 (-0.02% per year; p=0.822). Since 2021, a significant increase in both resistance proportion (+0.74% per year; p=0.006) and incidence (IRR=1.145, 95%CI: 1.017-1.288; p=0.025) of vanco-R *E. faecium* has been observed (Figure 14).

PART 2. VANCOMYCIN-RESISTANT ENTEROCOCCI (VRE)

Figure 14. Evolution of the crude resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of vancomycin resistance in *Enterococcus faecium* (clinical samples only), Belgian acute care hospital sites, 2014-2024



Note: Prior to 2016, vancomycin resistance was separated under vancomycin resistance (defined as vanco-R and susceptible to teicoplanin or susceptibility unknown) and glycopeptide resistance (defined as vanco-R and teicoplanin resistant). Since 2017, vancomycin resistance is questioned independently from the susceptibility to teicoplanin.

The evolution of the crude resistance proportion and incidence (per 1 000 hospitalisations) of vanco-R *E. faecium* by region and by level of specialty care can be found in Annex (Figure A4 – A7).

Thirty-seven hospital sites provided data on the resistance mechanism underlying vancomycin resistance. In total, these hospitals reported 149 vanco-R *E. faecium* isolates, of which 131 (87.9%) harboured *vanA* genes and 5 (3.4%) *vanB* genes.

In total, 22 cases of *E. faecium* resistant to linezolid (linezolid-R) were reported by 13 acute care hospital sites (n=13/104; 14.3%), with a range of 1 to 5 isolates per hospital site. The crude resistance proportion and incidence of linezolid-R *E. faecium* were 0.25% and 0.011 cases per 1 000 hospitalisations, respectively.

One hundred hospital sites provided complete data on vancomycin resistance, linezolid resistance and combined vancomycin-linezolid resistance in *E. faecium*. Among these hospitals, 53%, 87% and 96% reported no cases of vancomycin resistance, linezolid resistance and combined resistance, respectively. In total, 259 vanco-R *E. faecium* cases were reported (range: 1-50 isolates), along with 22 linezolid-R cases (range: 1-5 isolates), and 7 cases exhibiting combined vancomycin-linezolid resistance (range: 1-3 cases).

1.2 ENTEROCOCCUS FAECALIS

A total of 25 371 *E. faecalis* (median: 159 isolates per hospital; IQR: 84-289) isolated from clinical samples (excluding faeces samples) were reported in 2024. Among these, 12 cases of vanco-R *E. faecalis* were reported by 5 acute care hospital sites (4.7%; min-max: 1-6 isolates per hospital site). The crude resistance proportion and incidence of vanco-R *E. faecalis* were 0.05% and 0.006 cases per 1 000 hospitalisations, respectively.

Forty hospital sites provided data on the resistance mechanism underlying vancomycin resistance. In total, these hospitals reported 9 vanco-R *E. faecalis* isolates and all harboured *vanA* genes.

PART 2. VANCOMYCIN-RESISTANT ENTEROCOCCI (VRE)

In total, 57 cases of linezolid-R *E. faecalis* were reported by 24 acute care hospital sites (n=24/104; 23.1%), with a range of 1 to 8 isolates per hospital site. The crude resistance proportion and incidence of linezolid-R *E. faecalis* were 0.22% and 0.029 cases per 1 000 hospitalisations, respectively.

In *E. faecalis*, crude resistance to linezolid (0.22%) was comparable to that observed in *E. faecium* (0.25%), whereas vancomycin resistance was much lower (0.05% vs. 3.08%).

One hundred hospital sites provided complete data on vancomycin resistance, linezolid resistance and combined vancomycin-linezolid resistance in *E. faecalis*. Among these hospitals, 95%, 77% and 99% reported no cases of vancomycin resistance, linezolid resistance and combined resistance, respectively. In total, 12 vanco-R *E. faecalis* cases were reported (range: 1-6 isolates), along with 56 linezolid-R cases (range: 1-8 isolates), and one case exhibiting combined vancomycin-linezolid resistance.

1.3 VRE SCREENING

Seventy-seven hospitals provided complete data on VRE screening. In total, 68 700 patients were screened for VRE carriage in these hospitals (median: 407 screenings per hospital; IQR: 83-1 476). Four hospitals reported no VRE screenings. Overall, 664 vanco-R *E. faecium* and 6 vanco-R *E. faecalis* isolates were identified by these hospitals.

On average, 6.8% of admitted patients were screened per hospital (median: 3.6%; IQR: 0.5-10.6%). This percentage varied by region: 2.9% in Flanders (median: 0.9%; IQR: 0.3-5.0%), 9.7% in Wallonia (median: 7.2%; IQR: 2.7-15.6%), and 18.2% in Brussels (median: 11.5%; IQR: 7.4-33.4%).

Ninety-two hospital sites provided information on their VRE screening practices. Only one hospital (1.1%) systematically screened all patients on all wards upon admission. Among the remaining hospitals, 51 (56.0%) screened all patients admitted to high-risk wards upon admission, according to locally defined criteria (e.g. intensive care units, onco-haematological wards). In addition, 85 hospitals (93.4%) screened all known VRE carriers, and 72 (79.1%) screened all patients transferred from another Belgian hospital or from a foreign hospital with a known high prevalence upon admission.

During hospitalisation, 89 hospitals (96.7%) performed screening in the event of an outbreak on a specific ward, 54 (58.7%) conducted routine screening on high-risk wards, and 84 (91.3%) screened patients who had shared a room for more than 12 hours with a patient in whom VRE infection or carriage had been identified.

1.4 OUTBREAKS

In the 2024 data collection, the question on outbreak during the previous year specifically referred to vancomycin-resistant enterococci, rather than vanco-R and/or linezolid-R enterococci as in previous years. Despite this more restrictive definition, a higher proportion of hospitals reported an outbreak in 2024 (15.6%) compared with 2023 (13.8%). The reported number of patients involved is in particular high for the last two years. Table 10 details the number of clusters reported and the number of patients involved between 2014 and 2024.

PART 2. VANCOMYCIN-RESISTANT ENTEROCOCCI (VRE)

Table 10. Evolution of the number of outbreaks with vancomycin or linezolid resistant enterococci reported in the national surveillance in Belgian acute care hospital sites, 2014-2024

	Outbreaks of vancomycin-resistant (VRE) and/or linezolid-resistant enterococci										VRE only
	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hospital sites reporting an outbreak (%)	3/40 (7.5)	7/75 (9.3)	7/95 (7.4)	13/98 (13.3)	13/96 (13.5)	16/91 (17.6)	7/88 (8.0)	4/97 (4.1)	11/97 (11.3)	12/91 (13.8)	15/96 (15.6)
Hospital sites with no answer	0	0	1	4	13	10	14	9	10	16	10
Total N of reported clusters (min-max N of clusters reported by hospital)	3 (1-1)	11 (1-4)	12 (1-3)	21 (1-6)	28 (1-13)	19 (1-3)*	7 (1-2)*	3 (1-2)*	14 (1-3)*	26 (1-9)*	24 (1-4)
Patients involved	68	140	247	166	164	268	27	10	244	377	376
% patients colonised	79.4	87.7	88.8	89.8	88.4	94.4	77.8	90.0	87.3	92.3	88.1
% patients infected	20.6	12.3	11.2	10.2	11.6	5.6	22.2	10.0	12.7	7.7	11.9

*data missing for two (2020, 2021, 2022, 2023) or three (2019) hospital sites

2. VRE in chronic care hospitals

In 2024, a total of 126 *E. faecium* isolates (median: 8 per hospital; IQR: 5-15) and 314 *E. faecalis* isolates (median: 18 per hospital; IQR: 11-45) were reported from clinical samples (excluding faecal samples). Eleven cases of vanco-R *E. faecium* (all carrying *vanA* genes) were reported by two of the 11 chronic care hospital sites. No cases of vanco-R *E. faecalis* were observed. Two cases of linezolid-R *E. faecalis* were reported. For both species, no combined vancomycin-linezolid-resistant isolates were identified (as reported by nine hospital sites).

No outbreaks involving vanco-R and/or linezolid-R enterococci were reported by the participating chronic care hospital sites between 2014 and 2024.

Six hospitals provided complete data on VRE screening. In total, 967 patients were screened for VRE carriage in these hospitals (median: 22 screenings per hospital; IQR: 0-353). Two hospitals reported no VRE screenings. Overall, 10 vanco-R *E. faecium* and zero vanco-R *E. faecalis* isolates were identified.

Eight hospital sites provided information on their VRE screening practices. Two hospitals (25.0%) systematically screened all patients on all wards upon admission. Among the remaining hospitals (n=5; missing answers for one hospitals), 2 (40.0%) screened all patients admitted to high-risk wards upon admission, according to locally defined criteria. In addition, all hospitals screened all known VRE carriers, and all patients transferred from another Belgian hospital or from a foreign hospital with a known high prevalence upon admission.

During hospitalisation, all hospitals performed screening in the event of an outbreak on a specific ward, and screened all patients who had shared a room for more than 12 hours with a patient in whom VRE infection or carriage had been identified. Four hospitals (50.0%) conducted routine screening on high-risk wards.

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

In 2024, 98.0% (n=100/102) of all acute care hospital administrative groups (mergers) participated in the MRGN surveillance with at least one hospital site. Two hospital administrative groups were considered as chronic care hospitals (length of stay > 16 days).

The number of hospital sites participating in 2024 can be found in Table 11 by hospital care type, region and hospital level of specialty care.

Table 11. Participation in the surveillance of multi-resistant Gram-negative bacteria (MRGN) by hospital care type, region and hospital level of specialty care (for acute care hospitals only), Belgian acute and chronic care hospital sites, 2024 (Type D data only)

	2024			
	Flanders	Wallonia	Brussels	Belgium
N of acute care hospital sites	57	39	12	108
Primary hospital sites	46	28	7	81
Secondary hospital sites	8	10	2	20
Tertiary hospital sites	3	1	3	7
N of chronic care hospital sites	5	5	1	11

N = number

1. Resistant Gram-negative bacteria in acute care hospitals

1.1 RESISTANCE IN *ESCHERICHIA COLI*

In 2024, 9.2% of all *E. coli* isolated from clinical samples were resistant to 3rd generation cephalosporins (3GC-R). The crude incidence was 4.20 per 1 000 hospitalisations or 0.73 per 1 000 patient-days (Table 12).

After a significant decrease between 2019 and 2022 (-0.63% per year; $p < 0.001$), the resistance proportion of 3GC-R *E. coli* increased again over the past three years (+0.50% per year; $p = 0.005$). However, this comparison between periods should be interpreted with caution, as the exclusion of the EUCAST "I" category in the resistance definition may have influenced the observed trends. No significant trend was observed over the entire surveillance period from 2014 to 2024 (-0.01% per year; $p = 0.886$).

The incidence of 3GC-R *E. coli* remained stable between 2022 and 2024 (IRR=1.033, 95%CI: 0.993-1.076; $p = 0.111$), but decreased significantly over the full surveillance period (IRR=0.977, 95%CI: 0.969-0.985; $p < 0.001$) (Figure 15).

The evolution of the median resistance proportion and incidence (per 1 000 hospitalisations) of 3GC-R *E. coli*, by region and by level of specialty care, can be found in Annex (Figure A8 – A11).

A total of 51 cases of meropenem-resistant (mero-R) *E. coli* (0.06%) were reported by 28 (25.9%) acute care hospital sites (range: 1-5 isolates per hospital). The crude incidence and incidence density of mero-R *E. coli* was 0.026 cases per 1 000 hospitalisations and 0.005 per 1 000 patient-days, respectively (Table 12).

Figure 16 presents the evolution of the crude (median all zero values, except for 0.05%, 0.024 per 1 000 hospitalisation or 0.003 per 1 000 patient days in 2019) resistance proportion and incidences of mero-R *E. coli*.

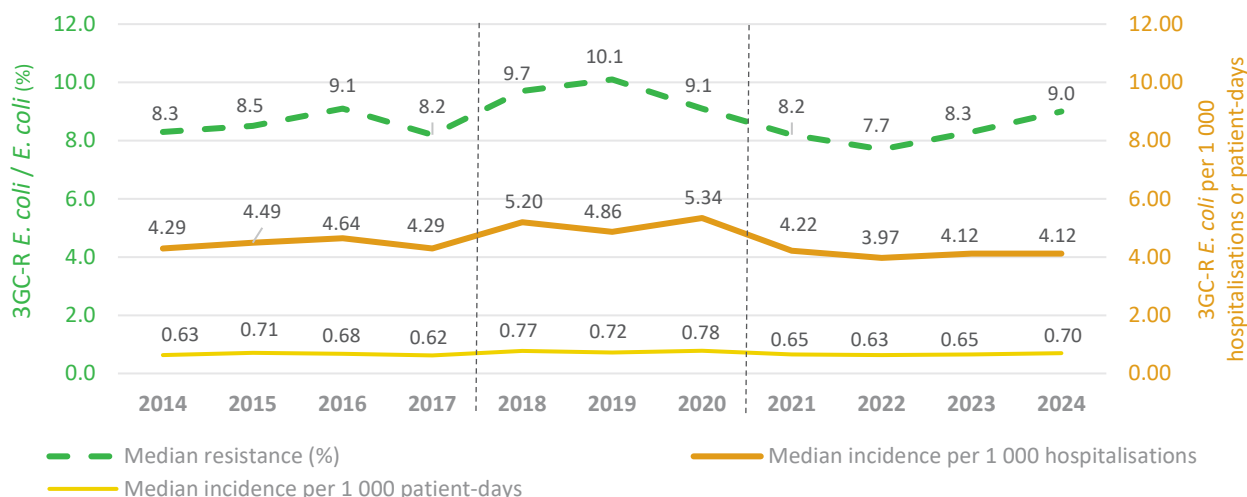
Table 12. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins or meropenem (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	<i>Escherichia coli</i> (clinical samples only)											
	Resistance to third generation cephalosporins						Resistance to meropenem					
	n	N	Crude	Mean	Median	P25 - P75	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)												
Belgium	8 257	89 988	9.2	9.3	9.0	7.1-10.9	51	89 988	0.06	0.06	0.00	0.00-0.03
Flanders	4 446	53 532	8.3	8.1	8.2	6.5-9.3	26	53 532	0.05	0.05	0.00	0.00-0.00
Wallonia	2 480	25 401	9.8	9.7	9.6	8.4-11.0	14	25 401	0.06	0.07	0.00	0.00-0.00
Brussels	1 331	11 055	12.0	13.9	14.7	12.3-15.9	11	11 055	0.10	0.09	0.00	0.00-0.16
Primary hosp	4 906	59 367	8.3	8.7	8.5	6.8-10.1	26	59 367	0.04	0.06	0.00	0.00-0.00
Secondary hosp	1 746	18 546	9.4	10.3	9.9	8.4-10.9	8	18 546	0.04	0.04	0.00	0.00-0.05
Tertiary hosp	1 605	12 075	13.3	13.6	12.7	11.8-15.6	17	12 075	0.14	0.16	0.16	0.00-0.24
Incidence per 1 000 hospitalisations												
Belgium	8 257	1 965 133	4.20	4.46	4.12	2.99-5.32	51	1 965 133	0.026	0.026	0.000	0.000-0.027
Flanders	4 446	1 179 548	3.77	3.97	3.90	2.76-5.16	26	1 179 548	0.022	0.019	0.000	0.000-0.000
Wallonia	2 480	570 800	4.34	4.67	4.36	3.06-5.78	14	570 800	0.025	0.032	0.000	0.000-0.000
Brussels	1 331	214 785	6.20	6.10	4.86	4.14-6.21	11	214 785	0.051	0.038	0.000	0.000-0.059
Primary hosp	4 906	1 128 461	4.35	4.50	4.07	2.86-5.44	26	1 128 461	0.023	0.026	0.000	0.000-0.000
Secondary hosp	1 746	466 880	3.74	4.13	4.59	3.11-5.21	8	466 880	0.017	0.013	0.000	0.000-0.027
Tertiary hosp	1 605	369 792	4.34	4.97	4.98	2.94-7.06	17	369 792	0.046	0.063	0.055	0.000-0.084
Incidence density per 1 000 patient-days												
Belgium	8257	11 299 042	0.73	0.72	0.70	0.49-0.87	51	11 299 042	0.005	0.004	0.000	0.000-0.004
Flanders	4 446	6 355 891	0.70	0.69	0.66	0.49-0.86	26	6 355 891	0.004	0.004	0.000	0.000-0.000
Wallonia	2 480	3 345 590	0.74	0.73	0.76	0.52-0.91	14	3 345 590	0.004	0.005	0.000	0.000-0.000
Brussels	1 331	1 597 561	0.83	0.78	0.68	0.50-0.84	11	1 597 561	0.007	0.005	0.000	0.000-0.008
Primary hosp	4 906	6 647 337	0.74	0.73	0.71	0.52-0.88	26	6 647 337	0.004	0.005	0.000	0.000-0.000
Secondary hosp	1 746	2 855 357	0.61	0.61	0.61	0.45-0.79	8	2 855 357	0.003	0.002	0.000	0.000-0.004
Tertiary hosp	1 605	1 796 348	0.89	0.87	0.84	0.77-1.03	17	1 796 348	0.009	0.009	0.010	0.000-0.013

Hosp = hospital; n = total number of *Escherichia coli* isolates resistant to 3rd generation cephalosporins or meropenem, N = total number of *Escherichia coli* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

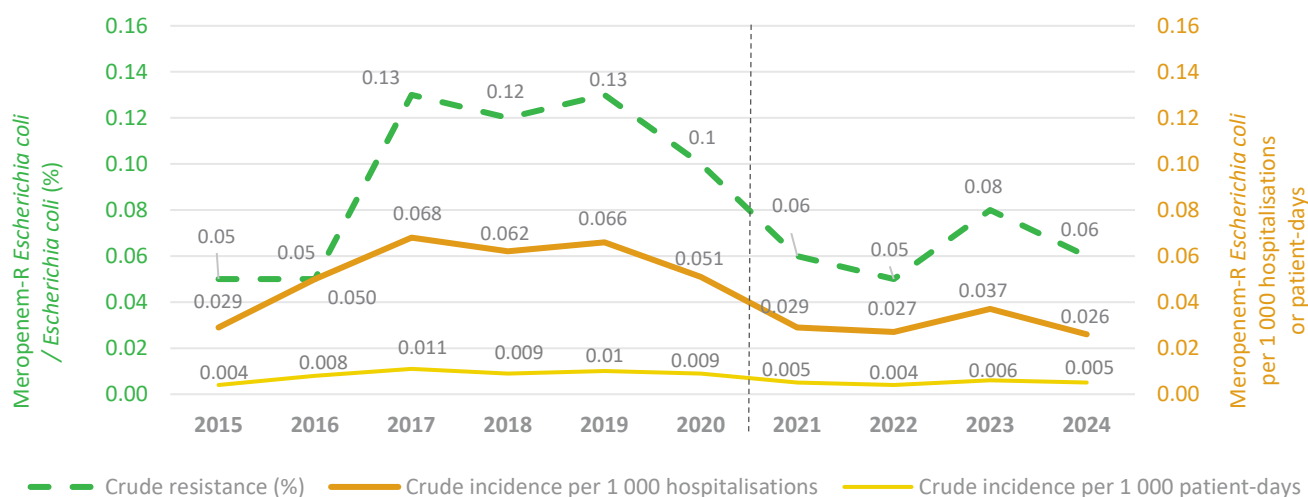
PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

Figure 15. Evolution of the median resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure – formerly intermediate result) is displayed.

Figure 16. Evolution of the crude resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to meropenem (clinical samples only), Belgian acute care hospital sites, 2014-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

1.2 RESISTANCE IN *KLEBSIELLA PNEUMONIAE*

In 2024, the crude resistance proportion 3GC-R *K. pneumoniae* was 18.2%, with a crude incidence of 1.96 cases per 1 000 hospitalisations or 0.34 cases per 1 000 patient-days (clinical samples only; Table 13).

Table 13. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins or meropenem (clinical samples only) by region and specialty care level within the hospital, Belgian acute care hospital sites, 2024

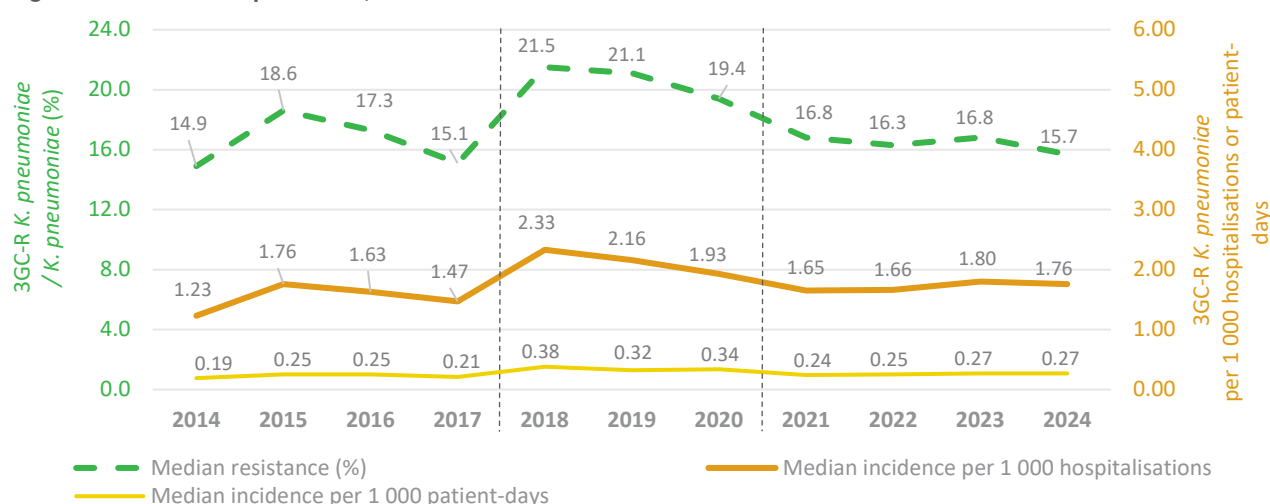
	<i>Klebsiella pneumoniae</i> (clinical samples only)											
	Resistance to third generation cephalosporins						Resistance to meropenem					
	n	N	Crude	Mean	Median	P25 - P75	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)												
Belgium	3 859	21 165	18.2	17.4	15.7	10.6-24.0	255	20 994	1.21	0.99	0.57	0.00-1.55
Flanders	1 583	11 513	13.7	11.4	11.7	8.3-14.9	107	11 342	0.94	0.65	0.31	0.00-1.23
Wallonia	1 684	6 567	25.6	24.7	25.0	21.2-31.0	100	6 567	1.52	1.16	0.76	0.00-2.33
Brussels	592	3 085	19.2	22.4	22.5	18.0-28.2	48	3 085	1.56	2.02	1.86	0.87-2.86
Primary hosp	2 034	12 668	16.1	16.5	14.0	9.9-23.3	95	12 497	0.76	0.82	0.33	0.00-1.43
Secondary hosp	927	4 502	20.6	19.6	17.3	14.8-27.4	60	4 502	1.33	1.19	1.02	0.24-1.72
Tertiary hosp	898	3 995	22.5	21.6	22.0	18.2-24.7	100	3 995	2.50	2.34	2.43	0.88-4.00
Incidence per 1 000 hospitalisations												
Belgium	3 859	1 965 133	1.96	2.09	1.76	0.89-2.56	255	1 954 412	0.130	0.111	0.068	0.000-0.153
Flanders	1 583	1 179 548	1.34	1.17	1.03	0.72-1.62	107	1 168 827	0.092	0.064	0.032	0.000-0.104
Wallonia	1 684	570 800	2.95	3.19	2.52	2.00-4.22	100	570 800	0.175	0.145	0.121	0.000-0.265
Brussels	592	214 785	2.76	2.89	2.58	1.96-3.09	48	214 785	0.223	0.215	0.161	0.136-0.301
Primary hosp	2 034	1 128 461	1.80	2.02	1.61	0.86-2.47	95	1 117 740	0.085	0.097	0.053	0.000-0.134
Secondary hosp	927	466 880	1.99	2.26	1.92	1.00-2.90	60	466 880	0.129	0.117	0.081	0.017-0.196
Tertiary hosp	898	369 792	2.43	2.47	2.21	1.88-3.26	100	369 792	0.270	0.251	0.242	0.132-0.366
Incidence density per 1 000 patient-days												
Belgium	3 859	11 299 042	0.34	0.33	0.27	0.16-0.43	255	11 207 539	0.023	0.018	0.012	0.000-0.027
Flanders	1 583	6 355 891	0.25	0.21	0.19	0.12-0.28	107	6 264 388	0.017	0.012	0.006	0.000-0.018
Wallonia	1 684	3 345 590	0.50	0.50	0.45	0.33-0.71	100	3 345 590	0.030	0.024	0.016	0.000-0.041
Brussels	592	1 597 561	0.37	0.36	0.30	0.25-0.46	48	1 597 561	0.030	0.028	0.027	0.017-0.040
Primary hosp	2 034	6 647 337	0.31	0.32	0.26	0.16-0.39	95	6 555 834	0.014	0.015	0.009	0.000-0.021
Secondary hosp	927	2 855 357	0.32	0.35	0.24	0.18-0.47	60	2 855 357	0.021	0.019	0.012	0.004-0.031
Tertiary hosp	898	1 796 348	0.50	0.47	0.47	0.30-0.59	100	1 796 348	0.056	0.051	0.048	0.023-0.067

Hosp = hospital; n = total number of *Klebsiella pneumoniae* isolates resistant to 3rd generation cephalosporins or meropenem, N = total number of *Klebsiella pneumoniae* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

Between 2014 and 2024, no significant trend was observed in the resistance proportion (-0.25% per year; $p=0.065$). However, since 2018, a significant decrease has been observed in both the resistance proportion (-1.23% per year; $p<0.001$) and incidence (IRR=0.938, 95%CI: 0.924-0.952; $p<0.001$) (Figure 17). The latter trend should be interpreted with caution, as the exclusion of the EUCAST “I” category in the resistance definition may have influenced comparisons over time.

Figure 17. Evolution of the median resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

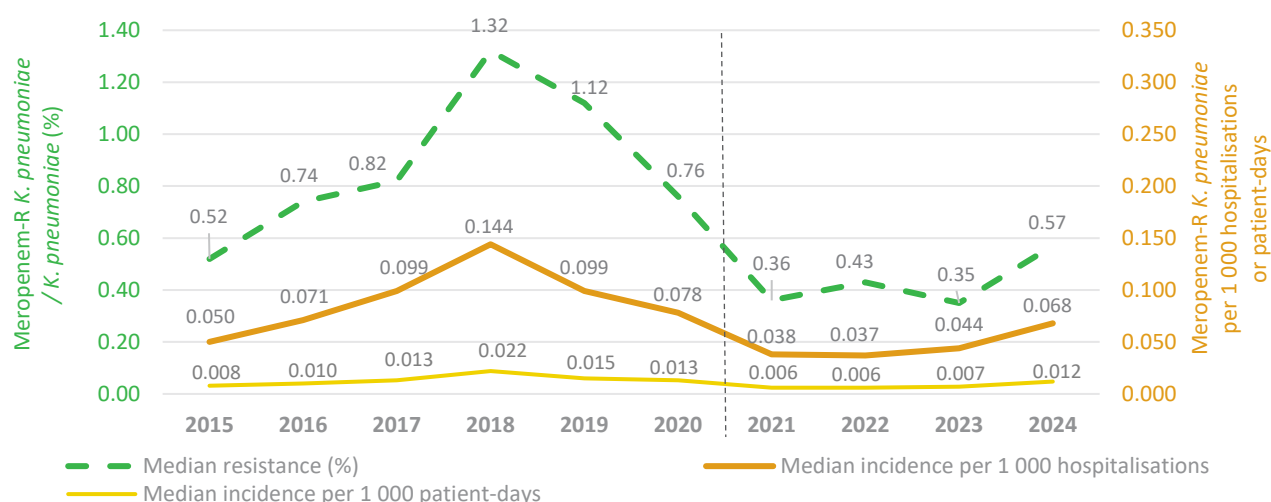
The evolution of the median resistance proportion and incidence (per 1 000 hospitalisations) of 3GC-R *K. pneumoniae*, by region and by level of specialty care, can be found in Annex (Figure A12 – A15).

Cases of mero-R *K. pneumoniae* were reported by 60.7% ($n=65/107$; min-max: 1-39 isolates) of the acute care hospital sites. The crude resistance proportion and incidence were 1.21% or 0.130 cases per 1 000 hospitalisations in 2024 (Table 13).

After an increase in both the resistance proportion (+0.34% per year; $p=0.002$) and incidence (IRR=1.083, 95%CI: 0.992-1.181; $p=0.075$) of mero-R *K. pneumoniae* between 2015 and 2018, both indicators decreased significantly until 2021: resistance proportion (-0.38% per year; $p=0.001$) and incidence (IRR=0.826, 95%CI: 0.765-0.892; $p<0.001$). Between 2021 and 2024, both the resistance proportion (-0.04% per year; $p=0.547$) and incidence (IRR=1.053, 95%CI: 0.966-1.147; $p=0.242$) showed non-significant increases (Figure 18). The latter trend should be interpreted with caution, as the exclusion of the EUCAST “I” category in the resistance definition may have influenced comparisons over time, particularly given the potentially substantial proportion of isolates categorized as intermediate within the combined I/R group prior to 2021.

The evolution of the median resistance proportion and incidence (per 1 000 hospitalisations) of mero-R *K. pneumoniae*, by region and level of specialty care, can be found in the Annex (Figures A16–A19).

Figure 18. Evolution of the median resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to meropenem (clinical samples only), Belgian acute care hospital sites, 2014-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

1.3 RESISTANCE IN *ACINETOBACTER BAUMANNII*

A minority of acute care hospital sites (n=26/108, 24.1%) reported at least one mero-R *A. baumannii* (range: 1-6 isolates) in 2024 (clinical samples only). The crude resistance proportion and incidence of mero-R *A. baumannii* in clinical samples were 6.21% and 0.027 cases per 1 000 hospitalisations, or 0.005 per 1 000 patient days, respectively (Table 14).

Figure 19 presents the evolution of the crude (median all zero values) resistance proportion and incidences of mero-R *A. baumannii*. Between 2015 and 2024, no significant trend was observed in the resistance proportion (-0.07% per year; $p=0.505$), whereas the incidence decreased significantly (IRR=0.926, 95%CI: 0.897-0.955; $p<0.001$) over the same time period. More recently, between 2021 and 2024, no significant change in resistance proportion was observed (0.11% per year; $p=0.777$), while the incidence rate remained stable (IRR=0.870, 95%CI: 0.741-1.022; $p=0.091$).

The evolution of the crude resistance proportion and incidence (per 1 000 hospitalisations) of mero-R *A. baumannii*, by region and by level of specialty care, can be found in Annex (Figure A20 – A23).

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

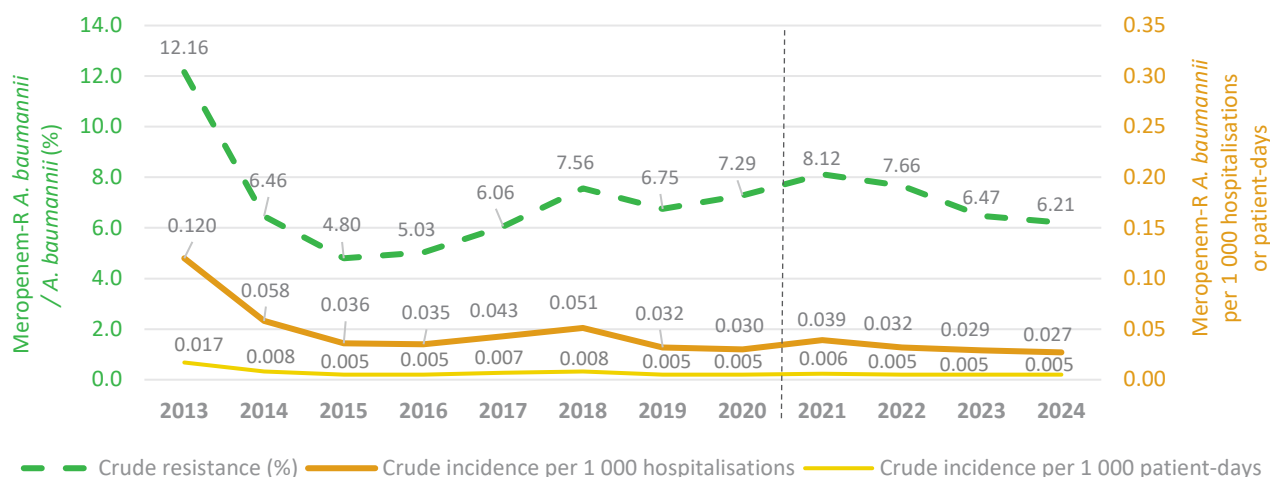
Table 14. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Acinetobacter baumannii* resistant to meropenem (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	<i>Acinetobacter baumannii</i> resistant to meropenem (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	53	854	6.21	5.13	0.00	0.00-0.00
Flanders	22	526	4.18	4.48	0.00	0.00-0.00
Wallonia	19	181	10.50	5.61	0.00	0.00-0.00
Brussels	12	147	8.16	6.67	5.26	0.00-12.50
Primary hospitals	16	455	3.52	3.52	0.00	0.00-0.00
Secondary hospitals	19	190	10.00	9.73	2.38	0.00-18.33
Tertiary hospitals	18	209	8.61	10.65	11.43	0.00-12.50
Incidence per 1 000 hospitalisations						
Belgium	53	1 965 133	0.027	0.023	0.000	0.000-0.000
Flanders	22	1 179 548	0.019	0.013	0.000	0.000-0.000
Wallonia	19	570 800	0.033	0.022	0.000	0.000-0.000
Brussels	12	214 785	0.056	0.071	0.019	0.000-0.094
Primary hospitals	16	1 128 461	0.014	0.015	0.000	0.000-0.000
Secondary hospitals	19	466 880	0.041	0.039	0.014	0.000-0.062
Tertiary hospitals	18	369 792	0.049	0.065	0.067	0.000-0.118
Incidence density per 1 000 patient-days						
Belgium	53	11 299 042	0.005	0.004	0.000	0.000-0.000
Flanders	22	6 355 891	0.003	0.003	0.000	0.000-0.000
Wallonia	19	3 345 590	0.006	0.004	0.000	0.000-0.000
Brussels	12	1 597 561	0.008	0.008	0.002	0.000-0.013
Primary hospitals	16	6 647 337	0.002	0.002	0.000	0.000-0.000
Secondary hospitals	19	2 855 357	0.007	0.006	0.002	0.000-0.011
Tertiary hospitals	18	1 796 348	0.010	0.011	0.013	0.000-0.013

n = total number of *Acinetobacter baumannii* resistant to meropenem isolates, N = total number of *Acinetobacter baumannii* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

Figure 19. Evolution of the crude resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Acinetobacter baumannii* resistant to meropenem (clinical samples only), Belgian acute care hospital sites, 2013-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

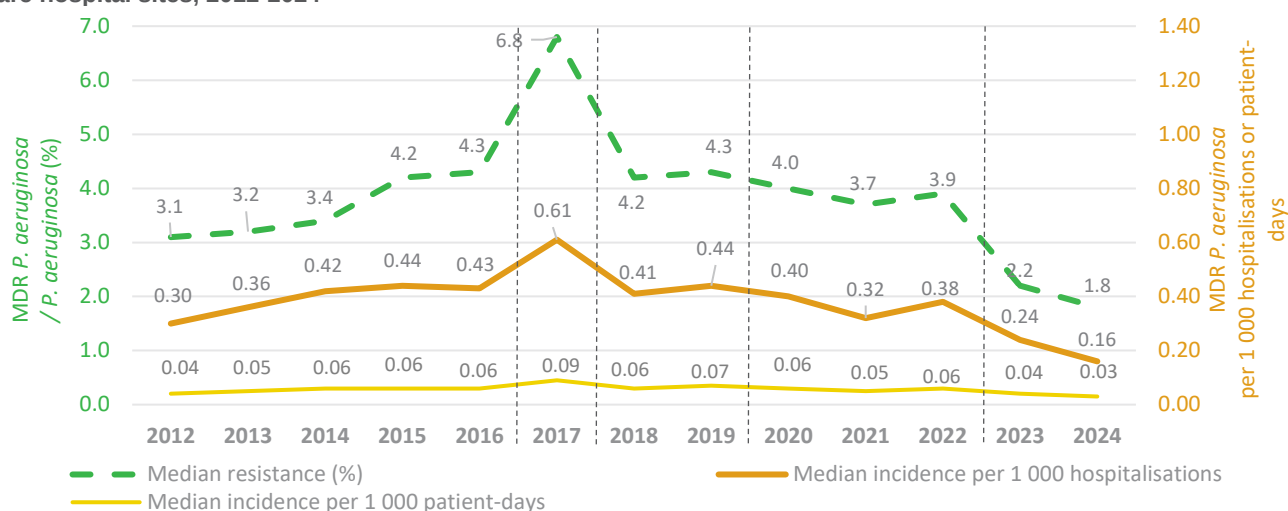
1.4 RESISTANCE IN *PSEUDOMONAS AERUGINOSA*

The crude resistance proportion of multidrug-resistant (MDR) *P. aeruginosa* was 3.1% in 2024. The crude incidence was 0.29 cases per 1 000 hospitalisations or 0.05 per 1 000 patient-days (Table 15).

Changes in definitions in 2017, 2018, 2020 and 2023 (see legend below Figure 20) make it challenging to interpret the evolution of MDR *P. aeruginosa*.

In the Annex (Figure A24 – A27), the evolution of the median resistance proportion and incidence per 1 000 hospitalisations of MDR *P. aeruginosa*, by region and by level of specialty care, can be consulted.

Figure 20. Evolution of the median resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* (clinical samples only), Belgian acute care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R - excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

Table 15. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* (clinical samples only) by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	multidrug-resistant (MDR) <i>Pseudomonas aeruginosa</i> (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	563	18 366	3.1	2.6	1.8	0.2-3.4
Flanders	244	9 806	2.5	2.1	1.5	0.0-2.2
Wallonia	154	5 744	2.7	2.8	2.3	0.7-3.8
Brussels	165	2 816	5.9	4.6	3.8	2.3-7.1
Primary hospitals	221	10 364	2.1	2.2	1.8	0.0-3.0
Secondary hospitals	114	3 995	2.9	3.6	2.2	1.2-4.1
Tertiary hospitals	228	4 007	5.7	5.4	4.9	1.9-8.3
Incidence per 1 000 hospitalisations						
Belgium	563	1 965 133	0.29	0.26	0.16	0.01-0.33
Flanders	244	1 179 548	0.21	0.19	0.09	0.00-0.22
Wallonia	154	570 800	0.27	0.25	0.21	0.06-0.41
Brussels	165	214 785	0.77	0.60	0.58	0.27-0.77
Primary hospitals	221	1 128 461	0.20	0.19	0.13	0.00-0.25
Secondary hospitals	114	466 880	0.24	0.34	0.17	0.09-0.47
Tertiary hospitals	228	369 792	0.62	0.78	0.64	0.21-1.49
Incidence density per 1 000 patient-days						
Belgium	563	11 299 042	0.05	0.04	0.03	0.00-0.05
Flanders	244	6 355 891	0.04	0.03	0.02	0.00-0.04
Wallonia	154	3 345 590	0.05	0.04	0.03	0.01-0.06
Brussels	165	1 597 561	0.10	0.08	0.08	0.02-0.12
Primary hospitals	221	6 647 337	0.03	0.03	0.02	0.00-0.05
Secondary hospitals	114	2 855 357	0.04	0.04	0.03	0.02-0.06
Tertiary hospitals	228	1 796 348	0.13	0.12	0.10	0.04-0.21

Hosp = hospital; n = total number of multidrug-resistant (MDR) *Pseudomonas aeruginosa* isolates, N = total number of *Pseudomonas aeruginosa* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

1.5 CARBAPENEMASE-PRODUCING ENTEROBACTERIALES

For the 2024 data collection, additional optional questions were introduced regarding the detection of carbapenemase-producing Enterobacterales (CPE) in *Klebsiella pneumoniae*, *Escherichia coli*, and other Enterobacterales.

A total of 60 hospital sites, representing approximately 53% of Belgian hospitals, voluntarily participated in this additional data collection, with an even regional distribution: 28 sites in Flanders (53%), 24 sites in Wallonia (53%) and 8 sites in Brussels (54%). Participating sites were mainly tertiary (n=28) and secondary (n=21) hospitals, while fewer primary hospitals (n=11) took part.

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Table 16. Incidence (per 1 000 hospitalisations) of *Escherichia coli* producing carbapenemase (CPE+) and resistant to meropenem in clinical and screening samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	<i>Escherichia coli</i>						<i>Escherichia coli</i>					
	Carbapenemase production (CPE)						Resistance to meropenem					
	n	N	Crude	Mean	Median	P25 - P75	n	N	Crude	Mean	Median	P25 - P75
Incidence per 1 000 hospitalisations – Clinical samples only												
Belgium*	125	1 098 117	0.114	0.110	0.070	0.000-0.177	28	1 098 117	0.025	0.022	0.000	0.000-0.000
Flanders	65	631 840	0.103	0.086	0.055	0.000-0.139	15	631 840	0.024	0.025	0.000	0.000-0.031
Wallonia	29	333 274	0.087	0.110	0.079	0.000-0.167	6	333 274	0.018	0.014	0.000	0.000-0.000
Brussels	31	133 003	0.233	0.191	0.129	0.042-0.375	7	133 003	0.053	0.038	0.000	0.000-0.052
Primary hosp	8	62 234	0.129	0.136	0.000	0.000-0.265	1	62 234	0.016	0.017	0.000	0.000-0.000
Secondary hosp	17	236 301	0.072	0.072	0.065	0.000-0.105	4	236 301	0.017	0.016	0.000	0.000-0.000
Tertiary hosp	100	799 582	0.125	0.128	0.085	0.031-0.192	23	799 582	0.029	0.028	0.000	0.000-0.043
Incidence per 1 000 hospitalisations – Screening samples												
Belgium**	327	867 705	0.377	0.310	0.153	0.000-0.419	77	867 705	0.089	0.020	0.000	0.000-0.000
Flanders	156	536 007	0.291	0.194	0.075	0.000-0.236	22	536 007	0.041	0.025	0.000	0.000-0.031
Wallonia	81	230 974	0.351	0.365	0.315	0.000-0.482	19	230 974	0.082	0.005	0.000	0.000-0.000
Brussels	90	100 724	0.894	0.728	0.593	0.000-0.629	36	100 724	0.357	0.060	0.000	0.000-0.105
Primary hosp	18	56 930	0.316	0.315	0.258	0.000-0.624	5	56 930	0.088	0.000	0.000	0.000-0.000
Secondary hosp	50	228 883	0.218	0.232	0.153	0.042-0.334	11	228 883	0.048	0.017	0.000	0.000-0.000
Tertiary hosp	259	581 892	0.445	0.373	0.090	0.000-0.538	61	581 892	0.105	0.032	0.000	0.000-0.057

*60 participating hospital sites; 22 sites (36.7%) reported zero cases; **54 participating hospital sites; 17 sites (31.5%) reported zero cases.

1.5.1 *ESCHERICHIA COLI*

In 2024, 125 CPE-positive *E. coli* clinical isolates were reported, corresponding to a crude incidence of 0.114 per 1 000 hospitalisations. In comparison, the same hospitals reported only 28 isolates resistant to meropenem (0.025 per 1 000 hospitalisations), indicating that a substantial proportion of CPE cases would not be detected if only resistance to meropenem were considered (Table 16).

The incidence of CPE-positive *E. coli* clinical isolates was highest in Brussels (0.233 per 1 000 hospitalisations), compared with Flanders (0.103) and Wallonia (0.087). By hospital type, the highest incidence was observed in tertiary hospitals (0.125), followed by primary (0.129) and secondary hospitals (0.072).

Screening data showed higher incidences than clinical samples. Overall, 327 CPE-positive screening isolates were reported, corresponding to 0.377 per 1 000 hospitalisations. As in clinical samples, Brussels showed the highest incidence (0.894), compared with Wallonia (0.351) and Flanders (0.291). Screening incidences were also highest in tertiary hospitals (0.445 per 1 000 hospitalisations).

Overall and across regions and hospital types, OXA-48 was the predominant carbapenemase type in both clinical and screening isolates (Figure 21).

1.5.2 *KLEBSIELLA PNEUMONIAE*

A total of 268 CPE-positive *K. pneumoniae* clinical isolates were reported in 2024, corresponding to a crude incidence of 0.248 per 1 000 hospitalisations. In comparison, the same hospitals reported 159 isolates resistant to meropenem (0.147 per 1 000 hospitalisations) (Table 17).

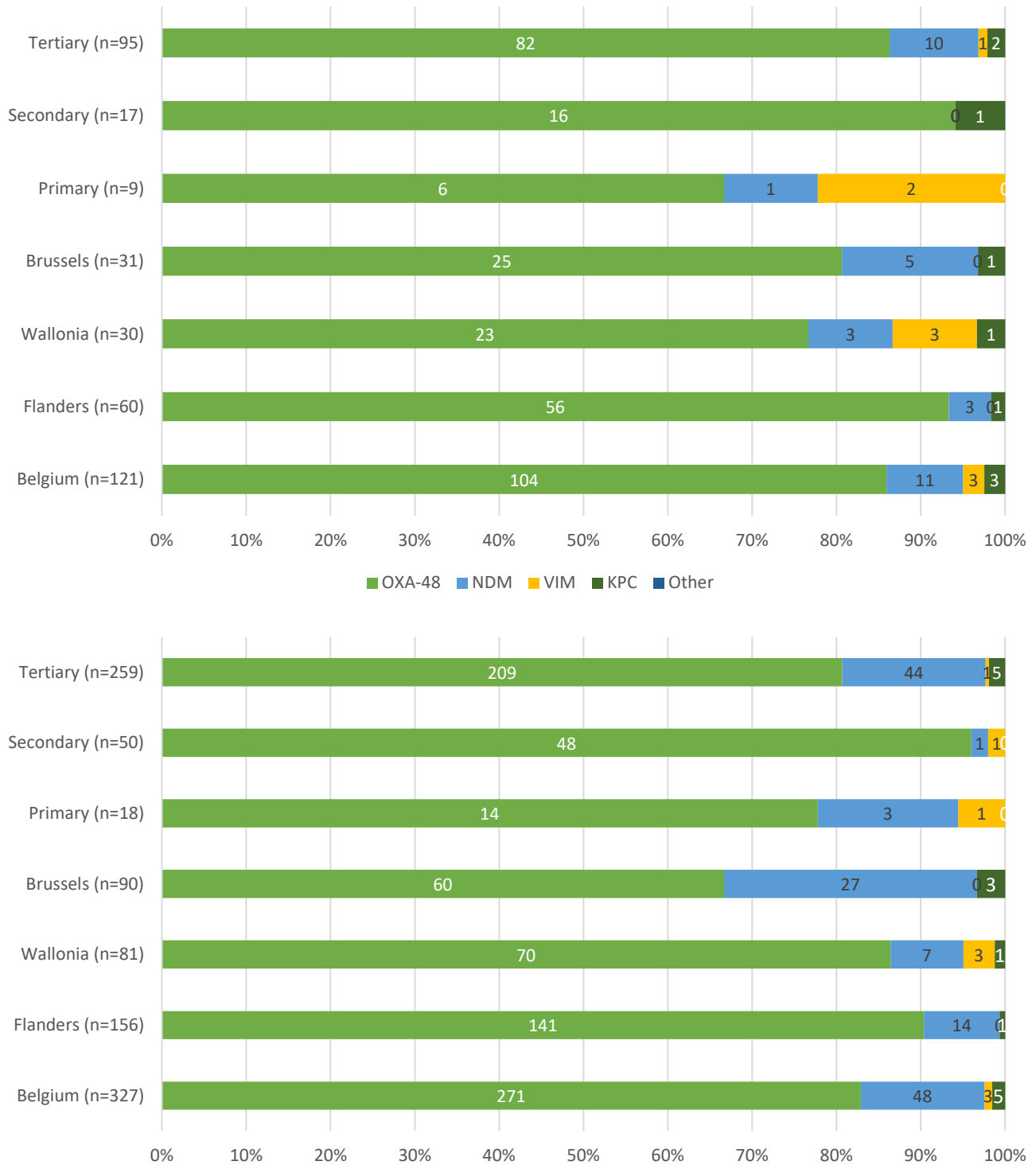
The incidence of CPE-positive clinical isolates was highest in Brussels (0.451 per 1 000 hospitalisations), followed by Wallonia (0.303) and Flanders (0.174). Tertiary hospitals reported the highest incidence (0.263), compared with primary (0.209) and secondary hospitals (0.207).

In screening samples, 303 CPE-positive isolates were reported nationally, corresponding to an incidence of 0.381 per 1 000 hospitalisations. As in clinical samples, the incidence was highest in Brussels (0.725), followed by Wallonia (0.565) and Flanders (0.217). Screening incidences were highest in tertiary hospitals (0.449).

OXA-48 was the dominant carbapenemase type overall and across regions and hospital levels in both clinical and screening isolates, with smaller proportions of NDM and KPC producers (Figure 22).

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Figure 21. Types of carbapenemase-producing *Escherichia coli* reported in clinical (upper panel) and screening (lower panel) samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024



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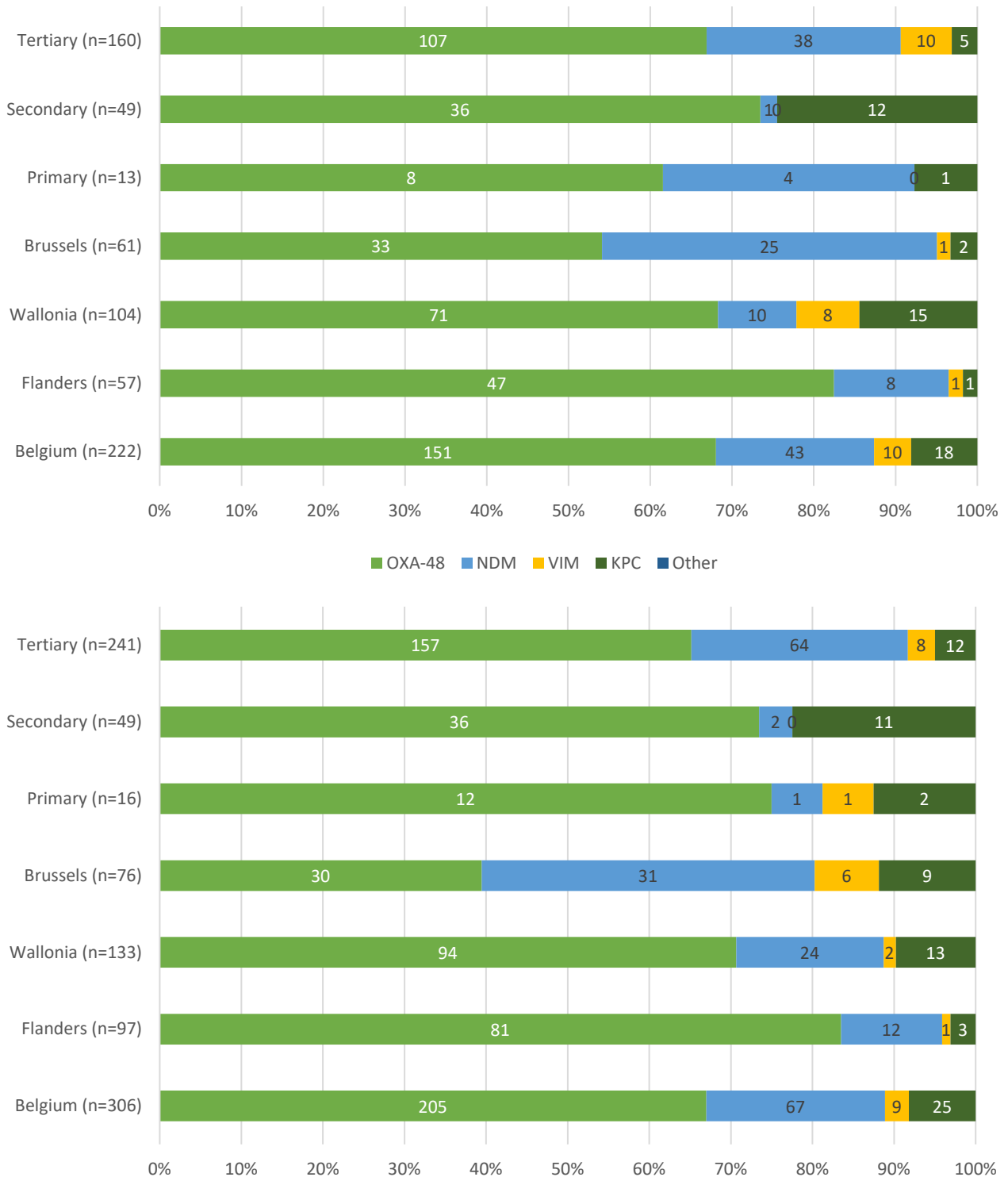
Table 17. Incidence (per 1 000 hospitalisations) of *Klebsiella pneumoniae* producing carbapenemase (CPE+) and resistant to meropenem in clinical and screening samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	<i>Klebsiella pneumoniae</i>											
	Carbapenemase production (CPE)						Resistance to meropenem					
	n	N	Crude	Mean	Median	P25 - P75	n	N	Crude	Mean	Median	P25 - P75
Incidence per 1 000 hospitalisations – Clinical samples only												
Belgium*	268	1 081 878	0.248	0.247	0.143	0.000-0.398	159	1 081 878	0.147	0.124	0.078	0.000-0.214
Flanders	107	615 601	0.174	0.095	0.064	0.000-0.143	75	615 601	0.122	0.069	0.000	0.000-0.100
Wallonia	101	333 274	0.303	0.353	0.216	0.064-0.584	54	333 274	0.162	0.146	0.125	0.000-0.255
Brussels	60	133 003	0.451	0.462	0.416	0.382-0.468	30	133 003	0.226	0.246	0.189	0.147-0.351
Primary hosp	13	62 234	0.209	0.293	0.107	0.000-0.530	7	62 234	0.112	0.153	0.000	0.000-0.377
Secondary hosp	49	236 301	0.207	0.215	0.128	0.000-0.235	24	236 301	0.102	0.103	0.000	0.000-0.136
Tertiary hosp	206	783 343	0.263	0.253	0.161	0.064-0.410	128	783 343	0.163	0.127	0.120	0.000-0.196
Incidence per 1 000 hospitalisations – Screening samples												
Belgium**	303	795 137	0.381	0.359	0.198	0.065-0.454	146	795 137	0.184	0.123	0.070	0.000-0.210
Flanders	101	465 920	0.217	0.135	0.072	0.000-0.183	46	465 920	0.099	0.062	0.000	0.000-0.100
Wallonia	129	228 493	0.565	0.546	0.386	0.189-0.795	44	228 493	0.193	0.156	0.126	0.000-0.265
Brussels	73	100 724	0.725	0.646	0.494	0.372-0.838	56	100 724	0.556	0.275	0.210	0.140-0.431
Primary hosp	16	62 234	0.257	0.318	0.214	0.000-0.468	10	62 234	0.161	0.153	0.000	0.000-0.377
Secondary hosp	48	200 177	0.240	0.249	0.158	0.097-0.331	17	200 177	0.085	0.107	0.000	0.000-0.136
Tertiary hosp	239	532 726	0.449	0.464	0.183	0.061-0.838	119	532 726	0.223	0.120	0.103	0.000-0.183

*60 participating hospital sites; 18 sites (30.0%) reported zero cases; **50 participating hospital sites; 8 sites (16.0%) reported zero cases.

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Figure 22. Types of carbapenemase producing *Klebsiella pneumoniae* reported in clinical (upper panel) and screening (lower panel) samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024



1.5.3 OTHER ENTEROBACTERIALES

Among Enterobacterales other than *E. coli* and *K. pneumoniae*, 156 CPE-positive clinical isolates were reported, corresponding to a crude incidence of 0.187 per 1,000 hospitalisations. The incidence was highest in Brussels (0.316), followed by Wallonia (0.251) and Flanders (0.132) (Table 18).

In screening samples, 427 CPE-positive isolates were reported, corresponding to an incidence of 0.512 per 1,000 hospitalisations. Again, the highest incidence was observed in Brussels (1.486), followed by Wallonia (0.749) and Flanders (0.196). Across hospital levels, screening incidences were highest in tertiary hospitals (0.555 per 1,000 hospitalisations) (Table 18).

As for *E. coli* and *K. pneumoniae*, OXA-48 was the predominant carbapenemase type in both clinical and screening isolates of other Enterobacterales (Figure 23).

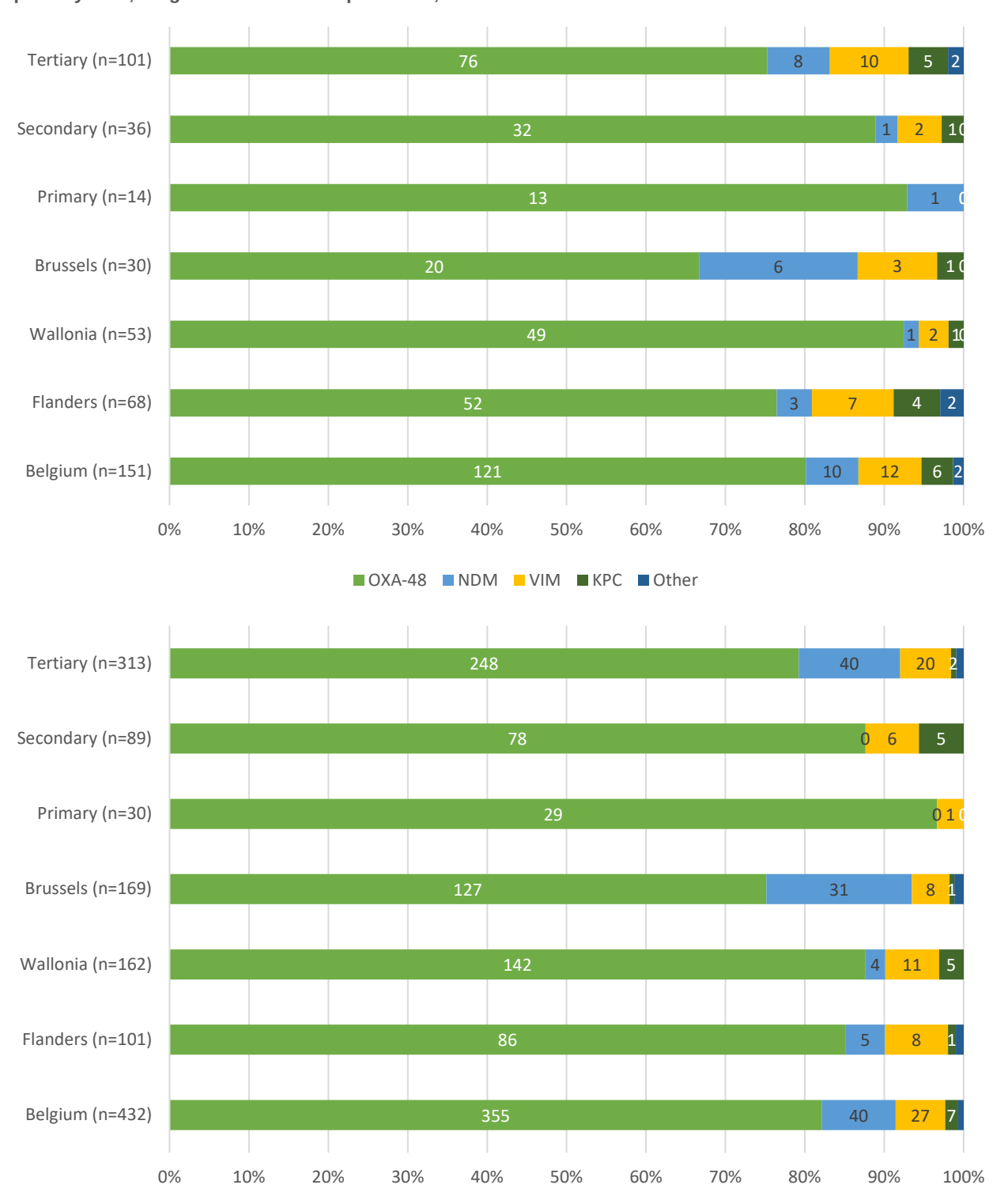
Table 18. Incidence (per 1 000 hospitalisations) of Enterobacterales (other than *E. coli* and *K. pneumoniae*) producing carbapenemase (CPE+) in clinical and screening samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024

	Enterobacterales (other than <i>E. coli</i> and <i>K. pneumoniae</i>) producing carbapenemase (CPE+)					
	n	N	Crude	Mean	Median	P25 - P75
Incidence per 1 000 hospitalisations – Clinical samples						
Belgium*	156	833 523	0.187	0.197	0.136	0.017-0.261
Flanders	67	508 929	0.132	0.111	0.122	0.000-0.156
Wallonia	53	210 833	0.251	0.277	0.241	0.101-0.397
Brussels	36	113 761	0.316	0.333	0.348	0.186-0.465
Primary hospitals	14	50 518	0.277	0.319	0.156	0.000-0.465
Secondary hospitals	36	221 098	0.163	0.156	0.130	0.000-0.269
Tertiary hospitals	106	561 907	0.189	0.184	0.171	0.104-0.245
Incidence per 1 000 hospitalisations – Screening samples						
Belgium**	427	833 523	0.512	0.589	0.229	0.075-0.449
Flanders	100	508 929	0.196	0.151	0.097	0.000-0.214
Wallonia	158	210 833	0.749	0.877	0.434	0.302-0.795
Brussels	169	113 761	1.486	1.727	0.393	0.330-1.804
Primary hospitals	30	50 518	0.594	0.689	0.214	0.000-0.302
Secondary hospitals	85	221 098	0.384	0.425	0.253	0.065-0.539
Tertiary hospitals	312	561 907	0.555	0.701	0.245	0.096-0.417

*52 participating hospital sites; 13 sites (25.0%) reported zero cases; **52 participating hospital sites; 9 sites (17.3%) reported zero cases.

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Figure 23. Types of carbapenemase producing Enterobacterales (other than *Escherichia coli* and *Klebsiella pneumoniae*) reported in clinical (upper panel) and screening (lower panel) samples by region and hospital level of specialty care, Belgian acute care hospital sites, 2024



1.6 ESBL AND CPE SCREENING

Eighty-one hospital sites reported both the number of patients screened for ESBL and CPE during the surveillance period. In total, 66 247 patients were screened for ESBL carriage (median per hospital: 225; IQR: 0-1 129), while 106 655 were screened for CPE (median per hospital: 724; IQR: 170-1 777). Twenty-seven hospitals reported no ESBL screenings, compared with only one hospital reporting zero CPE screenings.

On average, 6.4% of admitted patients were screened for ESBL per hospital (median: 1.8%; IQR: 0.0-9.1%). Screening coverage varied by region: 2.8% in Flanders (median: 0.0%; IQR: 0.0-2.2%), 9.1% in Wallonia (median: 6.9%; IQR: 1.7-14.4%), and 14.2% in Brussels (median: 8.1%; IQR: 1.9-27.7%).

For CPE, an average of 9.8% of admitted patients were screened per hospital (median: 5.2%; IQR: 1.6-11.5%). This percentage also varied by region: 4.4% in Flanders (median: 2.4%; IQR: 0.5-5.0%), 15.3% in Wallonia (median: 8.9%; IQR: 5.4-20.7%), and 15.9% in Brussels (median: 9.8%; IQR: 6.1-27.7%).

Eighty-nine hospital sites provided complete information on their ESBL and CPE screening practices.

ESBL screening was largely selective. Only one hospital (1.1%) screened all patients on all wards upon admission. Among the remaining hospitals, 44 (50.0%) screened all patients admitted to high-risk wards, defined locally (e.g. intensive care units, onco-haematological wards). In addition, 38 hospitals (43.2%) screened all known ESBL carriers, and 36 (40.9%) screened all patients transferred from another Belgian hospital or from a foreign hospital with a known high prevalence upon admission. During hospitalisation, ESBL screening was performed in response to outbreaks in 63 hospitals (70.8%), routinely on high-risk wards in 44 hospitals (49.4%), and for patients who had shared a room for more than 12 hours with a known ESBL carrier in 38 hospitals (42.7%).

CPE screening, in contrast, was more comprehensive and systematically applied to high-risk situations. Only one hospital (1.1%) screened all patients on all wards upon admission. The other hospitals, however, targeted high-risk groups: 60 hospitals (68.2%) screened all admissions to high-risk wards, 85 hospitals (96.6%) screened known CPE carriers, and 83 hospitals (94.3%) screened patients transferred from other Belgian hospitals or from foreign hospitals with high CPE prevalence. During hospitalisation CPE screening was conducted almost universally in response to outbreaks (88 hospitals, 98.9%), routinely on high-risk wards (68 hospitals, 76.4%), and for patients sharing a room for more than 12 hours with a known CPE carrier (88 hospitals, 98.9%).

2. Resistant Gram-negative bacteria in chronic care hospitals

2.1 RESISTANCE IN *ESCHERICHIA COLI*

The crude resistance proportion was 9.0% for 3GC-R *E. coli* and 0.08% for mero-R *E. coli* (clinical samples only) in 2024. The crude incidence density was 0.22 and 0.002 per 1 000 patient-days for 3GC-R *E. coli* and mero-R *E. coli*, respectively (Table 19).

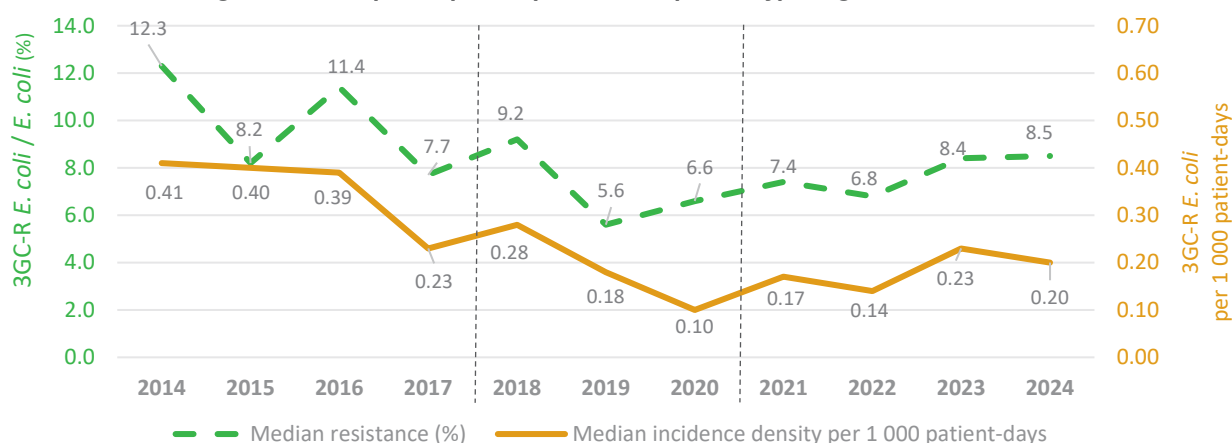
Table 19. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins or meropenem (clinical samples only) by region, Belgian chronic care hospital sites, 2024

	<i>Escherichia coli</i> (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	117	1 300	9.0	9.1	8.5	8.2-11.4
Incidence per 1 000 hospitalisations						
Belgium	117	19 583	5.97	8.36	6.66	2.27-12.13
Incidence density per 1 000 patient-days						
Belgium	117	532 218	0.22	0.25	0.20	0.11-0.34
Resistance to meropenem						
Resistance proportion (%)						
Belgium	1	1 300	0.08	0.61	0.00	0.00-0.00
Incidence per 1 000 hospitalisations						
Belgium	1	19 583	0.051	0.046	0.000	0.000-0.000
Incidence density per 1 000 patient-days						
Belgium	1	532 218	0.002	0.001	0.000	0.000-0.000

Hosp = hospital; n = total number of *Escherichia coli* isolates resistant to 3rd generation cephalosporins or meropenem, N = total number of *Escherichia coli* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

The overall evolution of the median resistance proportion and incidence density of 3GC-R *E. coli* in chronic care hospital sites is shown in Figure 24.

Figure 24. Evolution of the median resistance proportion and incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins (clinical samples only), Belgian chronic care sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

2.2 RESISTANCE IN *KLEBSIELLA PNEUMONIAE*

The crude resistance proportion of 3GC-R *K. pneumoniae* isolated from clinical samples was 32.3% in 2024. The crude incidence density was 0.24 per 1 000 patient-days (Table 20).

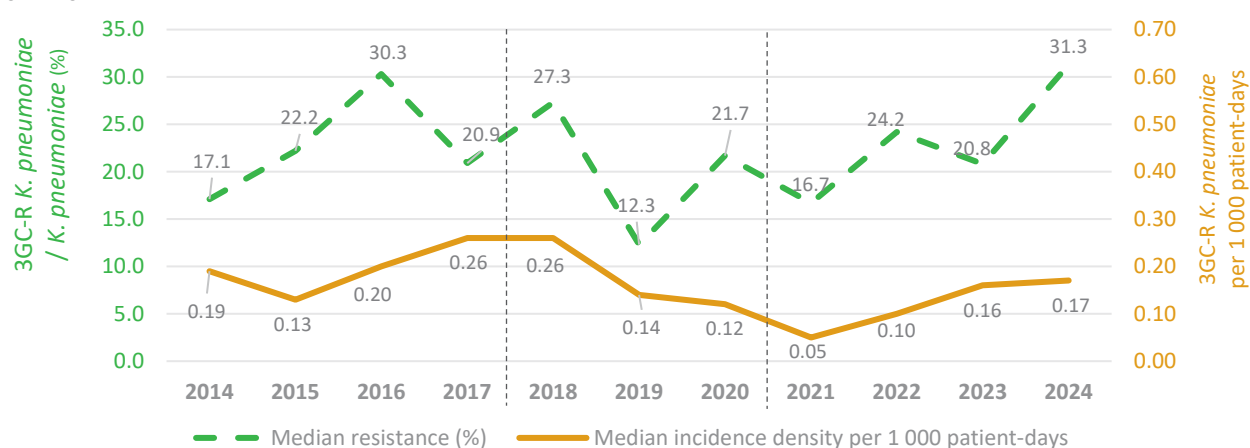
Table 20. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins or meropenem (clinical samples only) by region, Belgian chronic care hospital sites, 2024

	<i>Klebsiella pneumoniae</i> (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance to third generation cephalosporins						
Resistance proportion (%)						
Belgium	127	393	32.3	26.3	31.3	8.6-41.7
Incidence per 1 000 hospitalisations						
Belgium	127	19 583	6.49	8.70	5.71	2.54-10.20
Incidence density per 1 000 patient-days						
Belgium	127	532 218	0.24	0.29	0.17	0.05-0.59
Resistance to meropenem						
Resistance proportion (%)						
Belgium	5	393	1.27	0.90	0.00	0.00-2.70
Incidence per 1 000 hospitalisations						
Belgium	5	19 583	0.255	0.158	0.000	0.000-0.396
Incidence density per 1 000 patient-days						
Belgium	5	53 2218	0.009	0.008	0.000	0.000-0.000

Hosp = hospital; n = total number of *Klebsiella pneumoniae* isolates resistant to 3rd generation cephalosporins or meropenem, N = total number of *Klebsiella pneumoniae* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

The overall evolution of the median resistance proportion and incidence density of 3GC-R *K. pneumoniae* in chronic care hospital sites is shown in Figure 25.

Figure 25. Evolution of the median resistance proportion and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins (clinical samples only), Belgian chronic care hospital sites, 2014-2024



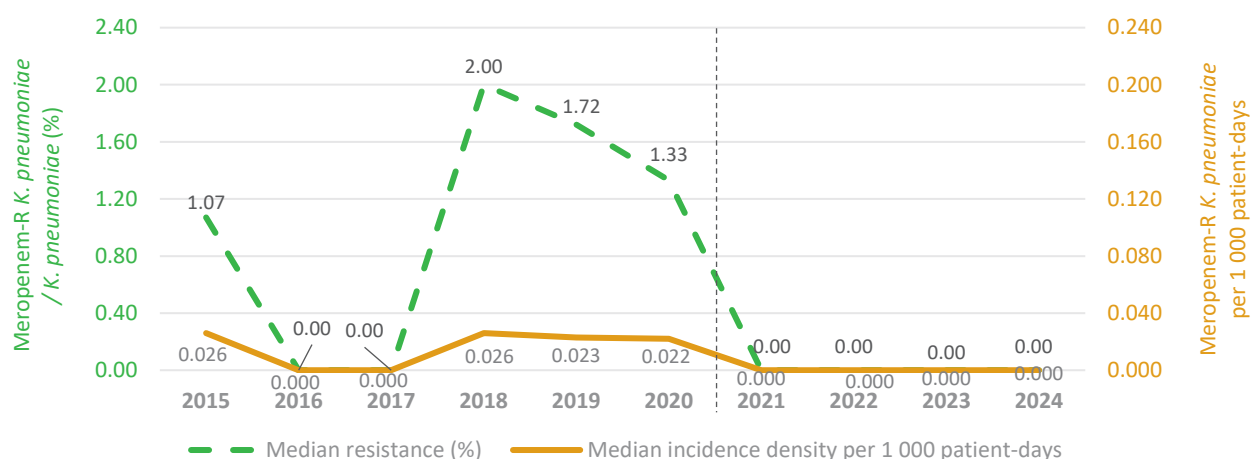
3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

PART 3. RESISTANCE IN GRAM-NEGATIVE BACTERIA

The crude resistance proportion and incidence density of mero-R *K. pneumoniae* were 1.27% and 0.009 cases per 1 000 patient-days in 2024, respectively (Table 20).

The evolution of the median resistance proportion and incidence density of mero-R *K. pneumoniae* in chronic care hospital sites can be seen in Figure 26.

Figure 26. Evolution of the median resistance proportion and incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to meropenem (clinical samples only), Belgian chronic care hospital sites, 2015-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

2.3 RESISTANCE IN ACINETOBACTER BAUMANNII

Similar to the previous year, participating chronic care hospital sites reported nine cases of *A. baumannii* isolated from clinical samples in 2024, none of which were mero-R.

2.4 RESISTANCE IN PSEUDOMONAS AERUGINOSA

The crude resistance proportion and incidence density of MDR *P. aeruginosa* were 2.8% and 0.02 cases per 1 000 patient-days in 2024 in the participating chronic care hospital sites, respectively (Table 21).

Table 21. Resistance proportion, incidence (per 1 000 hospitalisations) and incidence density (per 1 000 patient-days) of multidrug-resistant *Pseudomonas aeruginosa* (clinical samples only) by region, Belgian chronic care hospital sites, 2024

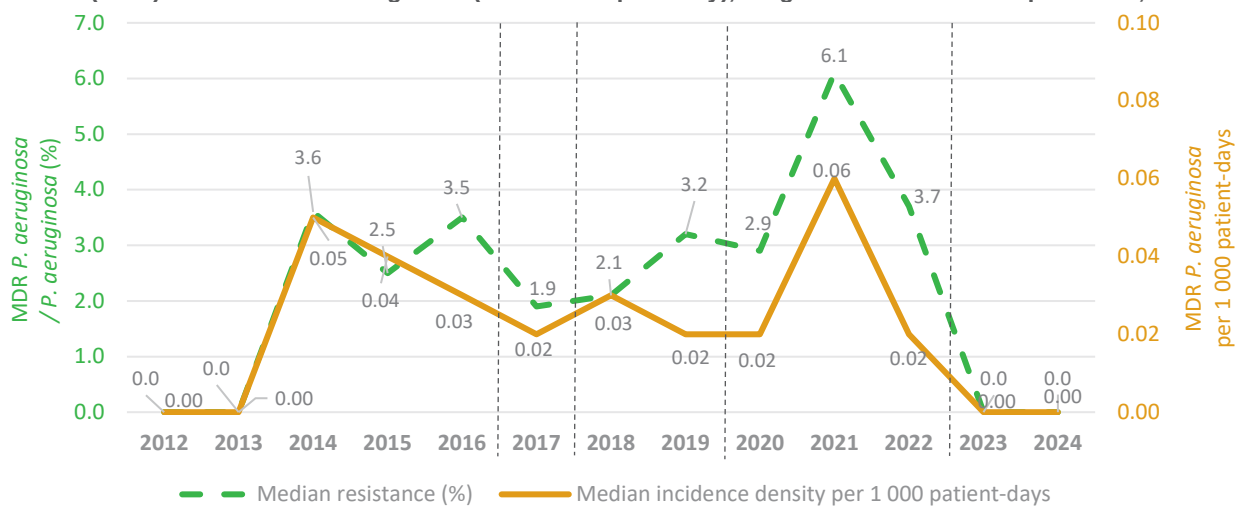
	multidrug-resistant (MDR) <i>Pseudomonas aeruginosa</i> (clinical samples only)					
	n	N	Crude	Mean	Median	P25 - P75
Resistance proportion (%)						
Belgium	9	325	2.8	2.4	0.0	0.0-4.5
Incidence per 1 000 hospitalisations						
Belgium	9	19 583	0.46	0.80	0.00	0.00-1.54
Incidence density per 1 000 patient-days						
Belgium	9	532 218	0.02	0.02	0.00	0.00-0.04

n = total number of multidrug-resistant *Pseudomonas aeruginosa* isolates, N = total number of *Pseudomonas aeruginosa* isolates for the calculation of the resistance proportion, total number of hospitalisations for the incidence or total number of patient-days for the incidence density calculations, crude = n/N

Figure 27 presents the evolution of the median resistance proportion and incidence density of MDR *P. aeruginosa*.

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Figure 27. Evolution of the median resistance proportion and incidence density (per 1 000 patient-days) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* (clinical samples only), Belgian chronic care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

MAIN FINDINGS AND RECOMMENDATIONS

This report presents the 2024 results of three national surveillance programs on AMR: (1) MRSA, (2) VRE, and (3) MRGN. Data collected for this report were gathered retrospectively and aggregated at the hospital level. The report distinguishes between acute and chronic care hospital sites, with acute care hospital sites that have a length of stay of ≥ 16 days classified as chronic care hospital sites. While both clinical and screening samples were collected for MRSA and VRE surveillance, only data from clinical samples are used in this report, unless otherwise stated. This approach was adopted to minimize inter-hospital variability, which could arise from differences in local screening practices.²

In 2024, participation in the national AMR surveillance systems remained very high across Belgian acute care hospital administrative groups. Coverage reached 95.1% for the mandatory MRSA surveillance and 98.0% for the MRGN surveillance, while the optional VRE surveillance also achieved strong uptake, with participation from 94.1% of hospital groups. This high level of engagement, including contributions from both acute and chronic care settings, combined with the internal validation through individual benchmark reports, ensures that the surveillance data provide a robust and representative overview of AMR trends in Belgium. It also reflects the sustained commitment of hospitals to national infection prevention and control (IPC) efforts and supports the reliability of the findings for guiding public health actions and policy decisions.

In 2024, MRSA continued to represent a notable proportion of *S. aureus* infections, accounting for 9% of clinical isolates, with a crude incidence of 1.34 cases per 1 000 hospitalisations. Long-term surveillance shows substantial reductions in both MRSA resistance and healthcare-associated MRSA (HA-MRSA) since 2004, reflecting the success of infection prevention strategies such as targeted screening, decolonization, and hand hygiene campaigns. However, the recent plateau in resistance decline and the continued presence of HA-MRSA - around 20% of all MRSA cases - highlight ongoing challenges in fully controlling MRSA transmission, particularly among patients with prior healthcare exposure.

Resistance trends among enterococci and Gram-negative bacteria present a mixed picture.

Vanco-R *E. faecium* has increased in both proportion (3.08% in 2024) and incidence since 2021, despite historically declining trends, while *E. faecalis* remains largely susceptible (0.05% resistance). The proportion of hospitals reporting at least one vanco-R *E. faecium* case increased markedly, from 9.3% in 2023 to 49.1% in 2024. In *E. faecalis*, crude resistance to linezolid (0.22%) was comparable to that observed in *E. faecium* (0.25%). 3GC-R *E. coli* has shown a slight upward trend in recent years, though incidence remains stable, and mero-R isolates are rare. In *K. pneumoniae*, 3GC-R continues to decline, but mero-R persists in a substantial proportion of hospitals. These trends should be interpreted with caution, as the exclusion of the EUCAST “I” category in the resistance definition may have influenced comparisons over time. Prior to 2021, isolates categorized as intermediate were included within the combined I/R group; according to estimates from the National Reference Centre, intermediate isolates may account for approximately 5–15% of the 3GC-I/R group and up to 50% of the mero-I/R group.

A. baumannii and MDR *Pseudomonas aeruginosa* show moderate resistance levels, with stable or decreasing incidence trends.

In 2024, the collection of data on CPE was re-introduced in the MRGN surveillance through additional optional questions. CPE represent a significant public health threat due to their resistance to last-line antibiotics, such as carbapenems, leaving few treatment options for infected patients. The reintroduction of CPE surveillance will provide crucial insights into its epidemiology, enabling timely interventions to prevent outbreaks and limit its

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spread. A strong surveillance program is vital to detecting emerging CPE cases, monitoring trends, and supporting infection control efforts across healthcare settings.

Participation in the CPE data collection was voluntary, yet uptake was good: 60 hospital sites, representing approximately 53% of Belgian acute-care hospitals, contributed data, with an even regional distribution and representation across hospital types. These results provide a national overview of CPE detection in both clinical and screening samples. In clinical samples, 268 CPE-positive *Klebsiella pneumoniae* and 125 CPE-positive *Escherichia coli* isolates were reported, compared with 159 and 28 meropenem-resistant isolates, respectively, indicating that a substantial proportion of CPE cases would not have been captured if only resistance to meropenem had been considered. Among other Enterobacterales, 156 CPE-positive clinical isolates were reported. Screening activities identified even higher numbers, including 303 CPE-positive *K. pneumoniae*, 327 *E. coli*, and 427 other Enterobacterales isolates.

Although OXA-48 was the most prevalent carbapenemase overall, its frequency was higher in *E. coli* (86%) and other Enterobacterales (80%), whereas in *K. pneumoniae* it was lower (68%), with other carbapenemase types, such as KPC and NDM, being more frequently detected. This likely explains the discrepancy between CPE detection and phenotypic carbapenem resistance, as certain carbapenemase types - particularly OXA-48-like enzymes - often confer low-level resistance, and a large proportion of such isolates remain susceptible to meropenem (e.g. around 80% of OXA-48-like CPE).

The voluntary nature of the CPE+ data collection means that the findings should be interpreted with caution. It is not possible to determine whether non-participating hospitals chose not to report or whether they were unable to do so because they do not routinely test for carbapenemase production or for specific carbapenemase types. For the next surveillance round, reporting of CPE cases and carbapenemase types will become mandatory. Hospitals that are unable to provide these data will be asked to indicate the reasons, in order to better understand potential gaps in diagnostic capacity or reporting. In addition, a short survey on CPE laboratory detection practices will be conducted to further assess testing approaches and capacities across hospitals.

Screening practices in 2024 were more comprehensively assessed across multiple multidrug-resistant organisms (MDROs) than in previous years. MRSA screening remained the most systematic, followed by CPE and VRE, while ESBL screening was less consistent. High-risk wards were commonly screened on admission for MRSA (82.4%) and CPE (68.2%), and nearly all hospitals screened known carriers and transferred patients for MRSA, CPE, and VRE. Outbreak-driven and high-risk ward screening during hospitalisation was similarly extensive for MRSA, CPE, and VRE, but less frequent for ESBL. Regional differences were observed in the proportion of patients screened, which may have influenced the number of detected cases, as more intensive screening efforts typically lead to higher detection rates. At the same time, systematic screening remains a key component of effective infection prevention and control, enabling early identification and timely implementation of measures to reduce transmission. The reporting of a large number of VRE clusters and patient involved remains an important aspect to consider targeted actions in the future.

Overall, the 2024 data underscore the sustained benefits of long-term IPC programs while highlighting emerging areas of concern. The resurgence of VRE, stabilization of 3GC-resistant *E. coli*, and the higher occurrence of CPE relative to meropenem-resistance demonstrate the need for continuous surveillance, adaptive strategies, refined screening policies, and robust antimicrobial stewardship to prevent the spread of multidrug-resistant pathogens in acute care settings.

AMR trends are also monitored by EARS-BE, the Belgian branch of the European Antimicrobial Resistance Surveillance Network (EARS-Net). This surveillance program retrospectively collects data from both hospital laboratories and non-hospital laboratories. EARS-BE differs from EARS-Net in the additional collection of data on antimicrobial susceptibility test (AST) results of isolates found in urine samples in addition to invasive samples, such as blood and cerebrospinal fluid (CSF).⁶

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According to the most recent EARS-Net report for Belgium in 2024, which covered data from 30 hospital laboratories, 7.4% of *S. aureus* isolates from blood and CSF were resistant to methicillin (MRSA). Among invasive *E. faecium* isolates, 3.4% exhibited resistance to vancomycin, with very low resistance to linezolid.⁷

Resistance trends in *E. coli* isolated from invasive samples in Belgium have shown an increase in resistance to 3GC in the past three years (11.9% in 2024). Carbapenem resistance remains extremely low at just 0.1%.⁷

For *K. pneumoniae*, no distinct trend in resistance was observed. Among isolates, 21.1% were resistant to 3GC. Carbapenem resistance, which had previously remained around 1%, rose to 2.3% in 2023 and stabilized at 2% in 2024.⁷

Belgium has maintained a stable rate of approximately 6–8% MDR *P. aeruginosa* isolates since 2014, with 7.1% of isolates classified as MDR in 2024. For *A. baumannii*, the small number of isolates in the EARS-BE dataset makes it difficult to discern clear trends. While resistance rates vary from year to year, the total number of resistant isolates has remained relatively constant.⁷

The AMR and EARS-BE findings should be interpreted in the context of the Belgian “One Health” National Action Plan (NAP) for the fight against Antimicrobial Resistance 2026–2030. Antimicrobial resistance (AMR) is a serious threat to public health. For the past 25 years, Belgium has been implementing coordinated actions to address this challenge and reduce the risks associated with AMR. With the new “One Health” NAP AMR 2026–2030, public authorities aim to further strengthen these efforts, with the objective of achieving concrete targets by 2030 aligned with national priorities and European ambitions.⁸

A key component of this plan will be the Sciensano-led initiative to establish a single, national, multi-centric AMR surveillance platform (NSIH-AMREXT), which aims to replace and integrate several existing hospital surveillance systems, including NSIH-AMR (MRSA, VRE and MRGN), EARS-BE, and the AMR component of NSIH-SEP (National Surveillance of Infections in Hospitals – Bloodstream infections), NSIH-ICU (National Surveillance of Infections in Hospitals – Intensive Care Unit), and NSIH-SSI (National Surveillance of Infections in Hospitals – Surgical Site Infections). By consolidating these parallel data collection streams, the integrated system is intended to reduce duplication of efforts, improve standardisation, and enhance the quality and comparability of AMR data. The new platform will also enable analyses at the level of individual antimicrobial susceptibility tests, improve representativeness, and maintain continuity with historical datasets. In addition, the planned expansion of surveillance to nursing homes and primary care settings reflects the One Health perspective of the national action plan, aiming to obtain more comprehensive AMR data across the continuum of care.

The current report acknowledges several limitations, including voluntary participation in some surveillance components and variability in reporting practices. Changes in surveillance definitions over time, especially for multidrug-resistant *P. aeruginosa*, complicate long-term trend analysis. Future efforts should focus on harmonizing definitions and increasing participation across hospital types.

In conclusion, while Belgium has made significant progress in combating AMR, continued vigilance and adaptive strategies are essential to address emerging challenges. Strengthening surveillance, enhancing infection prevention and control and antimicrobial stewardship, and integrating genomic technologies will be crucial in the fight against antimicrobial resistance in the coming years. The One Health approach remains pivotal in ensuring a comprehensive response across all sectors.

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ANNEX

Figure A1. Evolution of the median proportion of methicillin-resistant *Staphylococcus aureus* (MRSA) on the total number of reported *S. aureus* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 1994-2024

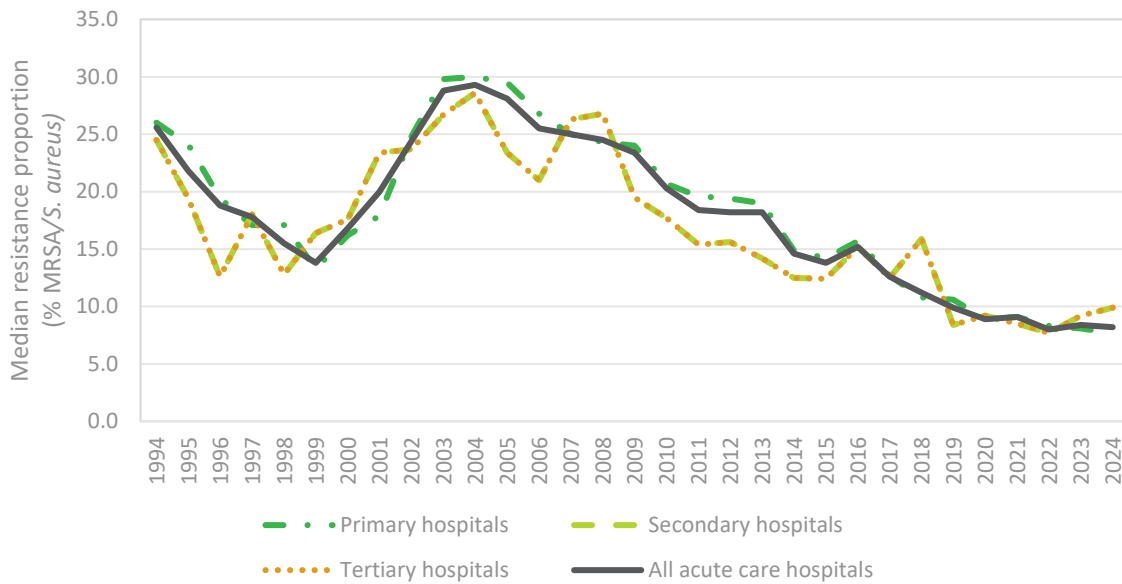


Figure A2. Evolution of the median incidence of methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 hospitalisations by level of specialty care (clinical samples only), Belgian acute care hospital sites, 1994-2024

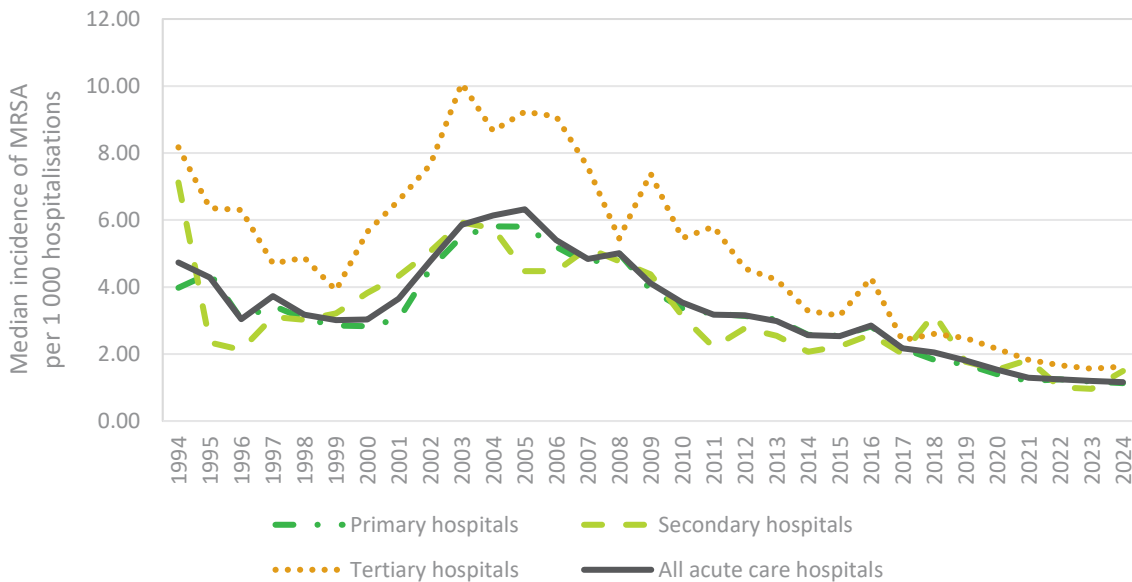
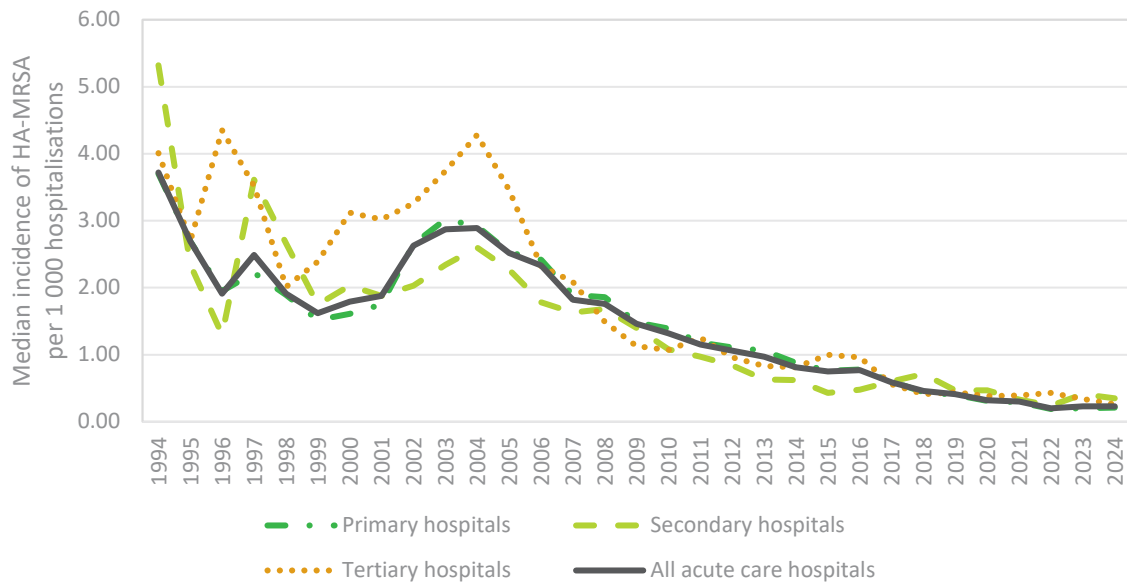


Figure A3. Evolution of the median incidence of healthcare-associated (HA-) methicillin-resistant *Staphylococcus aureus* (MRSA) per 1 000 hospitalisations by level of specialty care (clinical samples only), Belgian acute care hospital sites, 1994-2024



ANNEX

Figure A4. Evolution of the crude resistance proportion of vancomycin resistance in *Enterococcus faecium* by region (clinical samples only), Belgian acute care hospital sites, 2014-2024

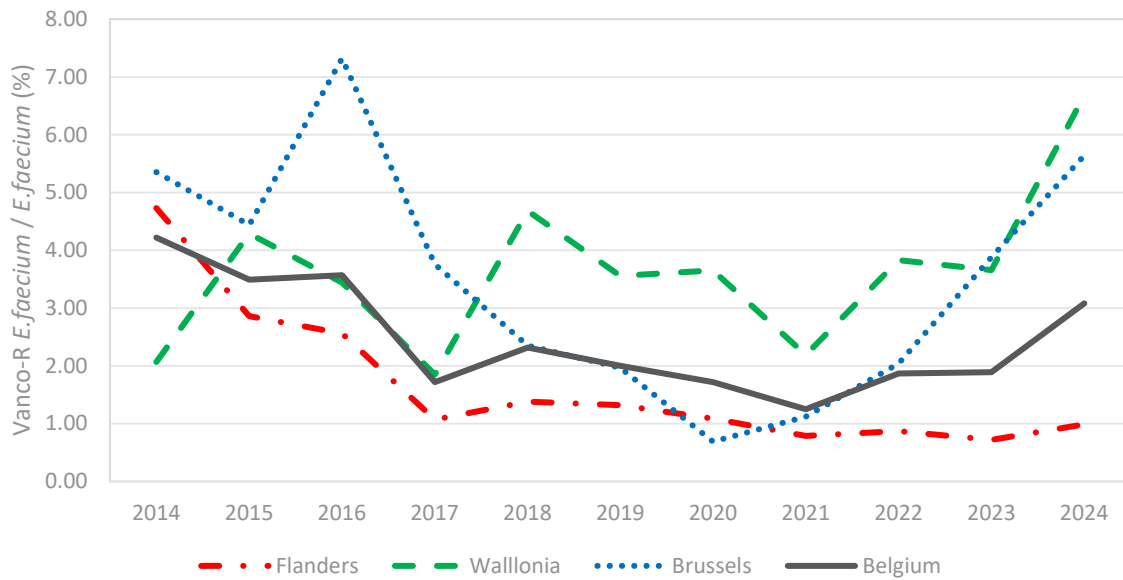
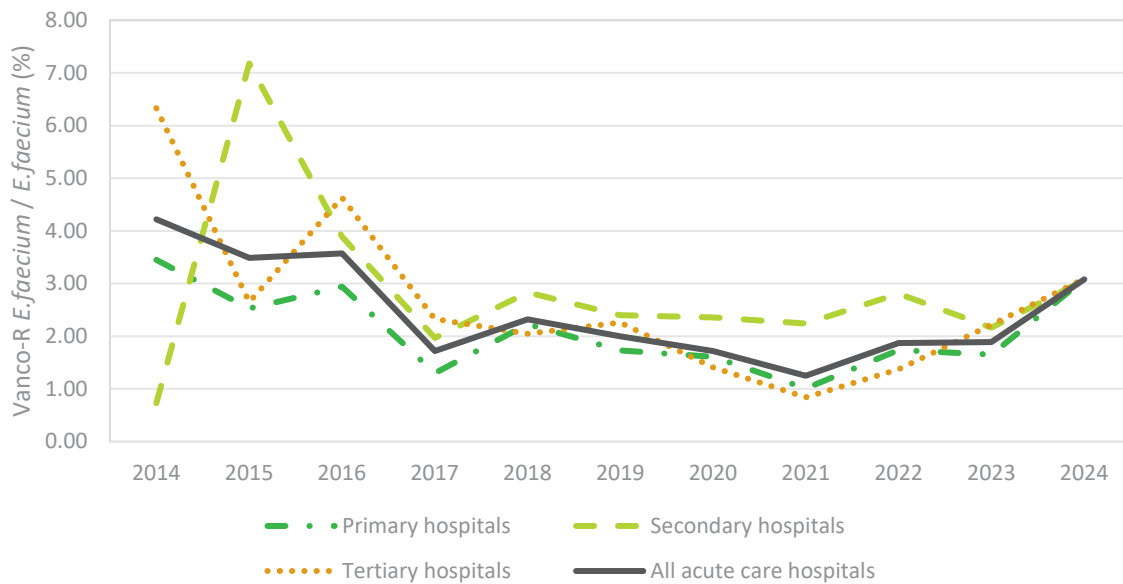


Figure A5. Evolution of the crude resistance proportion of vancomycin resistance in *Enterococcus faecium* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



ANNEX

Figure A6. Evolution of the crude incidence (per 1 000 hospitalisations) of vancomycin resistance in *Enterococcus faecium* by region (clinical samples only), Belgian acute care hospital sites, 2014-2024

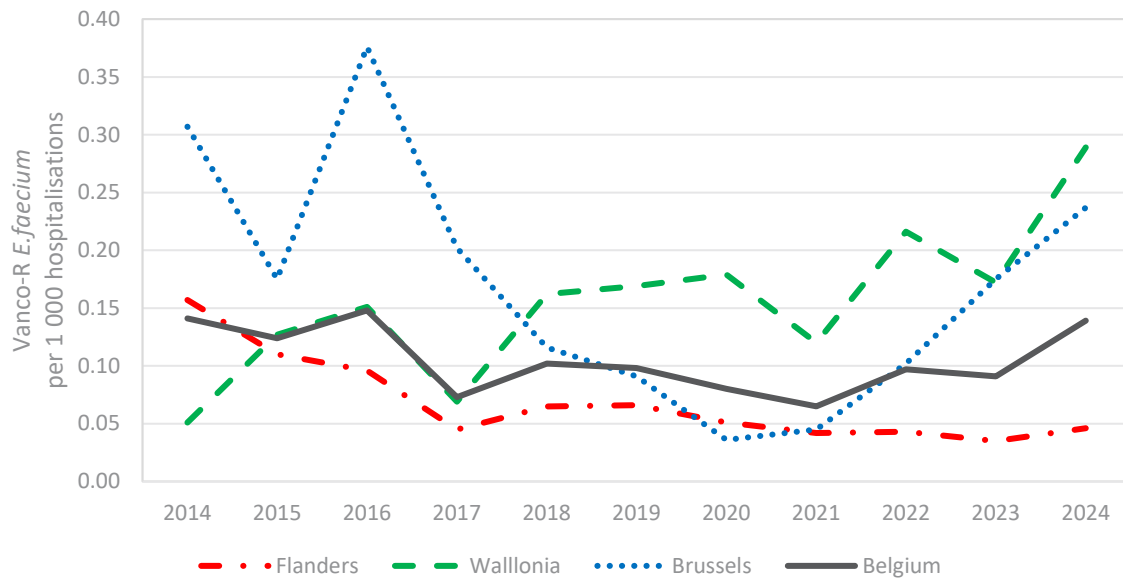
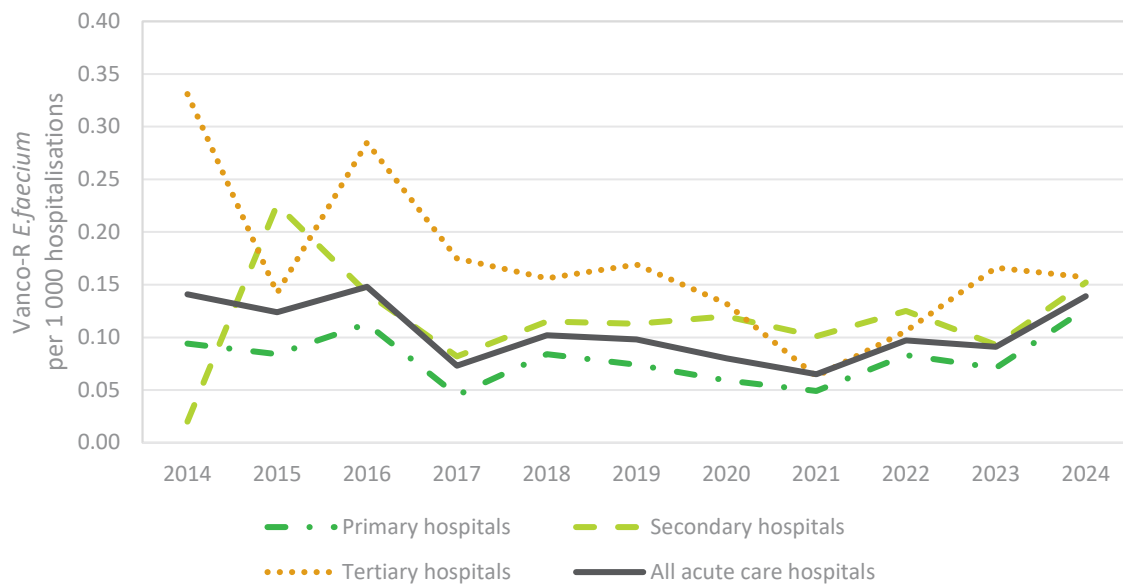


Figure A7. Evolution of the crude incidence (per 1 000 hospitalisations) of vancomycin resistance in *Enterococcus faecium* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



ANNEX

Figure A8. Evolution of the crude incidence density (per 1 000 patient-days) of vancomycin resistance in *Enterococcus faecium* by region (clinical samples only), Belgian acute care hospital sites, 2014-2024

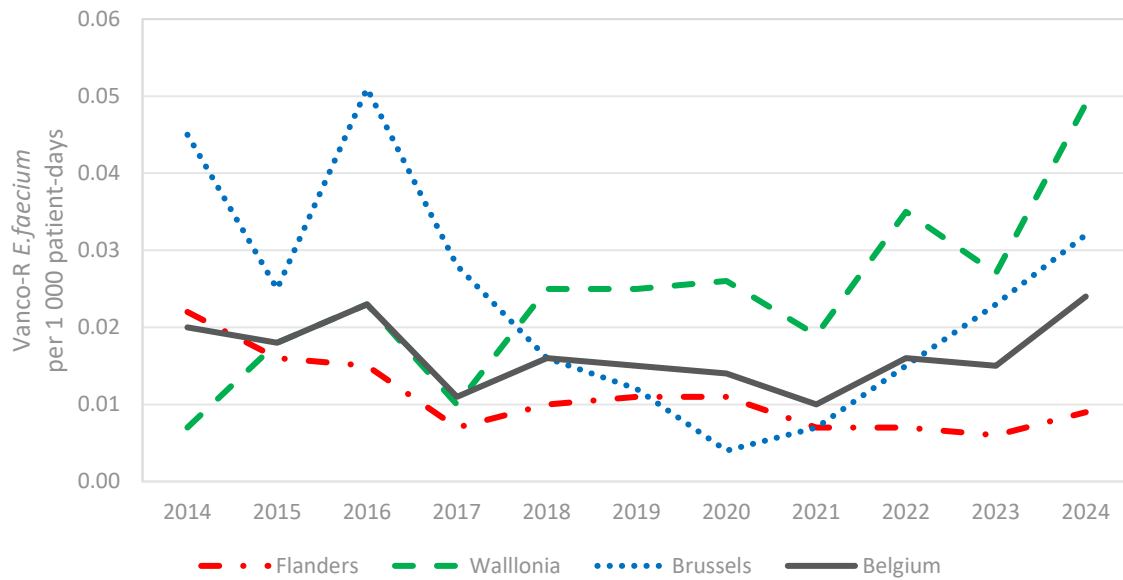
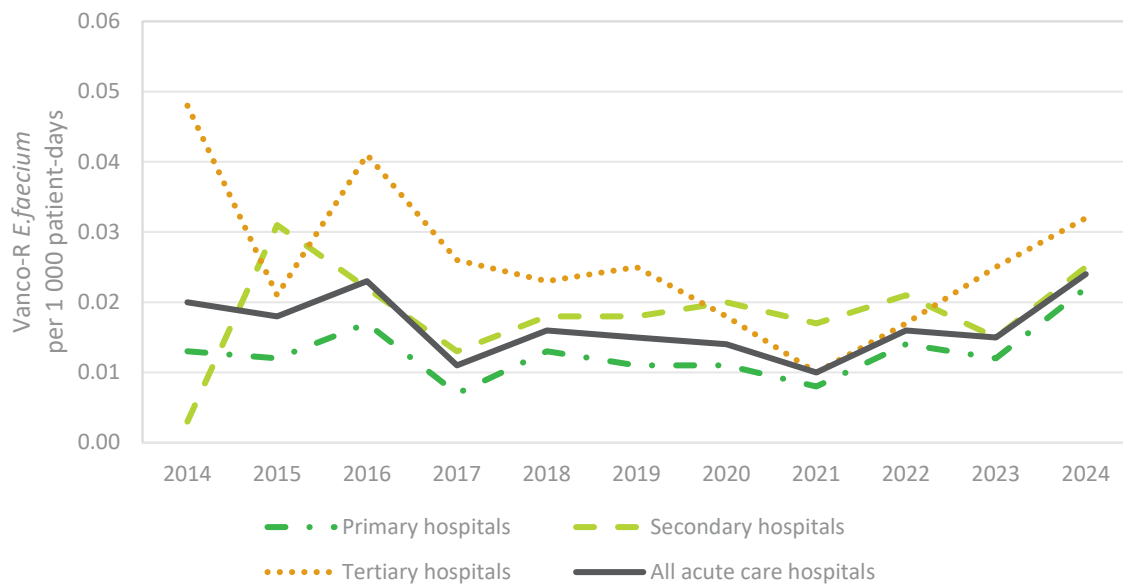
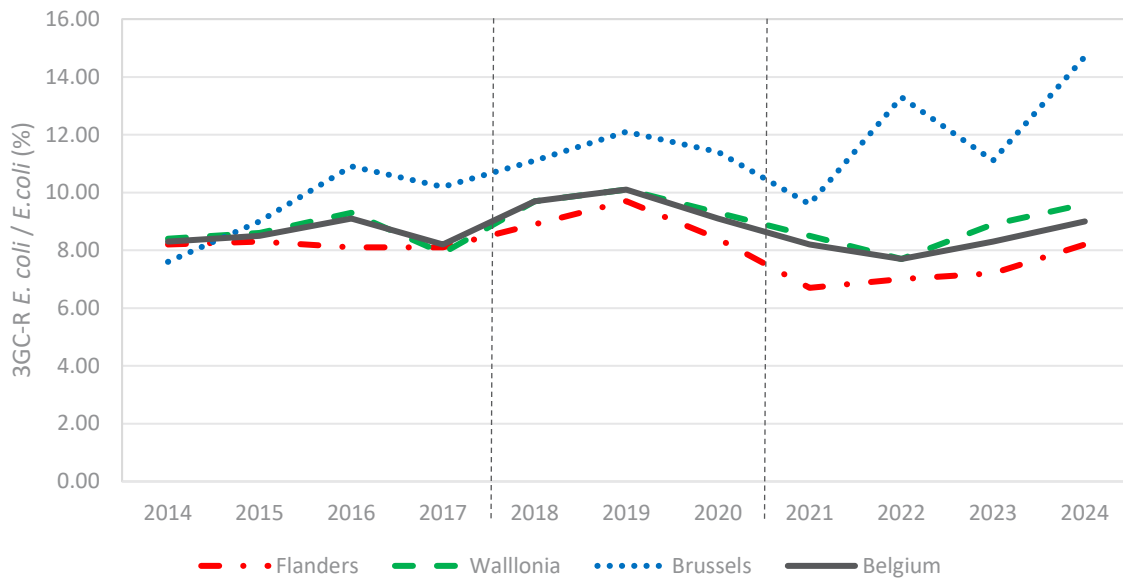


Figure A9. Evolution of the crude incidence density (per 1 000 patient-days) of vancomycin resistance in *Enterococcus faecium* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



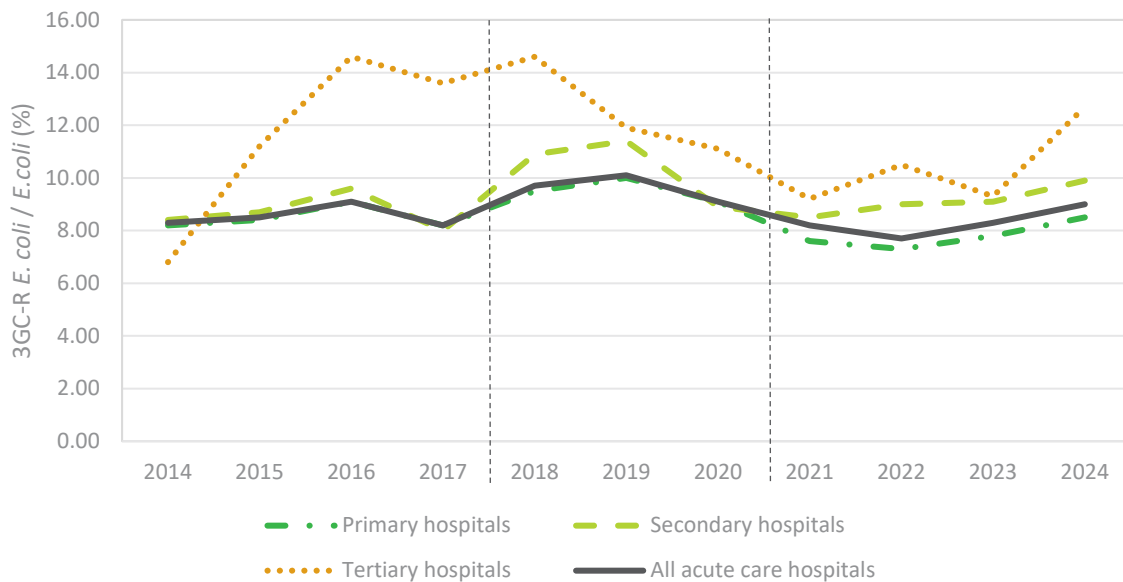
ANNEX

Figure A10. Evolution of the median resistance proportion of *Escherichia coli* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

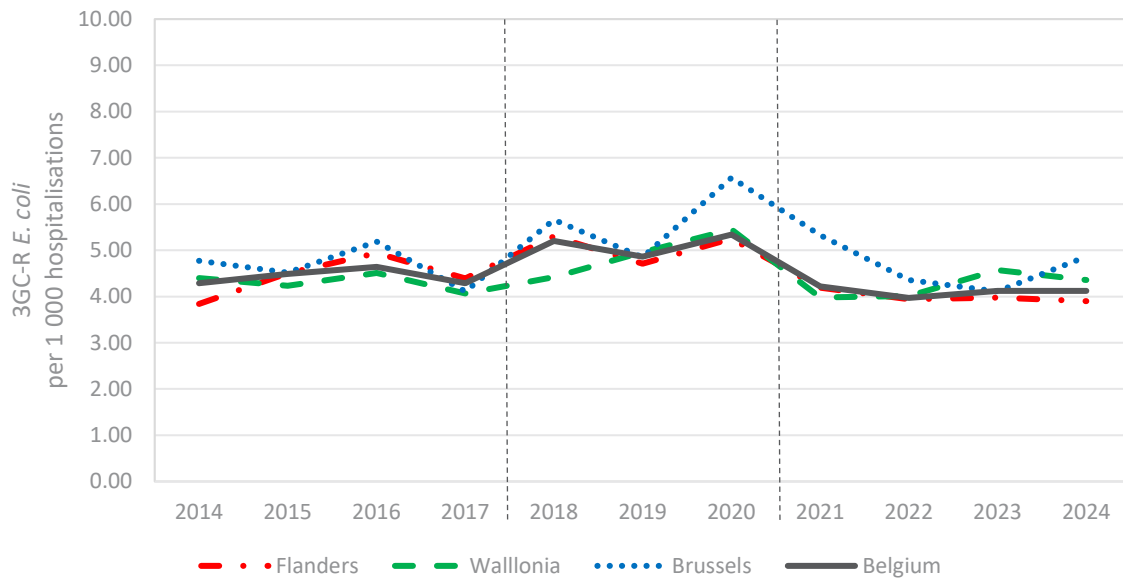
Figure A11. Evolution of the median resistance proportion of *Escherichia coli* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

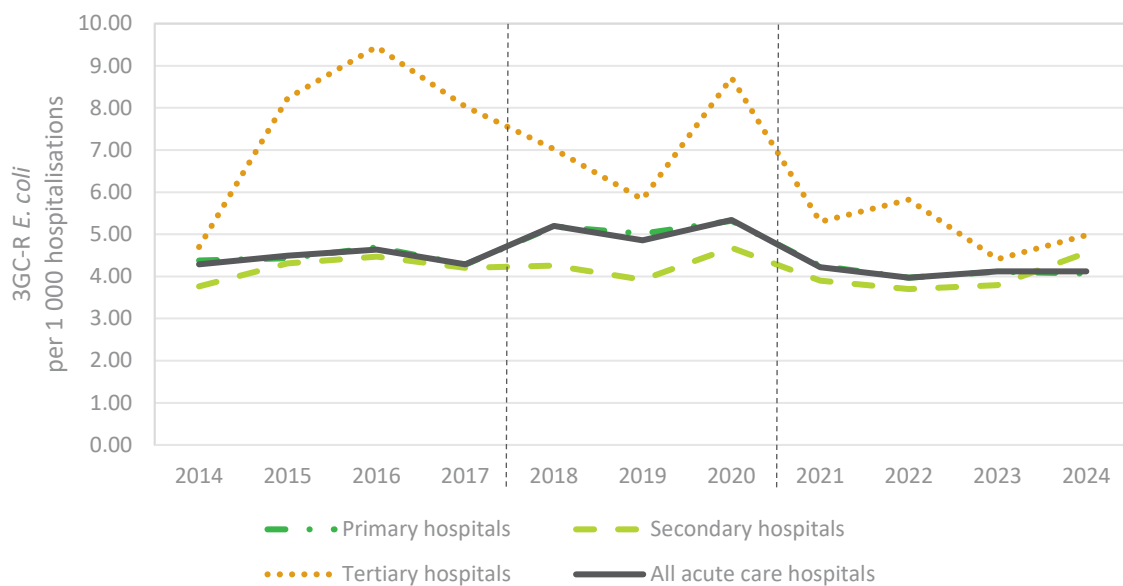
ANNEX

Figure A12. Evolution of the median incidence (per 1 000 hospitalisations) of *Escherichia coli* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

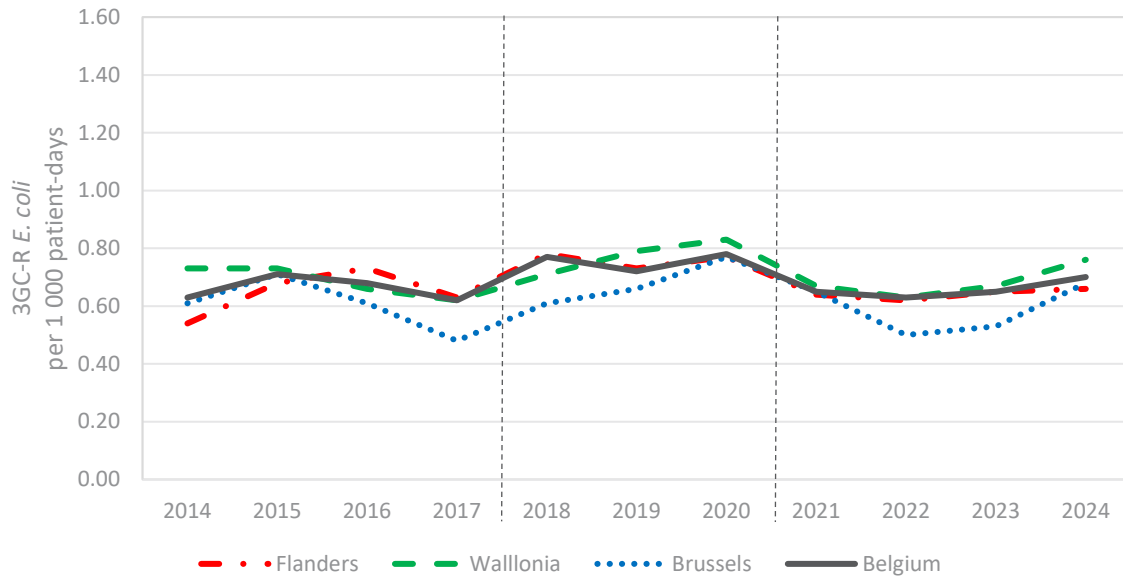
Figure A13. Evolution of the median incidence (per 1 000 hospitalisations) of *Escherichia coli* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

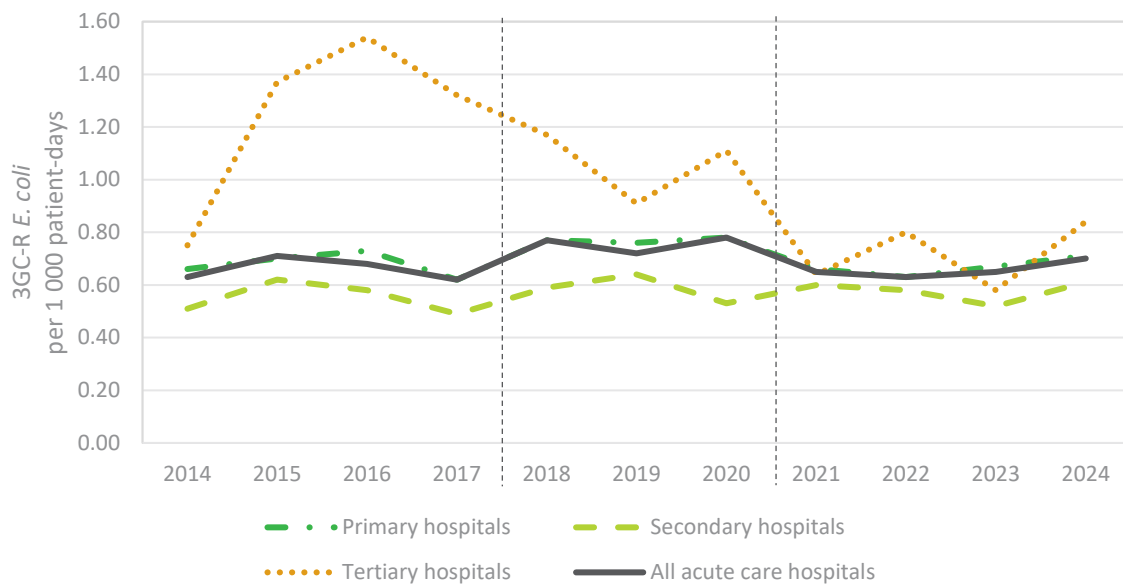
ANNEX

Figure A14. Evolution of the median incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

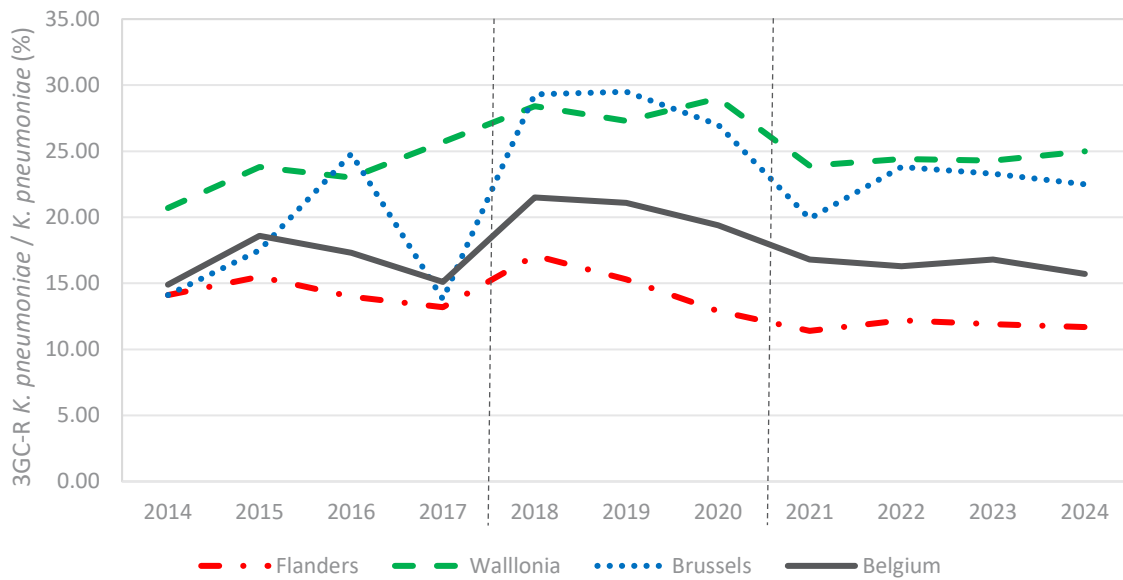
Figure A15. Evolution of the median incidence density (per 1 000 patient-days) of *Escherichia coli* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

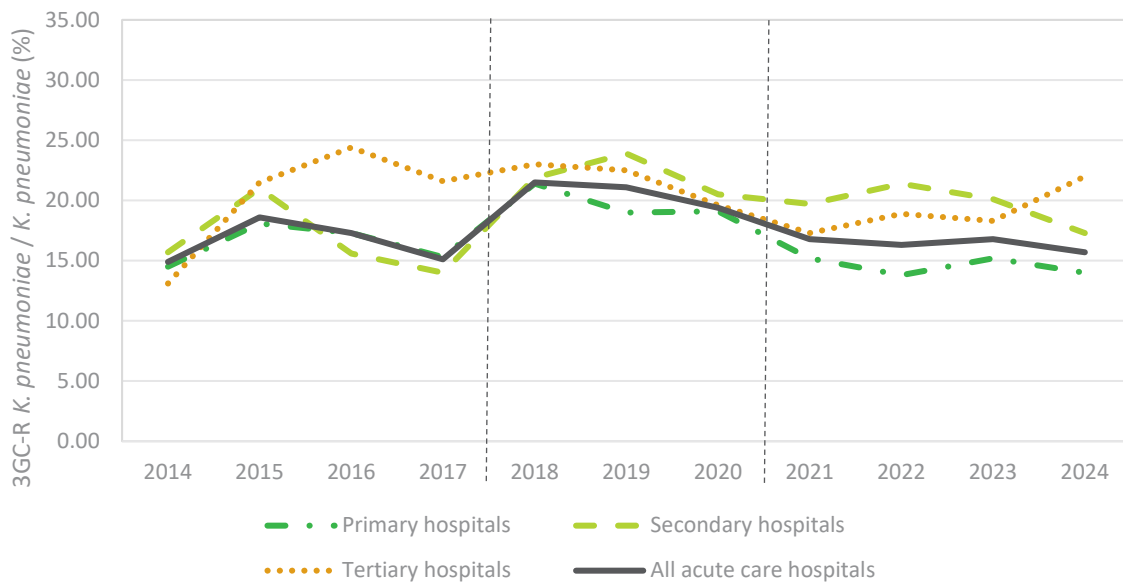
ANNEX

Figure A16. Evolution of the median resistance proportion of *Klebsiella pneumoniae* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

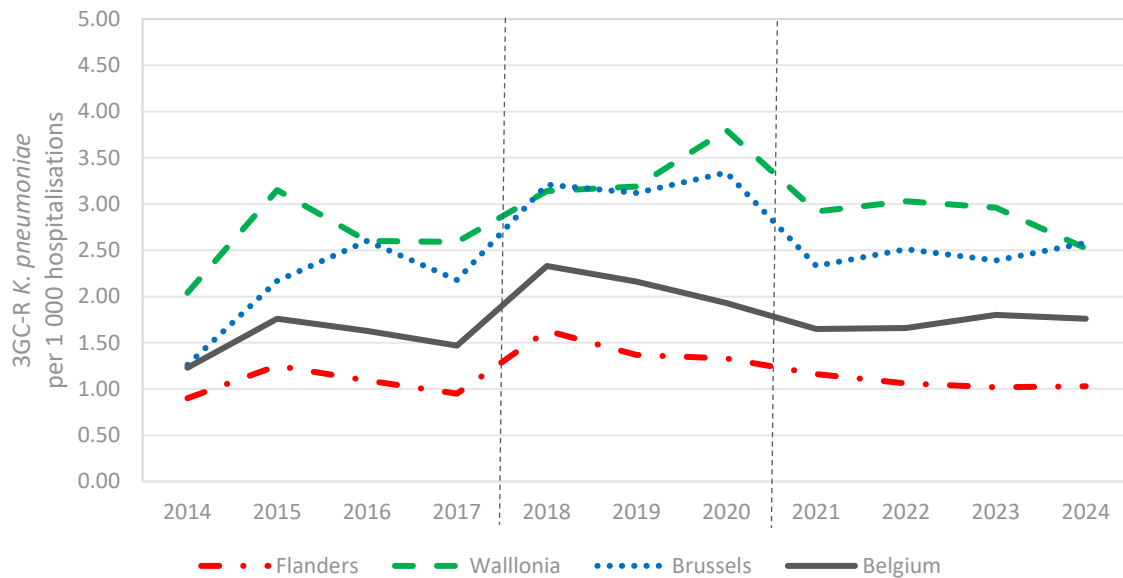
Figure A17. Evolution of the median resistance proportion of *Klebsiella pneumoniae* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

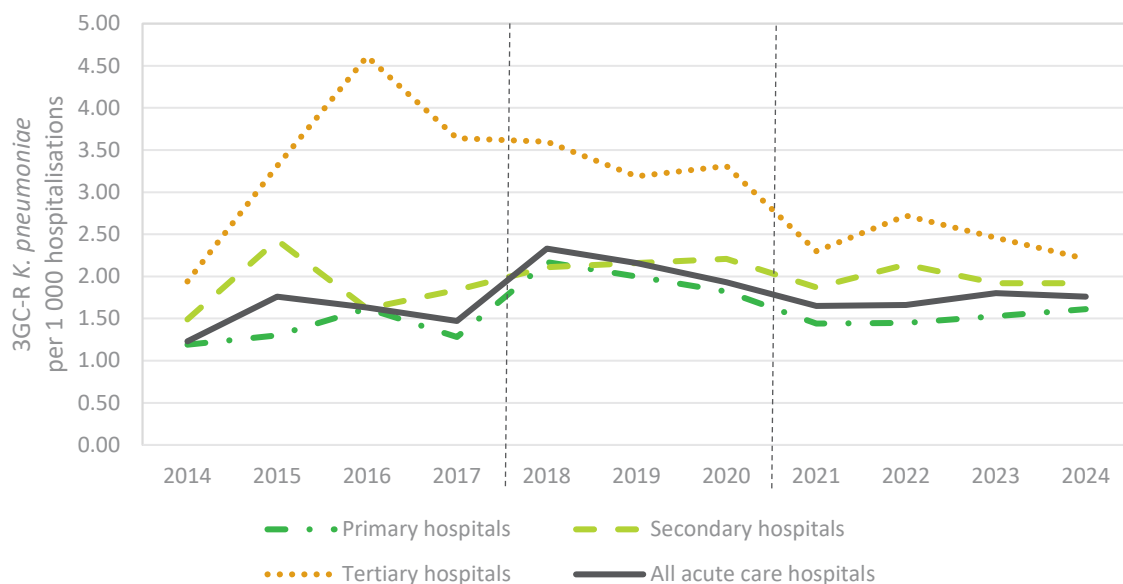
ANNEX

Figure A18. Evolution of the median incidence (per 1 000 hospitalisations) of *Klebsiella pneumoniae* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

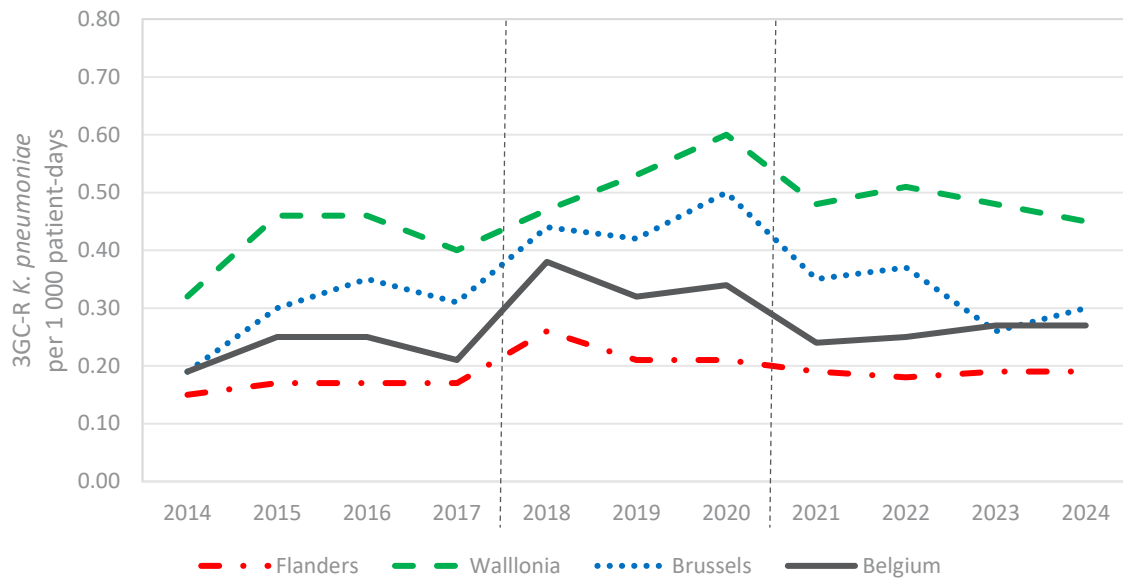
Figure A19. Evolution of the median incidence (per 1 000 hospitalisations) of *Klebsiella pneumoniae* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



3GC-R = resistant to 3rd cephalosporins; **Note:** the dashed vertical lines indicate methodological changes, i.e. a) prior to 2018 non-susceptibility to 4th generation cephalosporins was included, and b) prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

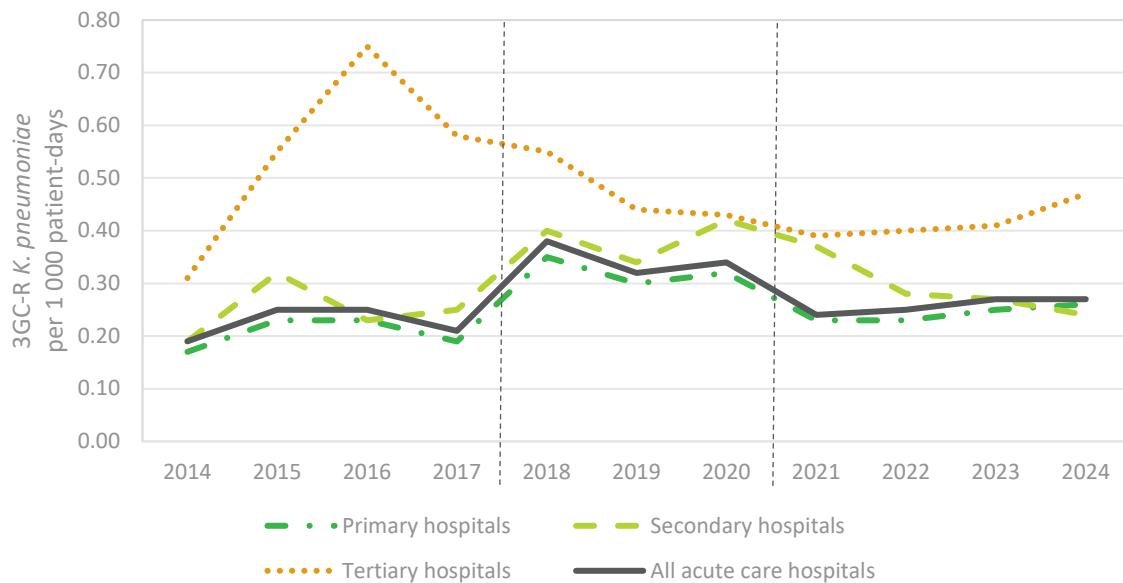
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Figure A20. Evolution of the median incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins by region (clinical samples only), Belgian acute care hospital sites, 2014-2024



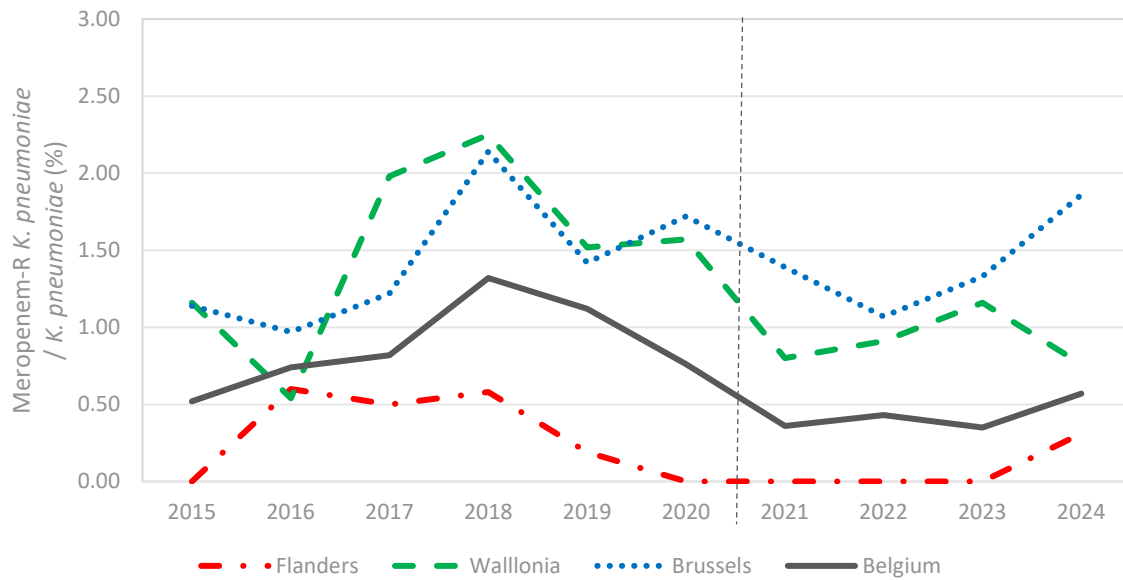
3GC-R = resistant to 3rd cephalosporins; **note:** prior to 2018 non-susceptibility to 4th generation cephalosporins was included, prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A21. Evolution of the median incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to third generation cephalosporins by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2014-2024



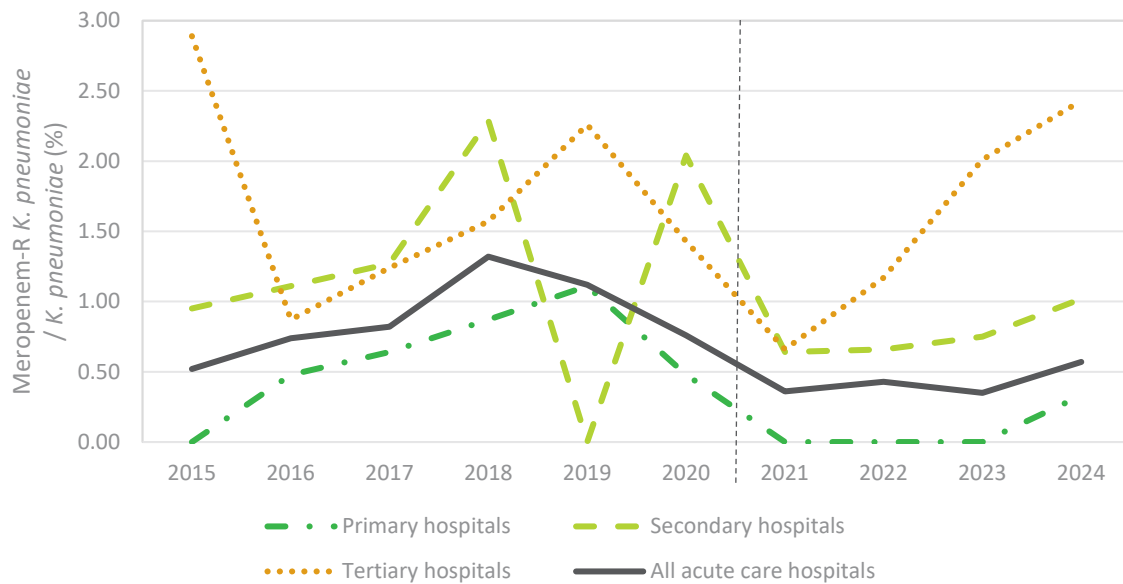
3GC-R = resistant to 3rd cephalosporins; **note:** prior to 2018 non-susceptibility to 4th generation cephalosporins was included, prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A22. Evolution of the median resistance proportion of *Klebsiella pneumoniae* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2015-2024



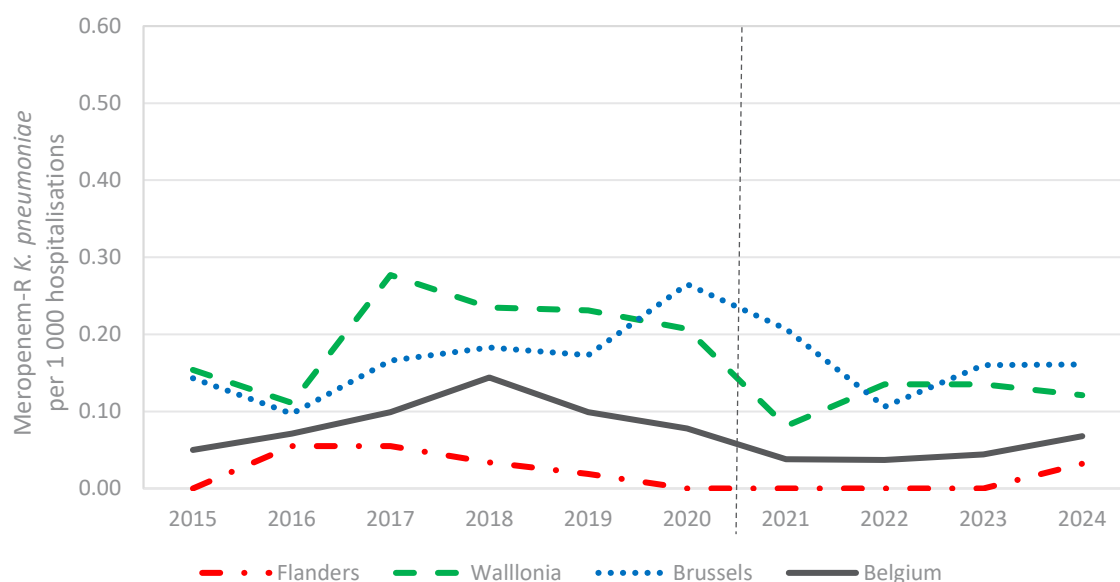
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A23. Evolution of the median resistance proportion of *Klebsiella pneumoniae* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2015-2024



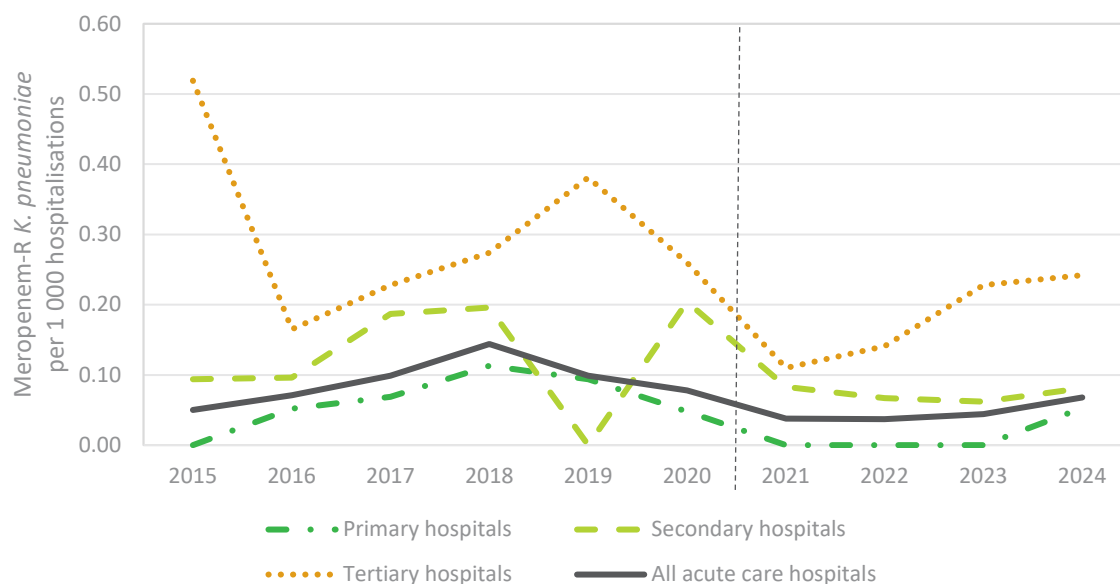
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A24. Evolution of the median incidence (per 1 000 hospitalisations) of *Klebsiella pneumoniae* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2015-2024



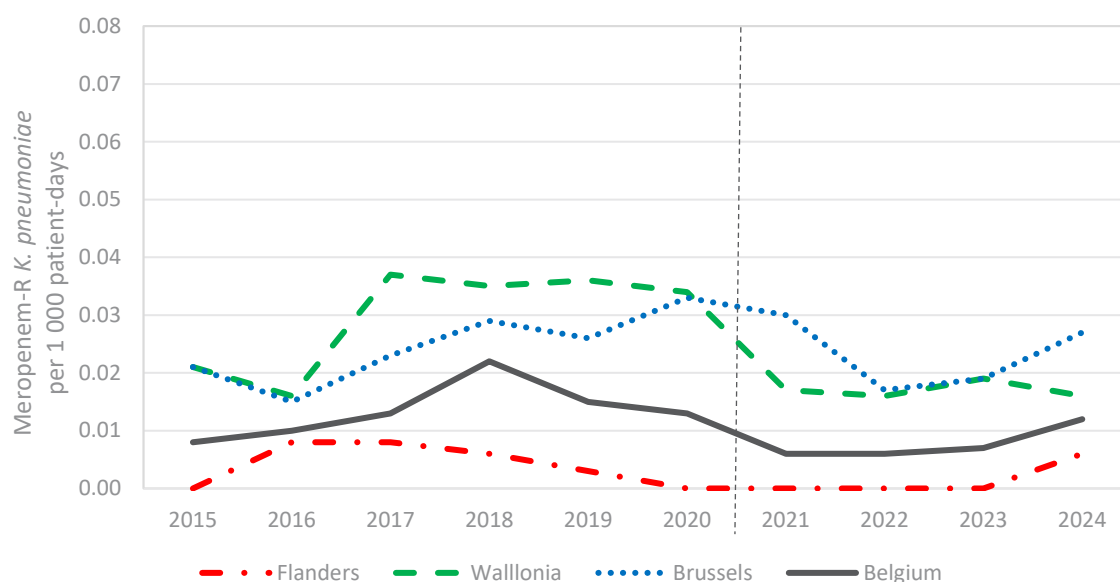
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A25. Evolution of the median incidence (per 1 000 hospitalisations) of *Klebsiella pneumoniae* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2015-2024



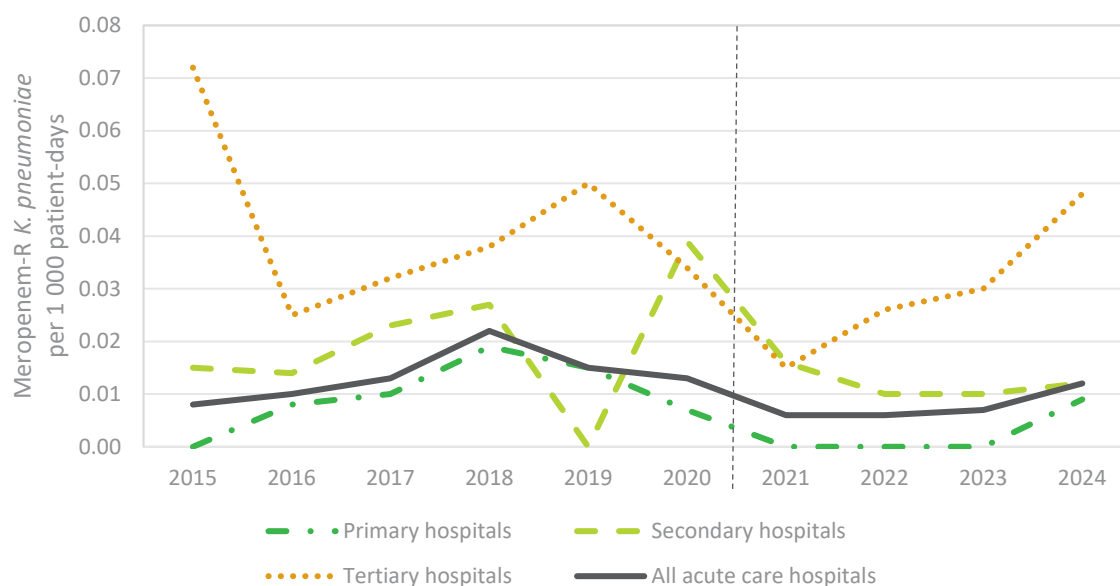
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A26. Evolution of the median incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2015-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

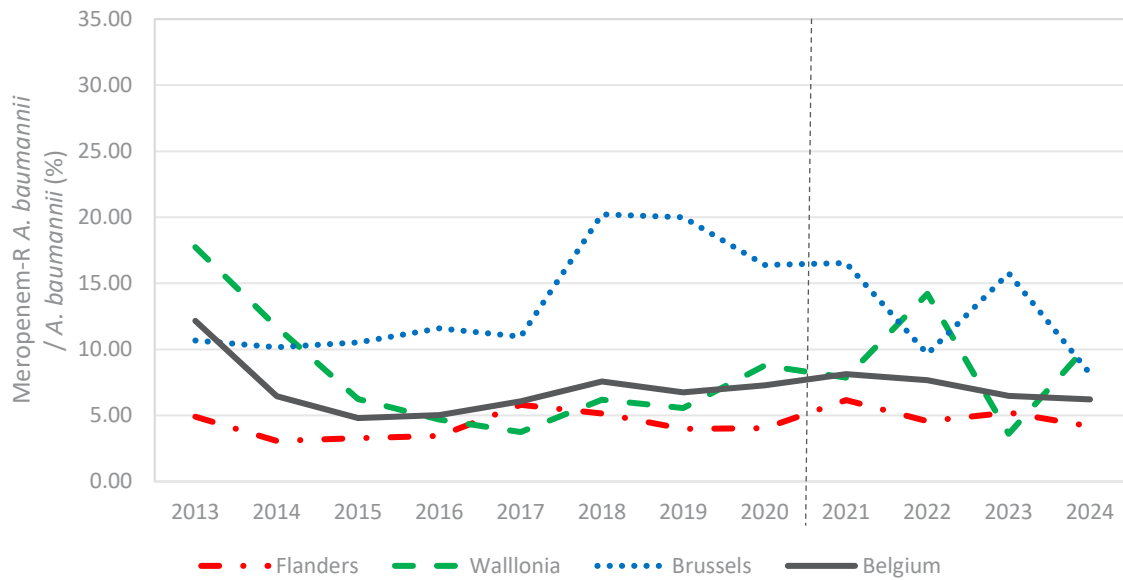
Figure A27. Evolution of the median incidence density (per 1 000 patient-days) of *Klebsiella pneumoniae* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2015-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

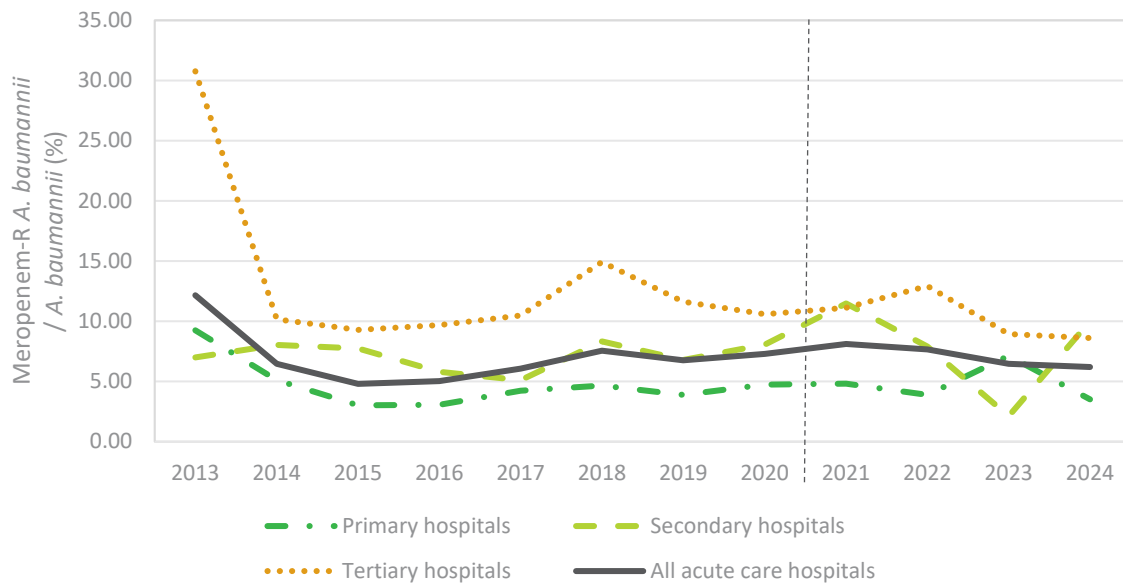
ANNEX

Figure A28. Evolution of the crude resistance proportion of *Acinetobacter baumannii* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2013-2024



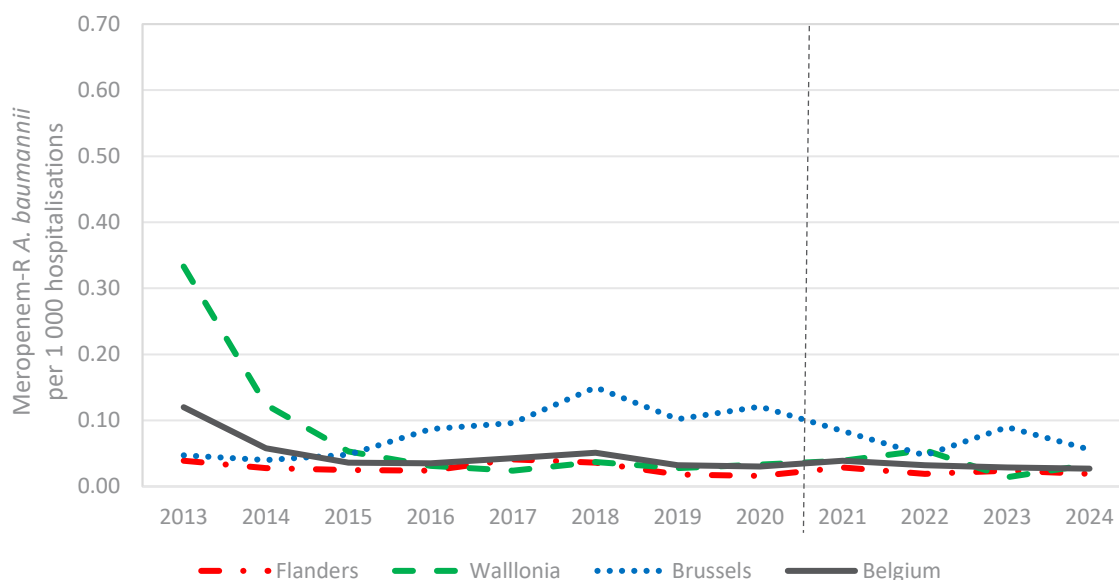
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A29. Evolution of the median resistance proportion of *Acinetobacter baumannii* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2013-2024



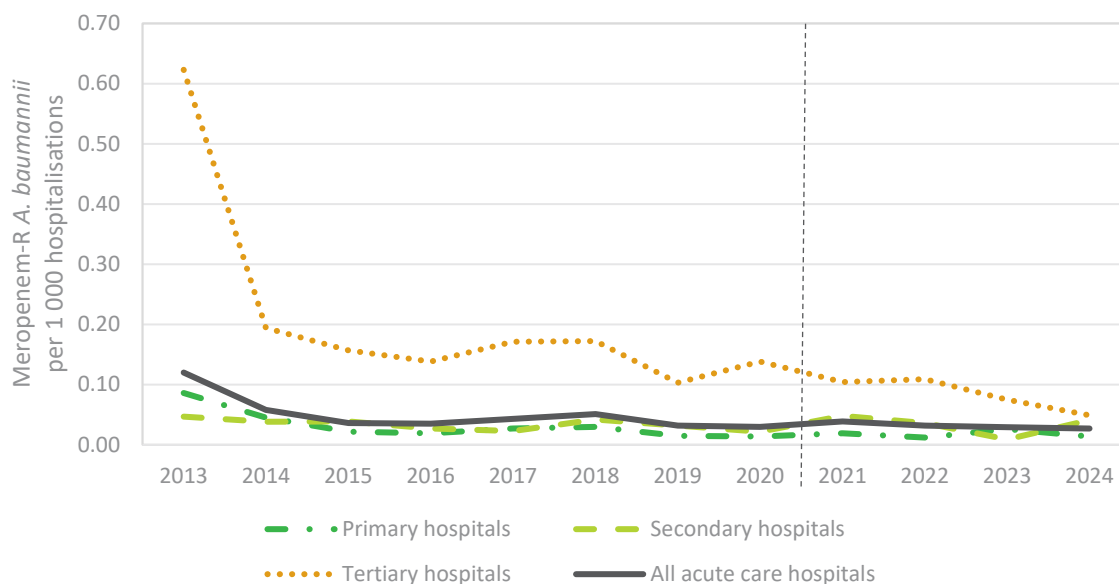
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A30. Evolution of the median incidence (per 1 000 hospitalisations) of *Acinetobacter baumannii* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2013-2024



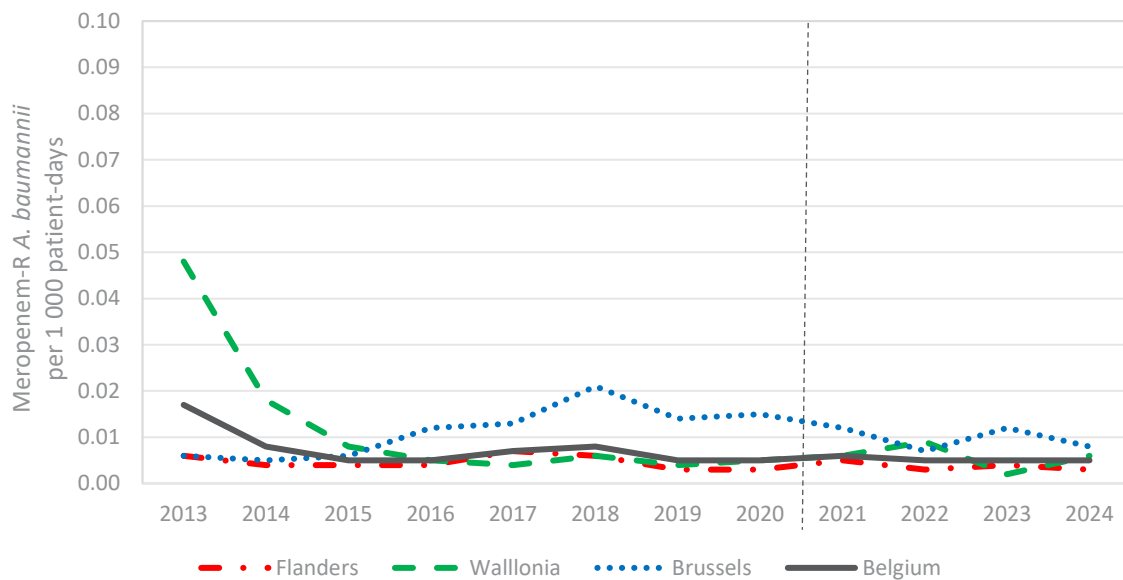
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A31. Evolution of the median incidence (per 1 000 hospitalisations) of *Acinetobacter baumannii* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2013-2024



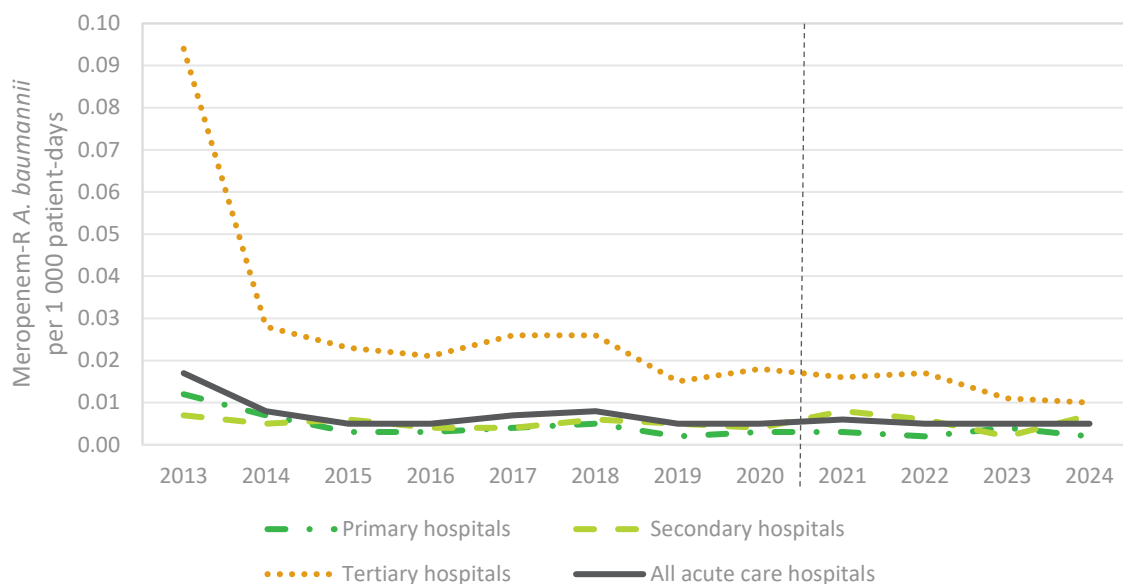
Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

Figure A32. Evolution of the median incidence density (per 1 000 patient-days) of *Acinetobacter baumannii* resistant to meropenem by region (clinical samples only), Belgian acute care hospital sites, 2013-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

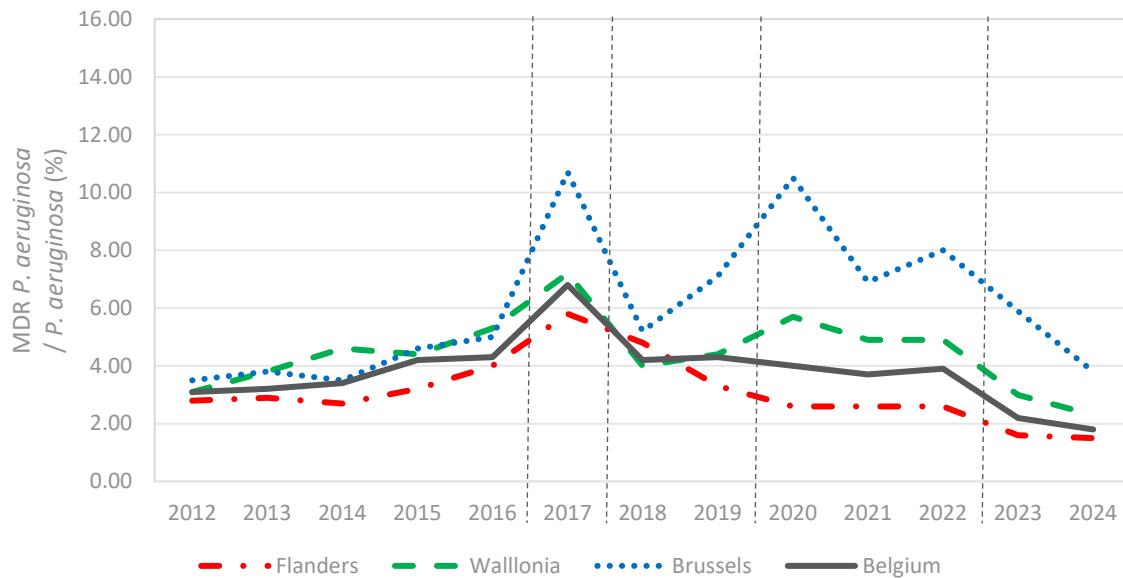
Figure A33. Evolution of the median incidence density (per 1 000 patient-days) of *Acinetobacter baumannii* resistant to meropenem by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2013-2024



Note: the dashed vertical line indicates a methodological change, i.e. prior to 2021 I/R (resistant, incl. also susceptible, increased exposure (intermediate result)) is displayed.

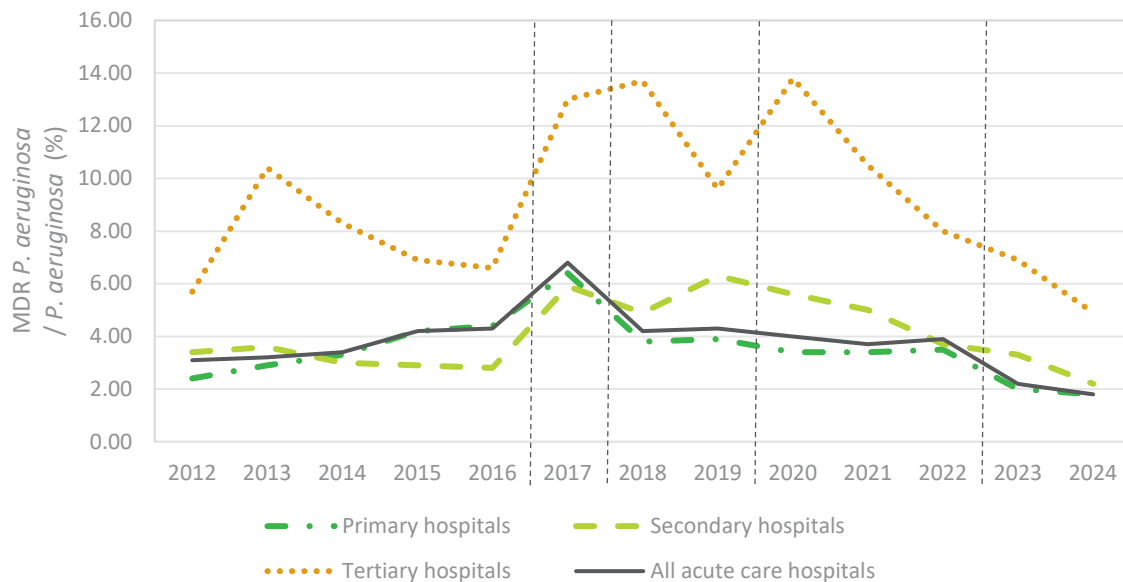
ANNEX

Figure A34. Evolution of the median resistance proportion of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by region (clinical samples only), Belgian acute care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

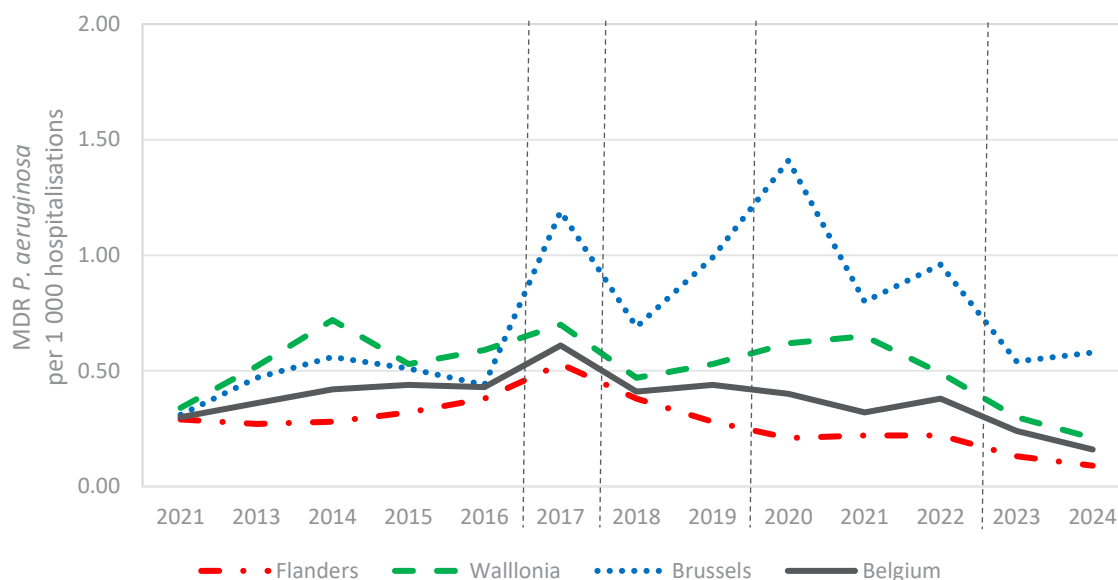
Figure A35. Evolution of the median resistance proportion of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

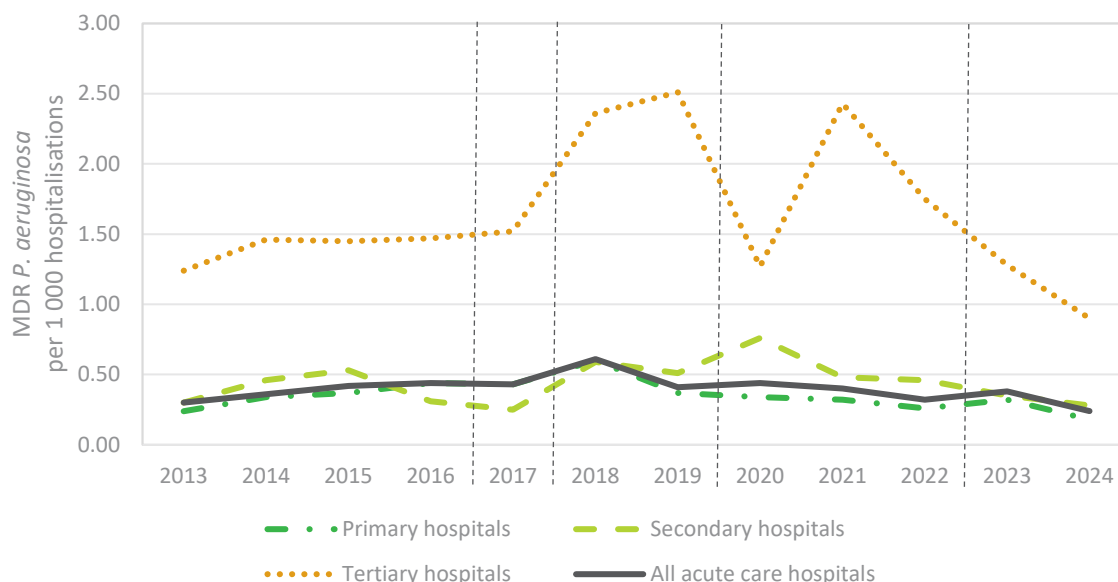
ANNEX

Figure A36. Evolution of the median incidence (per 1 000 hospitalisations) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by region (clinical samples only), Belgian acute care hospital sites, 2012-2024



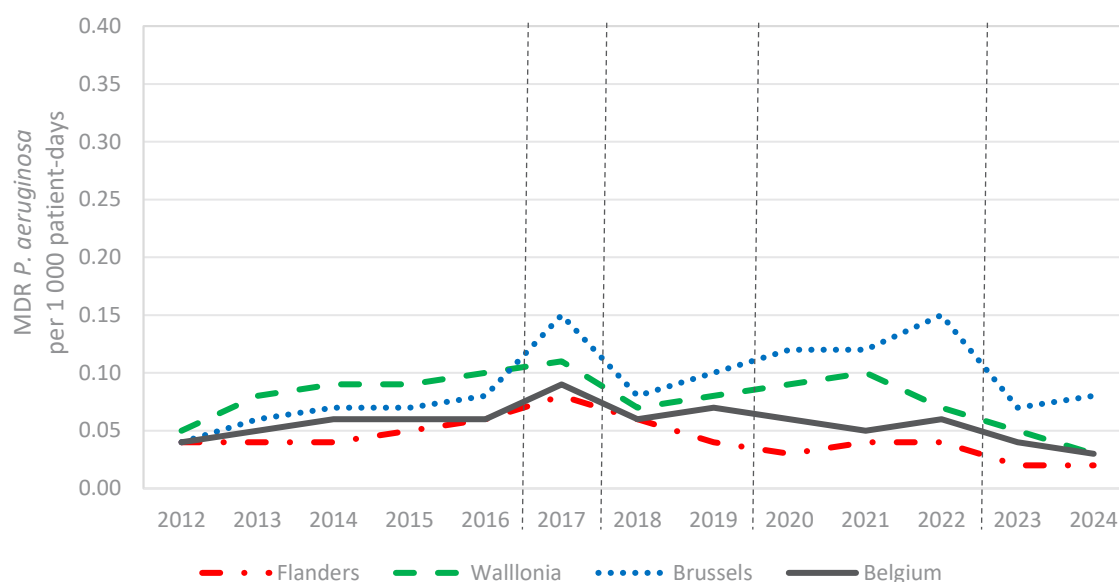
Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

Figure A37. Evolution of the median incidence (per 1 000 hospitalisations) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2012-2024



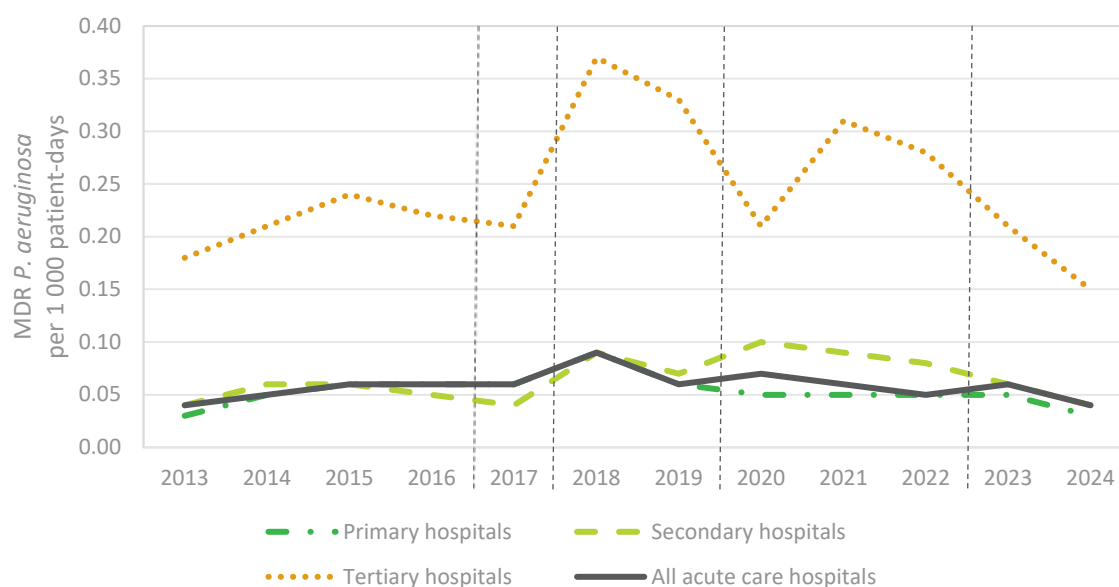
Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

Figure A38. Evolution of the median incidence density (per 1 000 patient-days) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by region (clinical samples only), Belgian acute care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

Figure A39. Evolution of the median incidence density (per 1 000 patient-days) of multidrug-resistant (MDR) *Pseudomonas aeruginosa* by level of specialty care (clinical samples only), Belgian acute care hospital sites, 2012-2024



Note: the dashed vertical lines indicate methodological changes, i.e. a) Between 2016 and 2017, the definition of MDR *P. aeruginosa* changed from reduced susceptibility (I or R) to at least one antibiotic in four out of the five following antibiotic classes to reduced susceptibility to at least three of the following antibiotic classes: fluoroquinolones (ciprofloxacin, levofloxacin), aminoglycosides (gentamicin, tobramycin, amikacin), carbapenems (meropenem, imipenem), 3rd and/or 4th generation cephalosporins (ceftazidime, cefepime) and anti-pseudomonas penicillins (piperacillin/tazobactam), b) In 2018, anti-pseudomonas penicillins (piperacillin/tazobactam) were dropped from the definition, c) Since 2020, only strict resistance (R- excluding susceptible, increased exposure (intermediate result)) is considered, and d) In 2023, gentamicin was removed from the list of aminoglycosides.

REPORT APPROVAL OF VARIOUS ENTITIES

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Table. Dates that the different entities have been sent and approved* this report

Entity	Received	Approved
Sciensano	N.A.	17/02/2026
NRC for resistant Gram-negative bacilli	9/02/2026	17/02/2026
NRC for resistant enterococci	9/02/2026	13/02/2026
NRC for <i>Staphylococcus aureus</i> and other <i>Staphylococci</i>	9/02/2026	16/02/2026
BAPCOC (Belgian Antibiotic Policy Coordination Committee)	18/03/2026	19/04/2026
TC-MDRO (Technical committee multidrug resistant organisms)	18/03/2026	No comments received by the proposed deadline
Regional authorities		
Flanders	18/03/2026	17/04/2026
Wallonia	18/03/2026	10/04/2026
Brussels	18/03/2026	No comments received by the proposed deadline

*includes passive approval

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